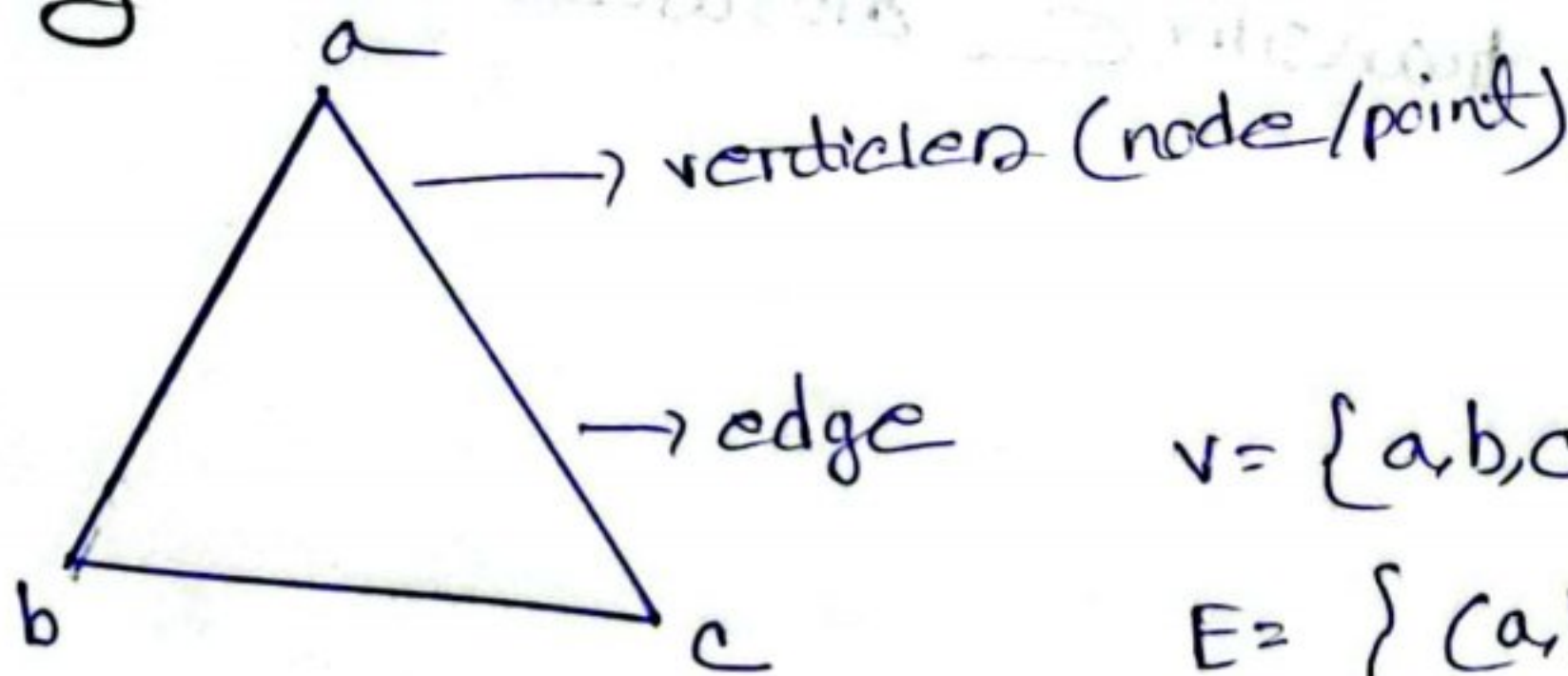


Graph

A graph is defined as an ordered pair of a set of vertices and a set of edges

$$G = (V, E) \text{ or } G(V, E)$$

Here, V is the set of vertices and E is the set of edges connecting the vertices



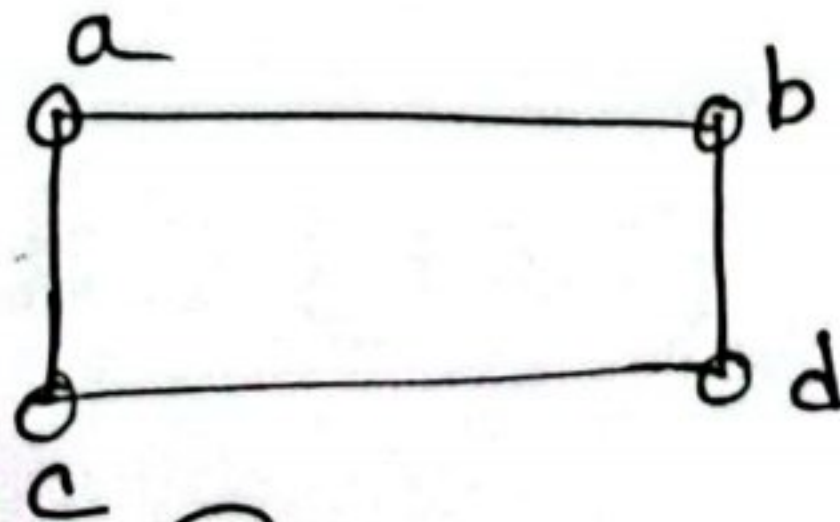
$$V = \{a, b, c\}$$

$$E = \{(a, b), (b, c), (a, c)\}$$

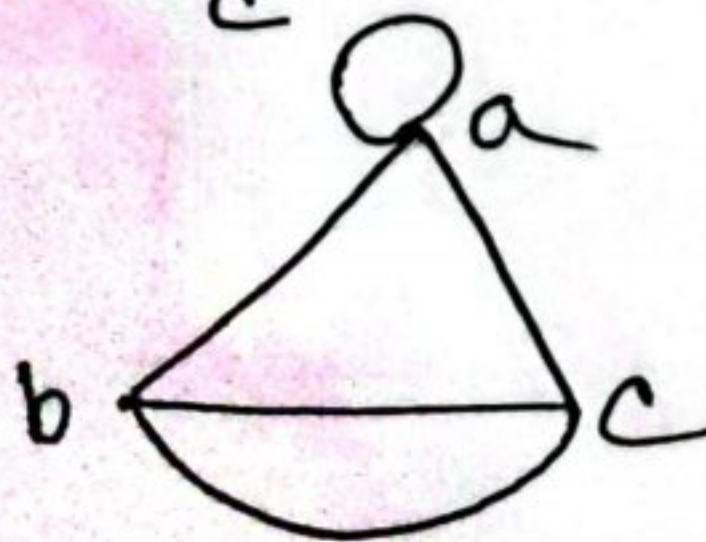
Classification of graph:

* Based on the existence of loop or multiple edges -

1 Simple graph

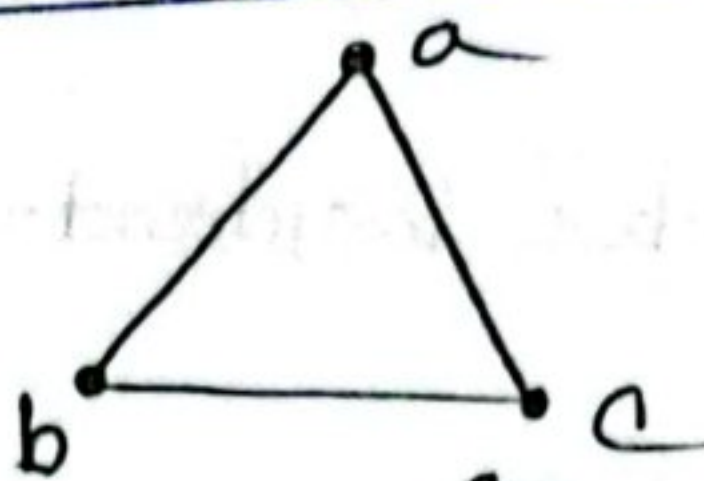


2 Multigraph



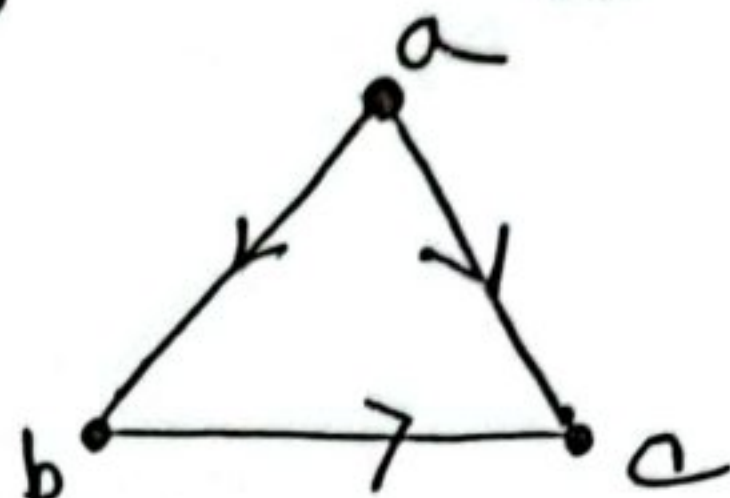
* Based on the orientations of the edges:-

1 Undirected graph



$(a, b) (b, a)$
unordered ~~graph~~ pair

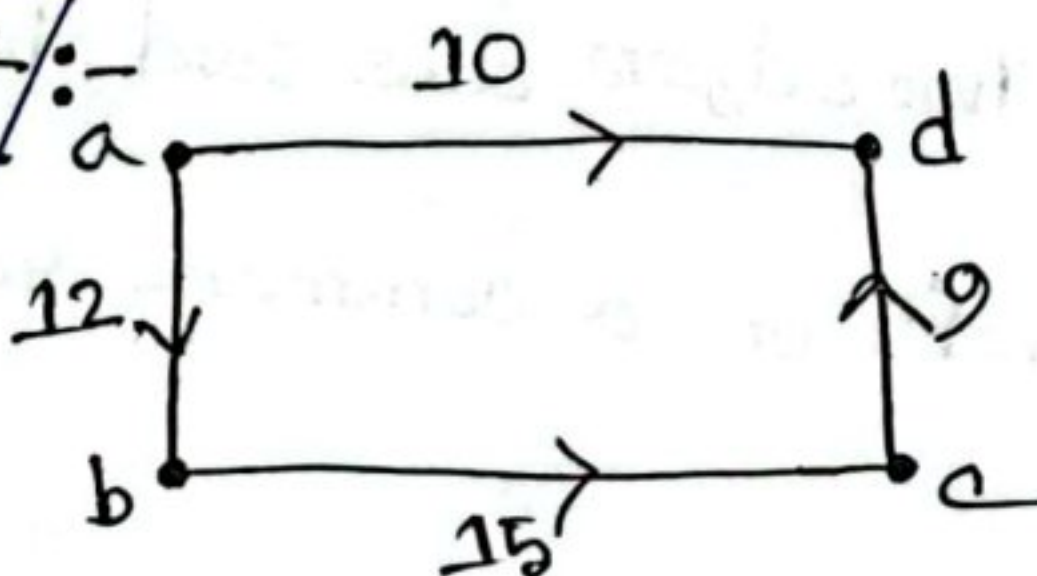
2 Directed or digraph



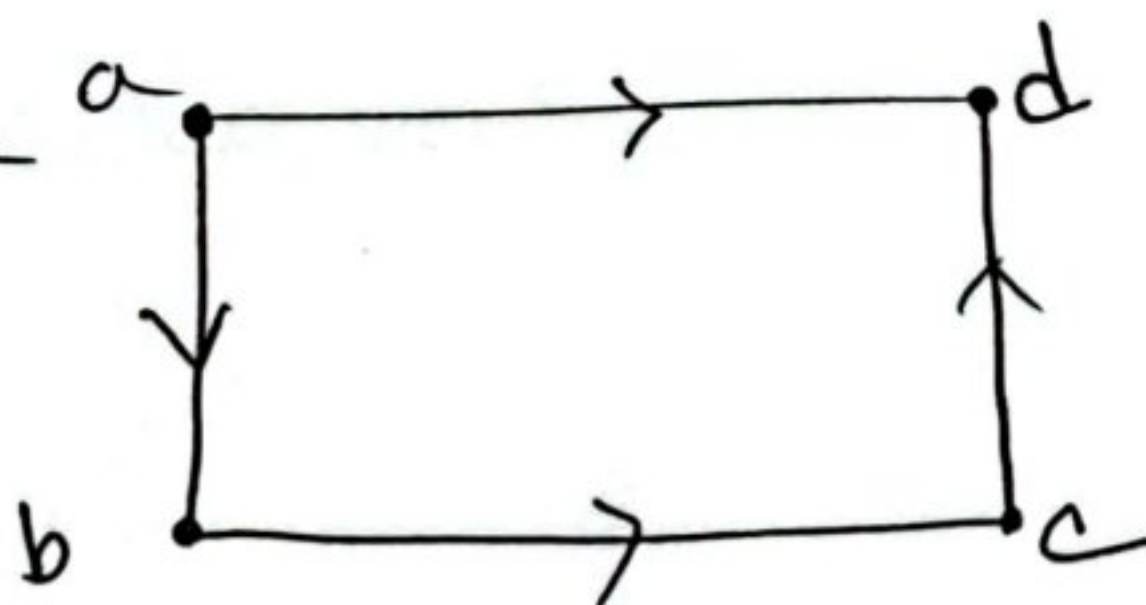
$(a, b) (b, a)$
ordered pair

* Based on height:-

1 Weighted graph

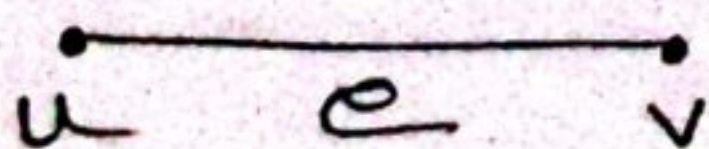


2 Unweighted graph

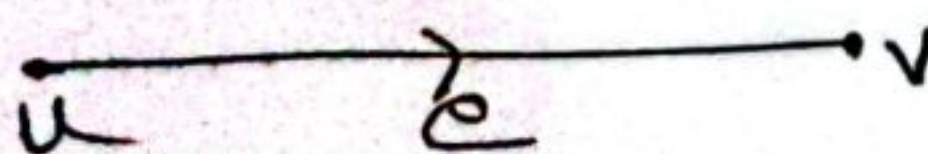


Important definitions

① Adjacent vertex: vertex u and v said to be adjacent if there is an edge. = $\{u, v\}$



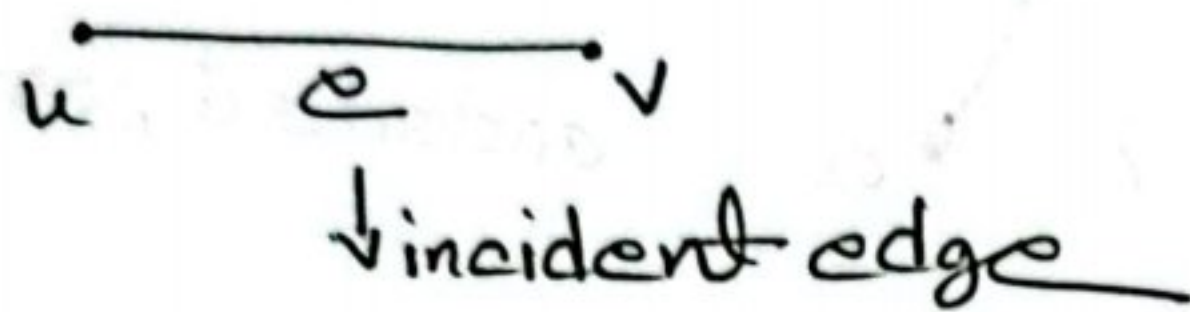
$\{u, v\}$ is adjacent with $e \in \{u, v\}$



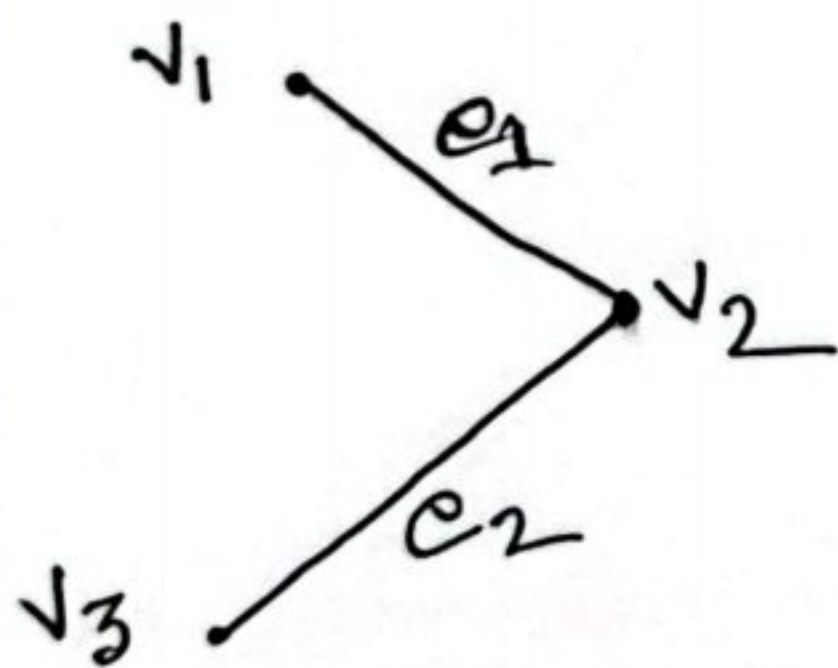
u is adjacent to v , and
 v is adjacent from u

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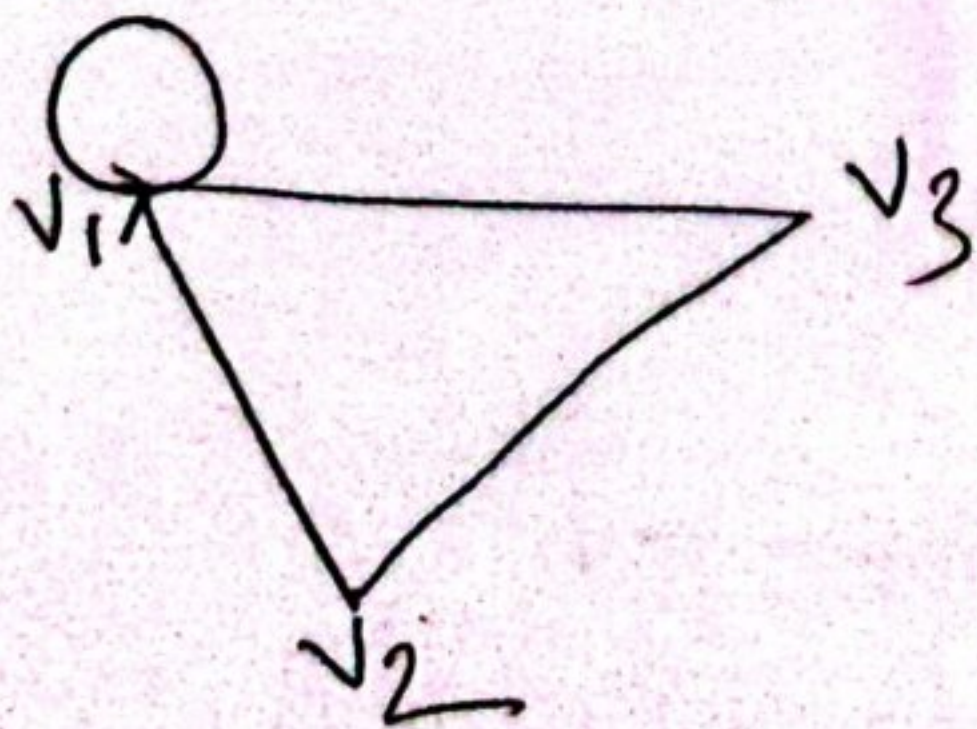
2) Incident edge: The edge (e) that joins the nodes u and v is said to be incident on each of its end points u and v .



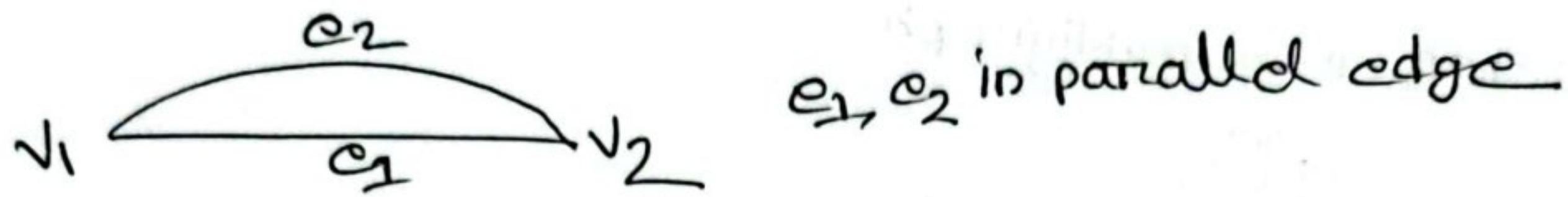
3) Adjacent edge: Two edges are said to be adjacent if they are incident on a common vertex.



4) Self loop: If there is an edge from v_i to v_i then that edge is called self loop.



5. Parallel edge: If two edges have same ~~end~~ end points then the edges are parallel edges.



6. Isolated vertex: A vertex having no edge incident on it is an isolated vertex.

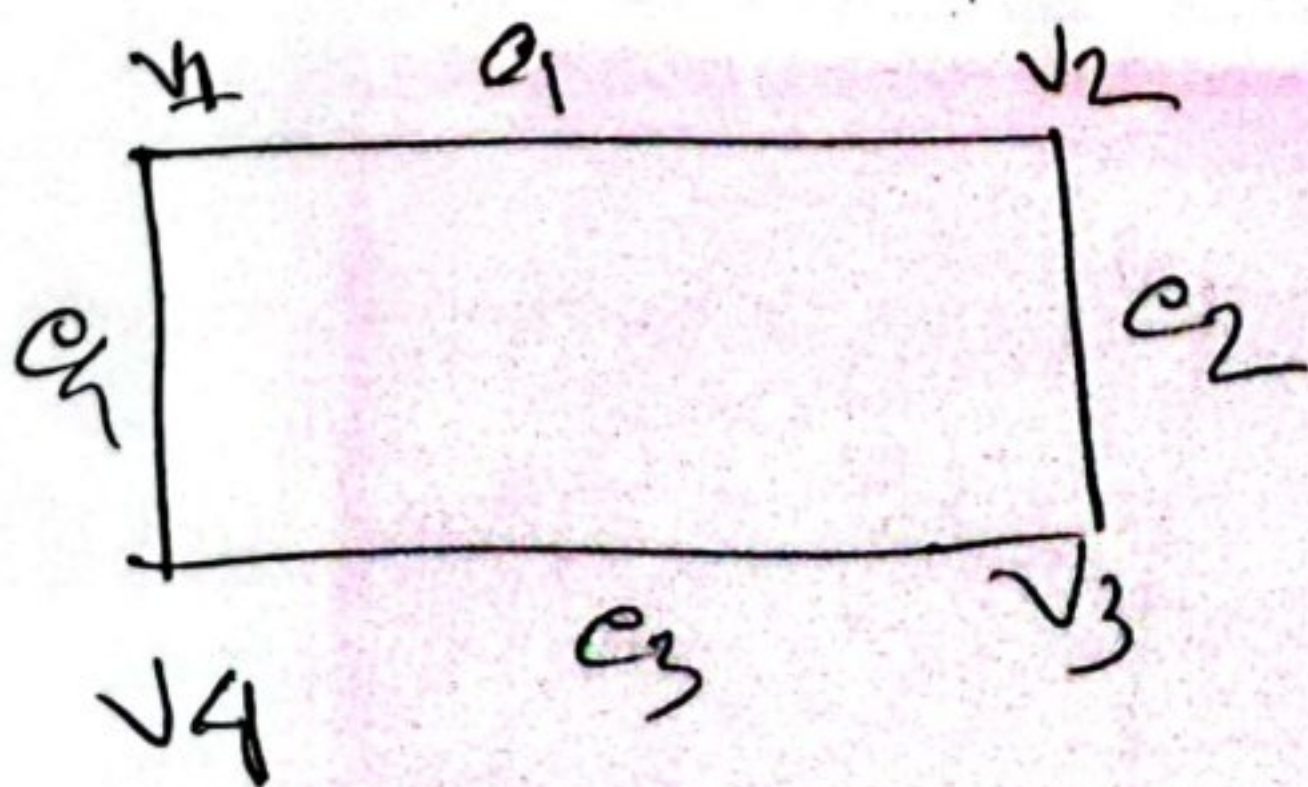


v_3

v_3 isolated vertex

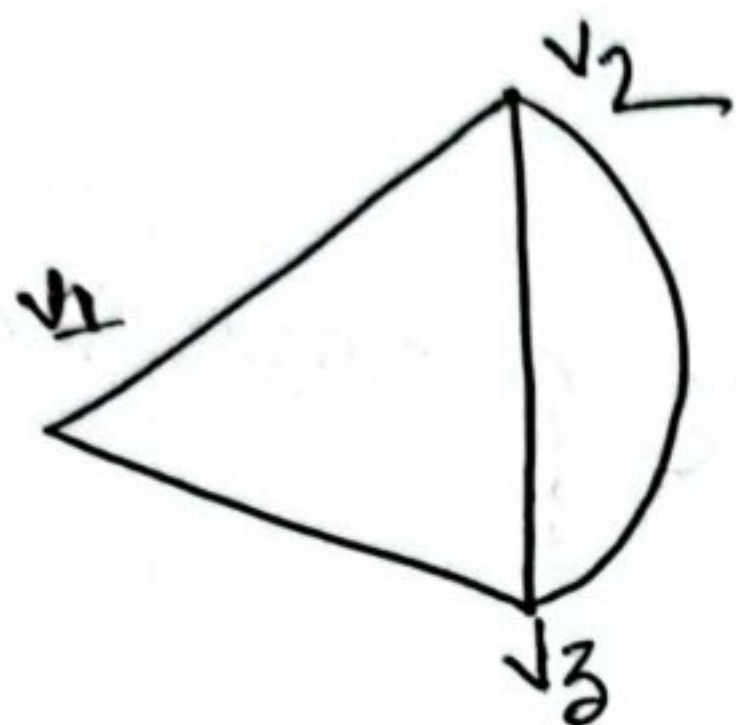
7. Simple graph

A graph which has no self-loops or parallel edge is called simple graph.



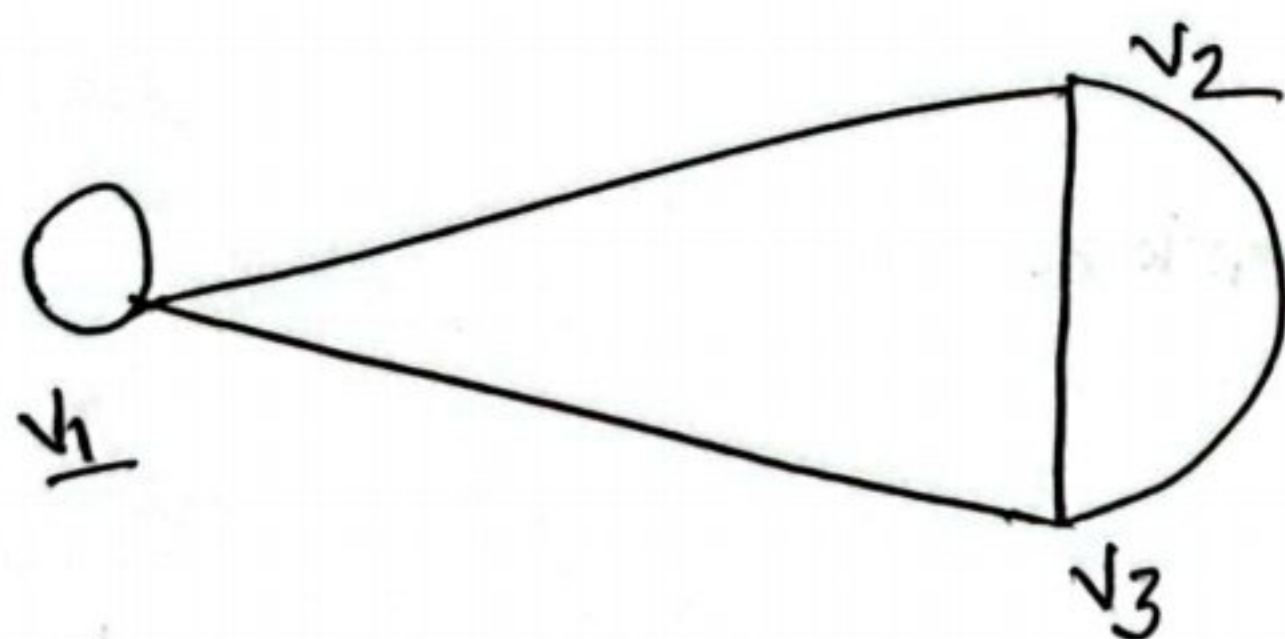
8) Multigraph:

A graph which has more than one edge between a pair of vertices is a multigraph.



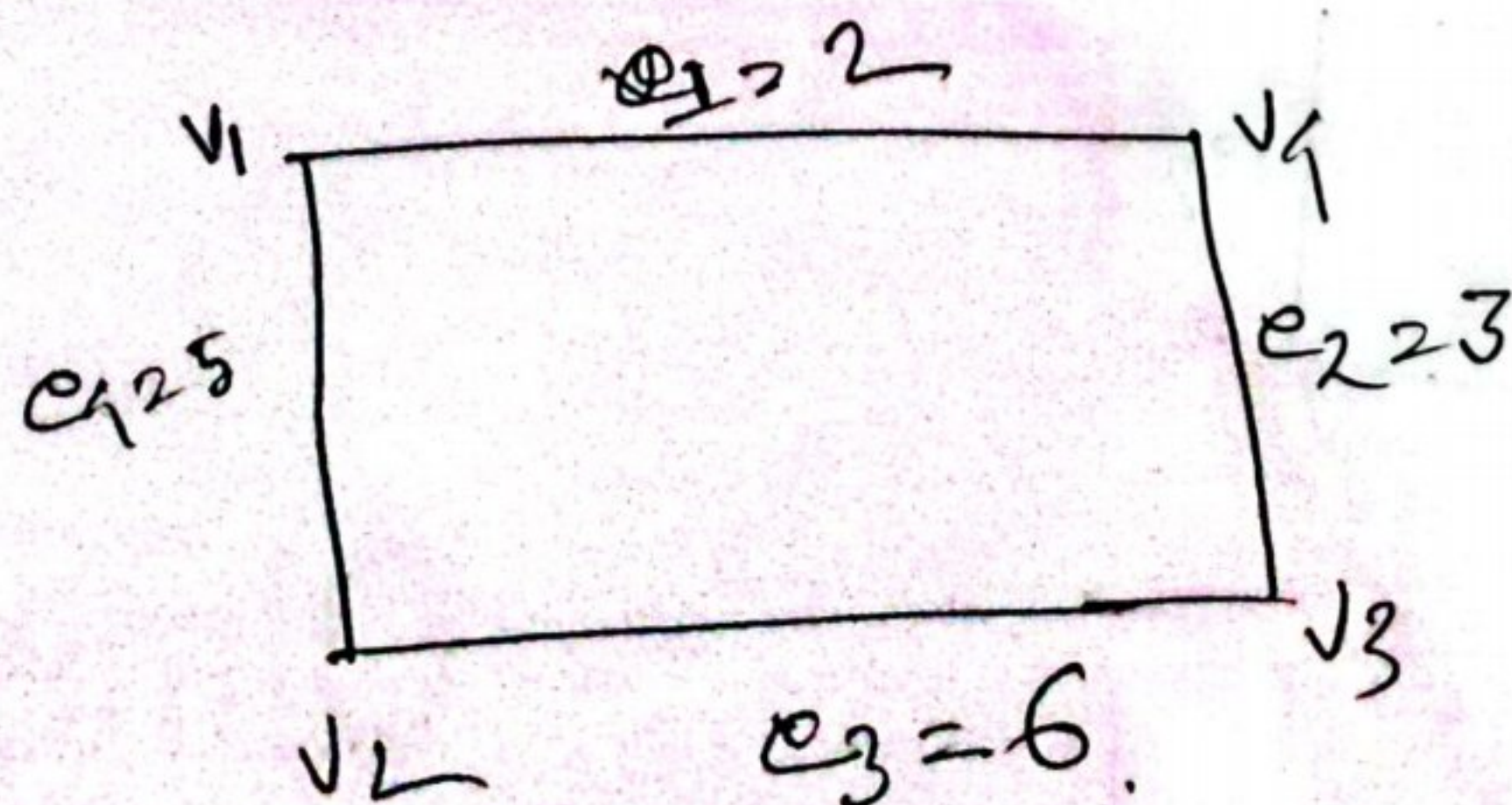
9. Pseudograph: A graph in which self loops and parallel

edges/multiple edges are allowed is called pseudograph



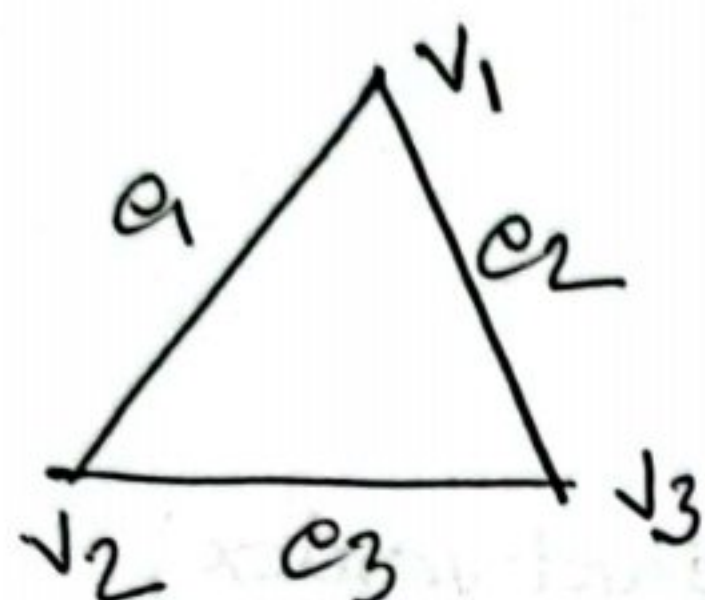
10. Weighted graph:

A graph, in which weights are assigned to every edge is called weighted graph.



11. finite graph A graph $G = (V, E)$ in which V and E

are finite set is called a finite graph.



$$V = \{v_1, v_2, v_3\}$$

$$E = \{e_1, e_2, e_3\}$$

Degree of a vertex

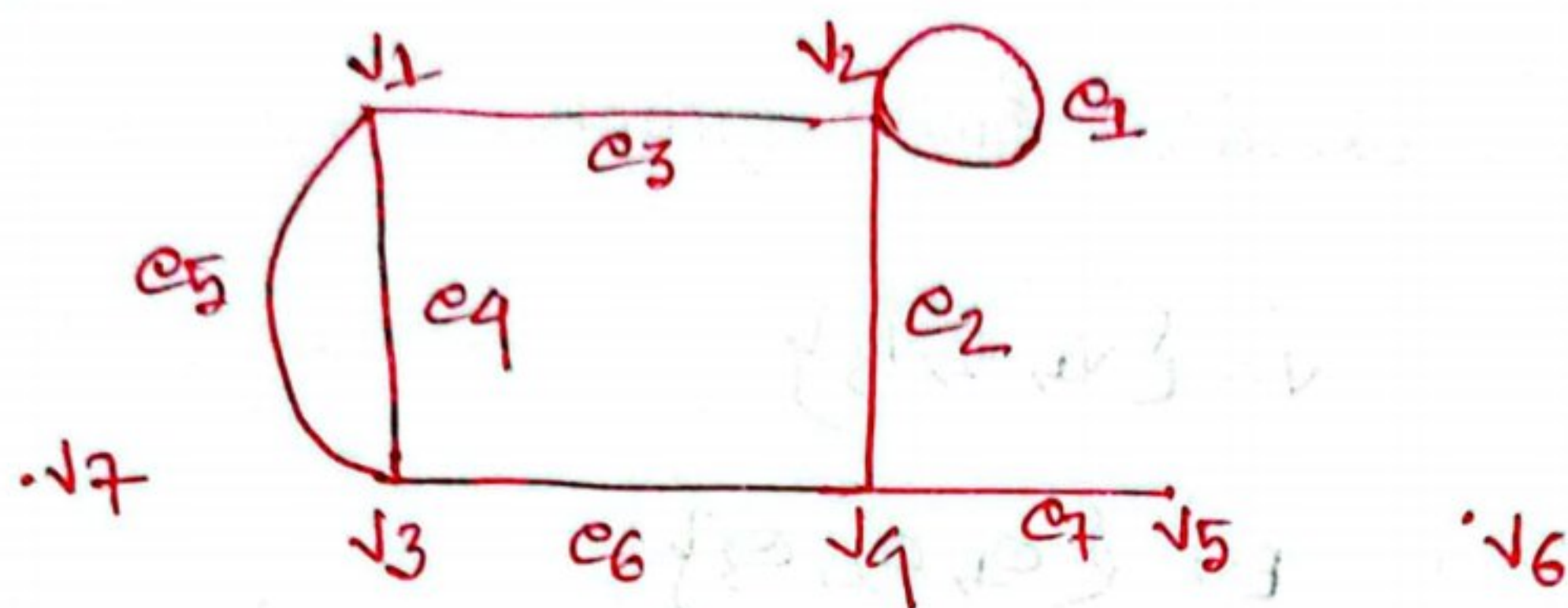
The number of edges incident on a vertex v , with self loops counted twice, is called degree of the vertex v and denoted by $\deg(v)$ or $\deg(v)$ or $d(v)$.

Isolated vertex: A vertex having no incident edge is called an isolated vertex. In other words, isolated vertices are those with zero degree.

Pendant vertex: A vertex of degree 1, is called a pendant or an end vertex.

Internal vertex: A vertex which is neither a pendant vertex nor an isolated vertex, is called an internal vertex or an intermediate vertex.

Q.1 find the degree of all vertex



$$\deg(v_1) = 3$$

$$\deg(v_2) = 4$$

$$\deg(v_3) = 3$$

$$\deg(v_4) = 3$$

internal vertex

$$\deg(v_5) = 1 \text{ (pendant vertex)}$$

$$\deg(v_6) = 0$$

$$\deg(v_7) = 0$$

isolated vertex

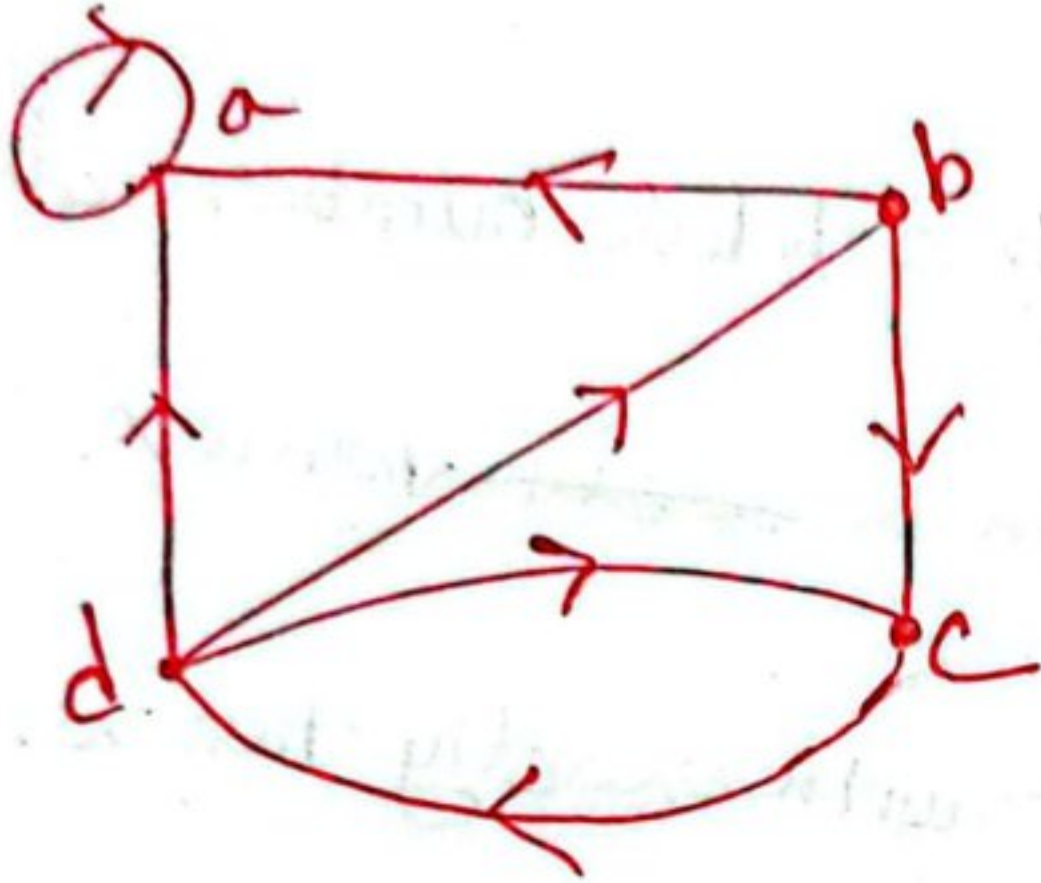
Degree's in Digraph

Indegree: The indegree of vertex v in a directed graph is the number of edges which are coming into the vertex (that is, the number of incoming edges) and it is denoted by $\bar{d}(v)$, $\deg^-(v)$ or $\text{Indeg}(v)$.

Outdegree: The outdegree of vertex v in a directed graph is the number of edges which are going out from the vertex v (that is, the number of outgoing

edges) and is denoted by $d^+(v)$, $\deg^+(v)$ or $\text{outdeg}(v)$.

Q-2: find the degree of vertex of the directed graph



Vertex	Indegree	Outdegree	Total
a	3	1	4
b	1	2	3
c	2	1	3
d	1	3	4

Handshake Theorem

Let $G = (V, E)$ be a directed graph with e edges. Then

$$\sum \deg(v) = 2e$$

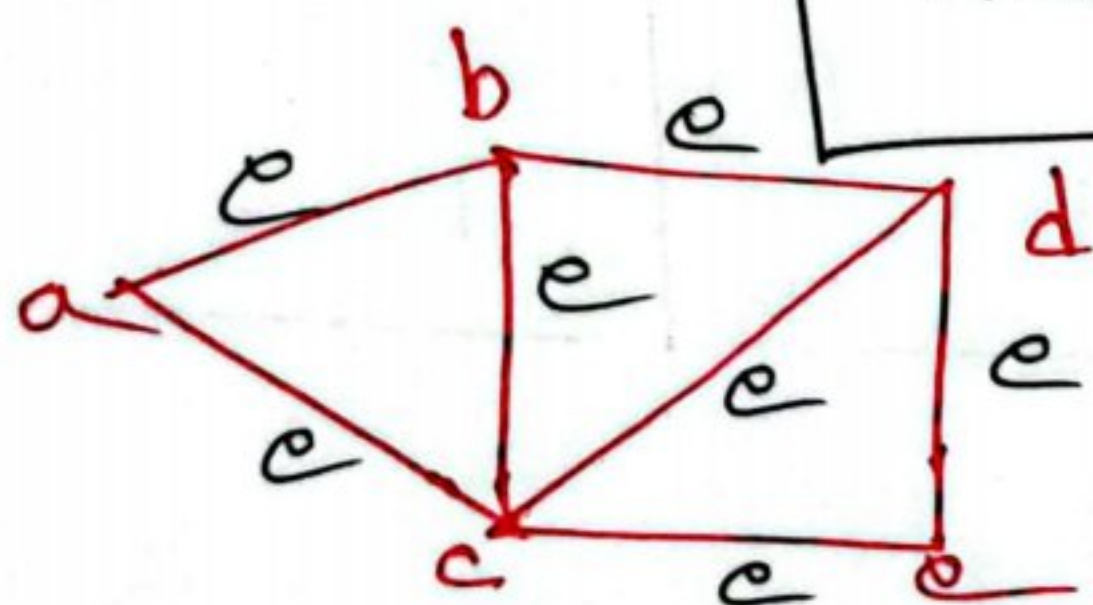
i.e., the sum of the degrees of the vertices in an undirected graph is even.

proof:

- ① The degree of a vertex is the number of edges incident with that vertex.
- ② The sum of degree counts the total number of an edge is incident with a ~~vertex~~ vertex.
- ③ Since every edge is incident with exactly two vertices, each edge gets counted twice, once at each end.
- ④ Thus the sum of the degrees is equal to twice the number of edges.

$$\sum \deg(v) = 2e$$

Ex-3



$$d(a) = 2$$

$$d(c) = 1$$

$$d(d) = 3$$

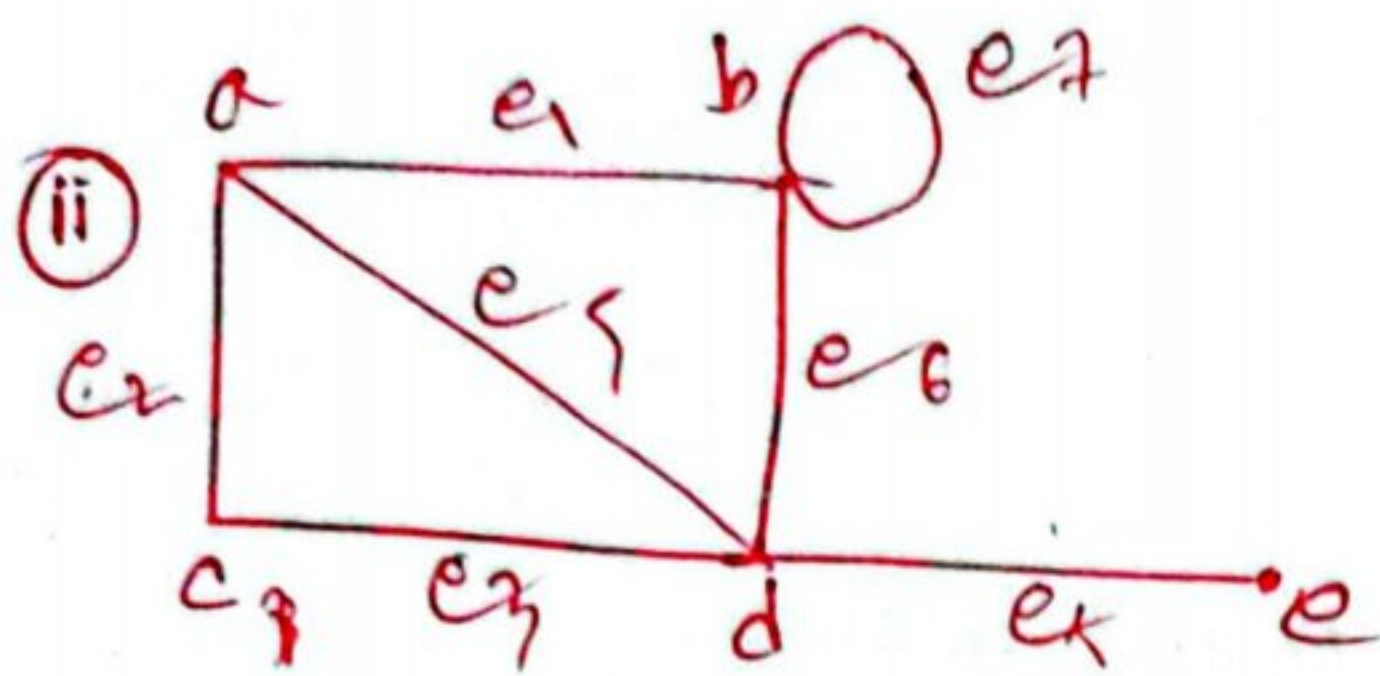
$$d(b) = 3$$

$$d(e) = 2$$

$$\sum d(v) = 2e$$

$$3 + 3 + 2 + 2 + 1 = 2 \times 7$$

$$\sum d(v) = 2e$$



$$d(a) = 3$$

$$d(e) = 1$$

$$d(b) = 4$$

$$d(c) = 2$$

$$d(d) = 4$$

$$\sum d(v) = 2e$$

$$3+4+2+4+1 = 2 \times 7$$

$$2 \times 7$$

$$\therefore \sum d(v) = 2e \quad \text{proved}$$

Q-4 In a undirected graph, the total number of odd degree vertices is even.

proof: Let $G = (V, E)$ a undirected graph. Let U denote the set of even degree vertices in G and W denote the set of odd degree vertices.

Then,

$$\sum_{v_i \in V} \deg(v_i) = \sum_{v_i \in U} \deg(v_i) + \sum_{v_i \in W} \deg(v_i)$$

$$\begin{matrix} 2e & - & \sum_{v_i \in U} \deg(v_i) & = & \sum_{v_i \in W} \deg(v_i) \\ \text{(even)} & & \text{(even)} & & \end{matrix}$$

Now, $\sum_{v_i \in W} \deg(v_i)$ is also even

Therefore, from (i),

$$\sum_{v_i \rightarrow w} \deg(v_i) \text{ is even.}$$

$\therefore$ The number of odd vertices in G is even.

Types of graph

1. Null graph

2. Regular graph

3. Complete graph

4. Cycle

5. Wheel

6. N-cube

7. Bipartite graph

8. Complete Bipartite graph.

Null graph

A graph without edges is graph or a null graph.

A graph which contains only isolated vertex is called a null graph. i.e. the set of edges in a null graph is empty.

Null graph is denoted on n vertices by N_n .

eg:

fig:

v_1

v_2

v_3

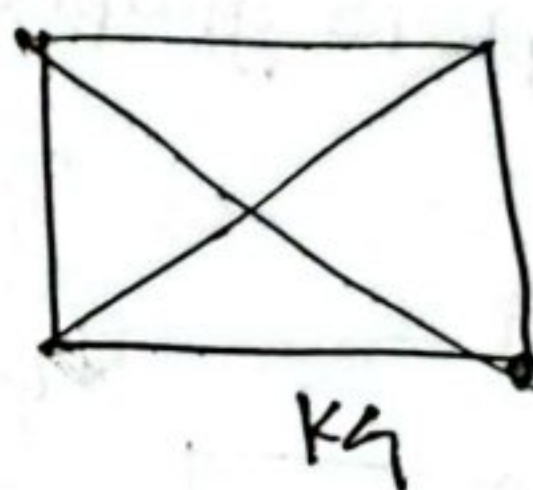
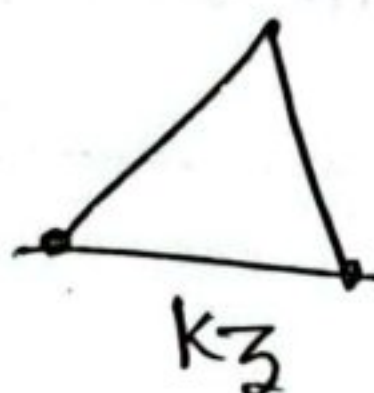
Complete graph

A simple graph G is said to be complete if every vertex in G is connected with every other vertex i.e. if G contains exactly one edge between each pair of distinct vertices.

A complete graph on n vertices is denoted by K_n .

K_1

K_2



Note: A complete graph K_n has $\frac{n(n-1)}{2}$ edges

$$K_4 = \frac{4(4-1)}{2} = 6 \text{ edges}$$

Size of every graph of order n is at most $\frac{n(n-1)}{2}$

K_n has exactly $\frac{n(n-1)}{2}$ edges.

Regular graph

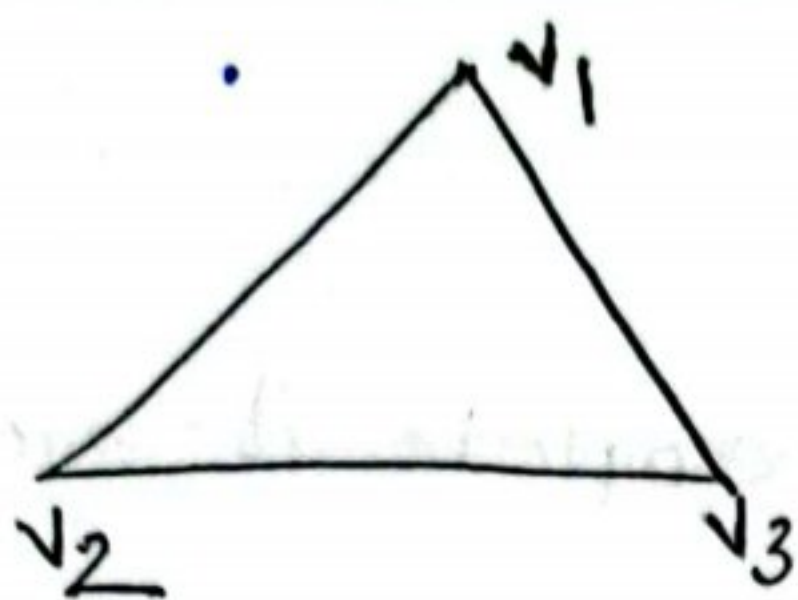
A graph in which all vertices have same or equal degree.

If the degree of each vertex in G is r , then the graph

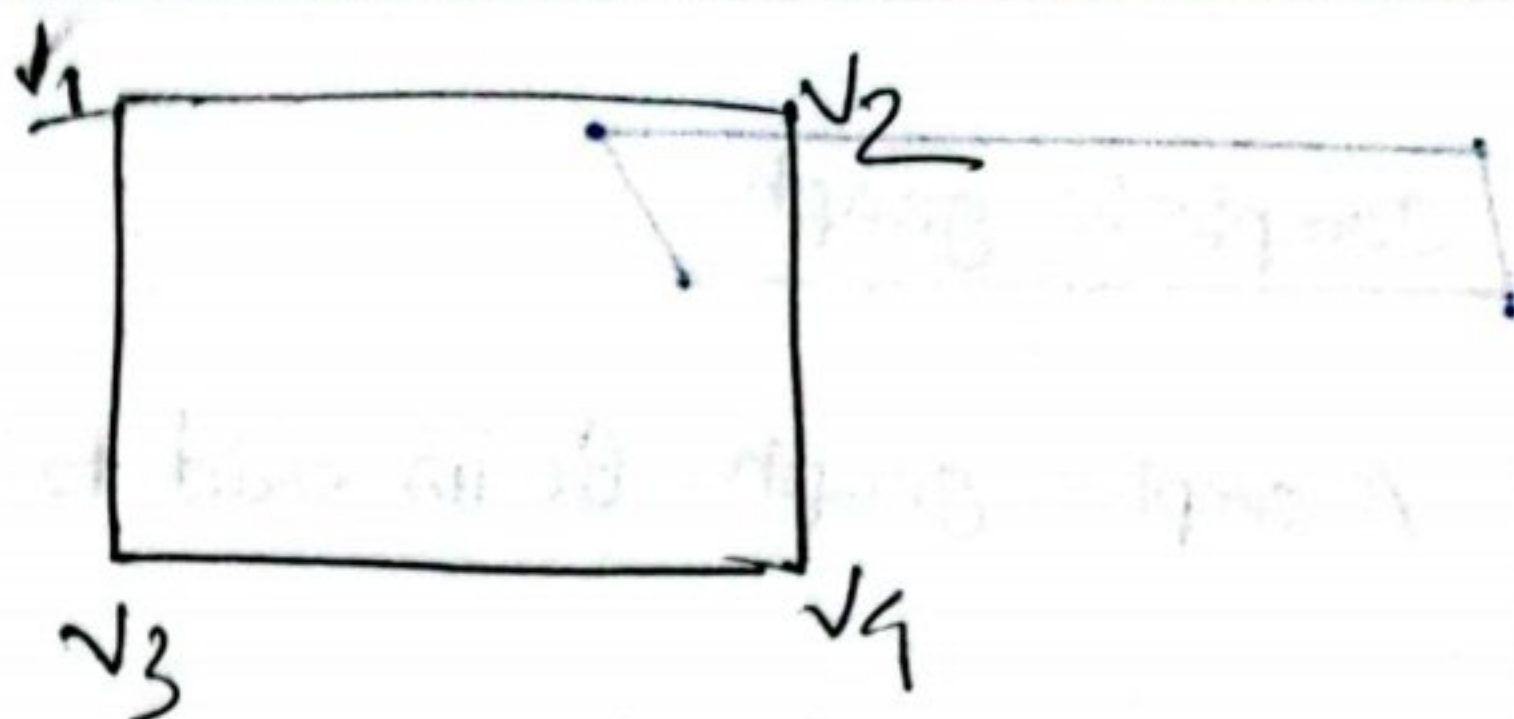
is called regular graph of degree r or r -regular graph.

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Regular Ex:



$$d(v_1) = d(v_2) = d(v_3) = 2$$



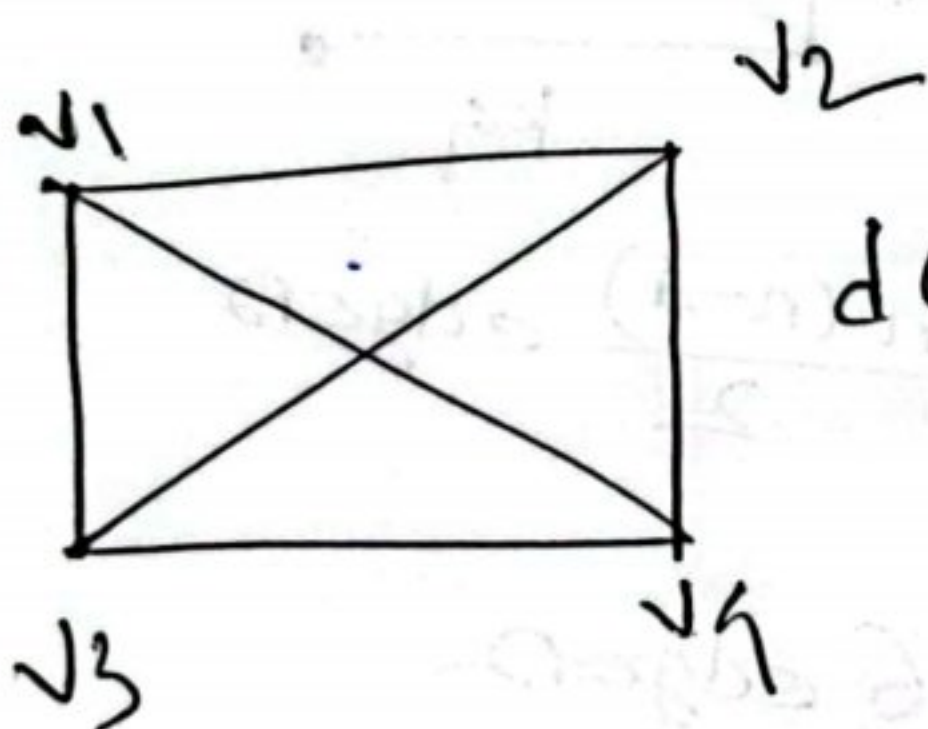
$$d(v_1) = d(v_2) = d(v_3) = d(v_4) = 2$$

<regular graph>

Note: (1) The complete graph K_n is a regular graph of degree

$n-1$

Ex:



$$d(v_1) = d(v_2) = d(v_3) = d(v_4) = 3$$

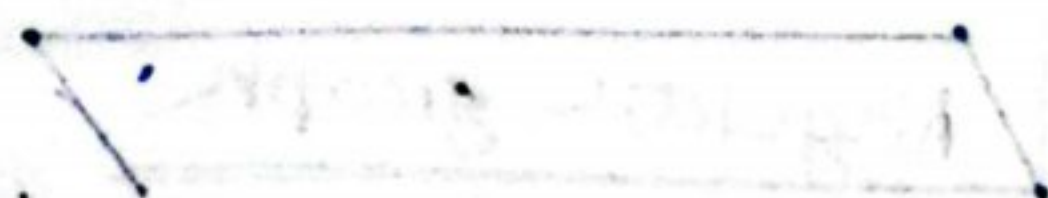
3-regular graph

(2) If G has n vertices and is regular of degree r ,

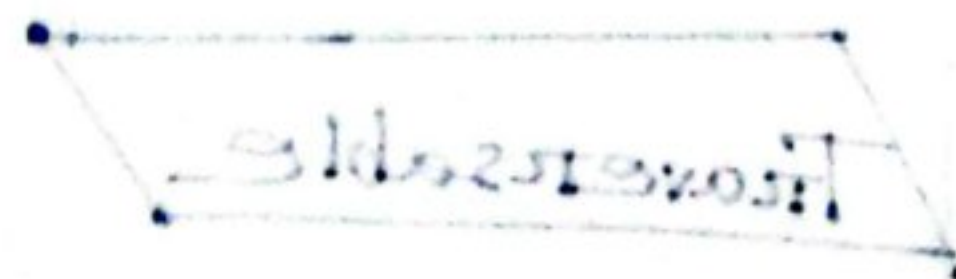
then G has $\frac{nr}{2}$ edges

Ex: $n=4, r=3$

$$\text{No. of edges} = \frac{nr}{2} = 6 \text{ edges}$$



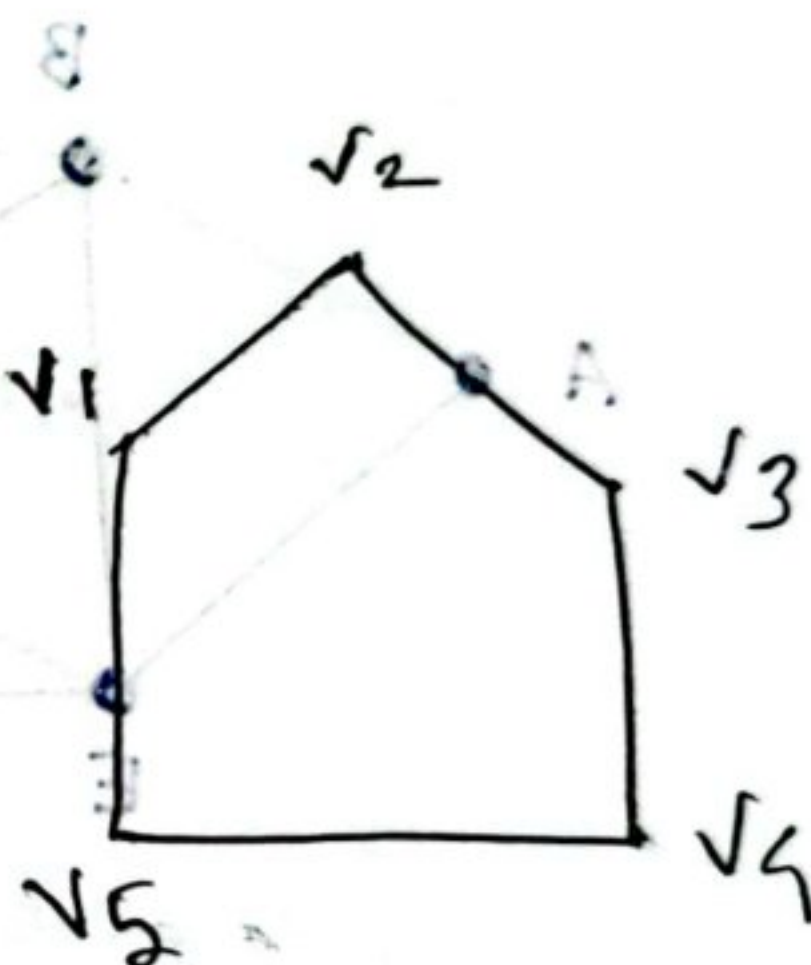
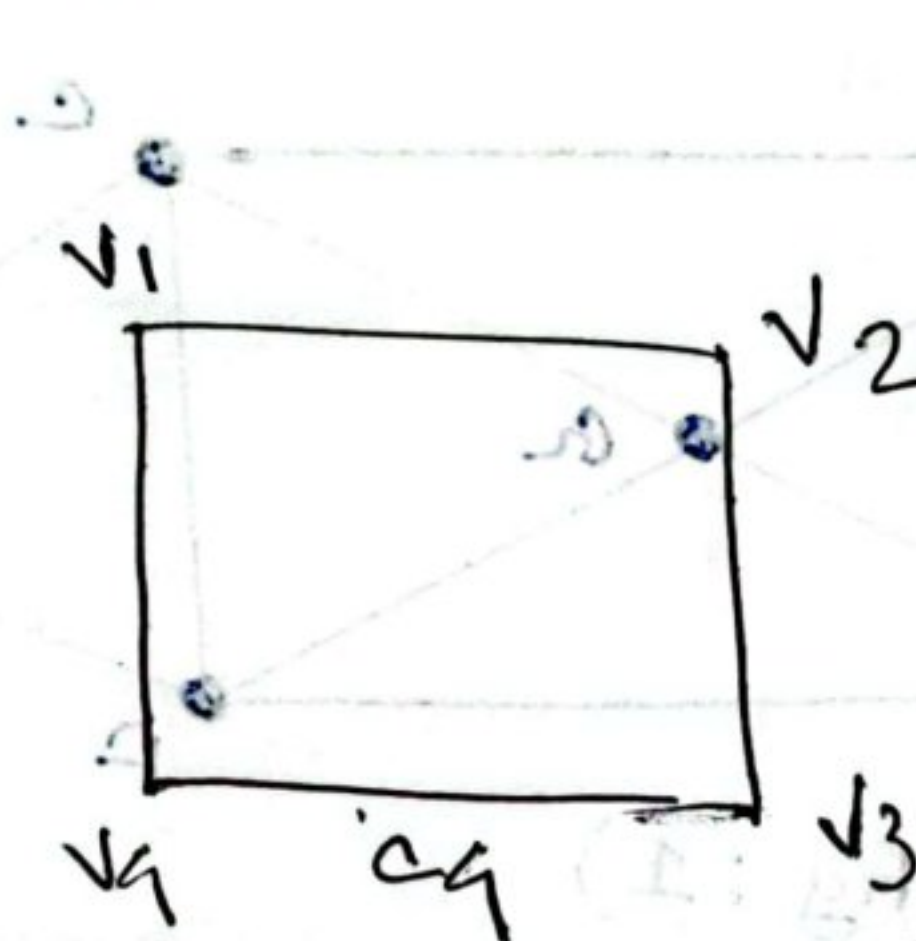
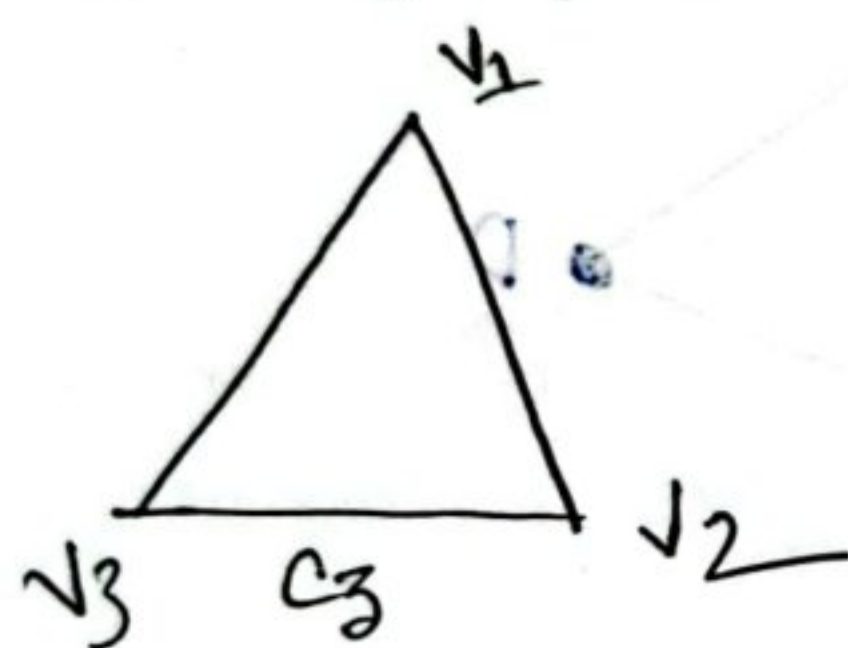
Cycles - Cyclic graph



The cycle C_n , $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\} \dots \{v_{n-1}, v_n\}$ and $\{v_n, v_1\}$

A graph containing atleast one cycle in it is known as

cyclic graph

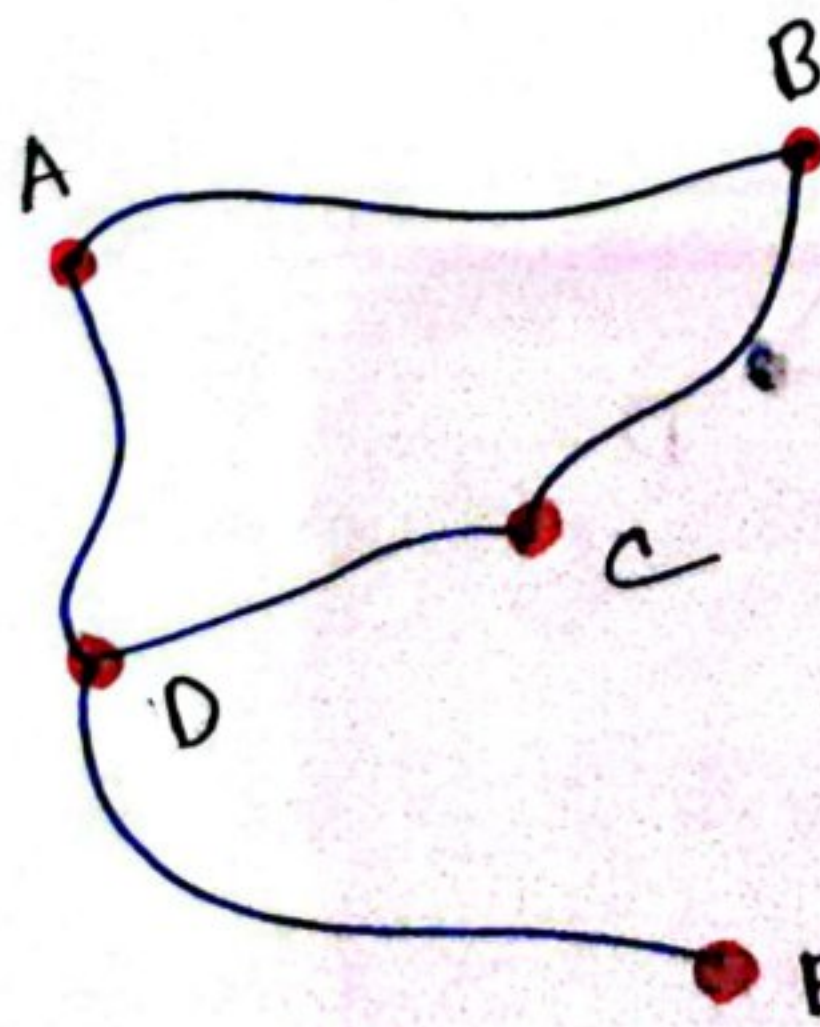


(i) Note: n vertices = n edges.

degree of each vertex is two.

Network

A network has points and lines



An even degree vertex has an even number of sides connected to it.

$$A = B = C = 2 \text{ sides}$$

An odd degree vertex has odd

number of sides connected to it.

D 3 sides E 1 side

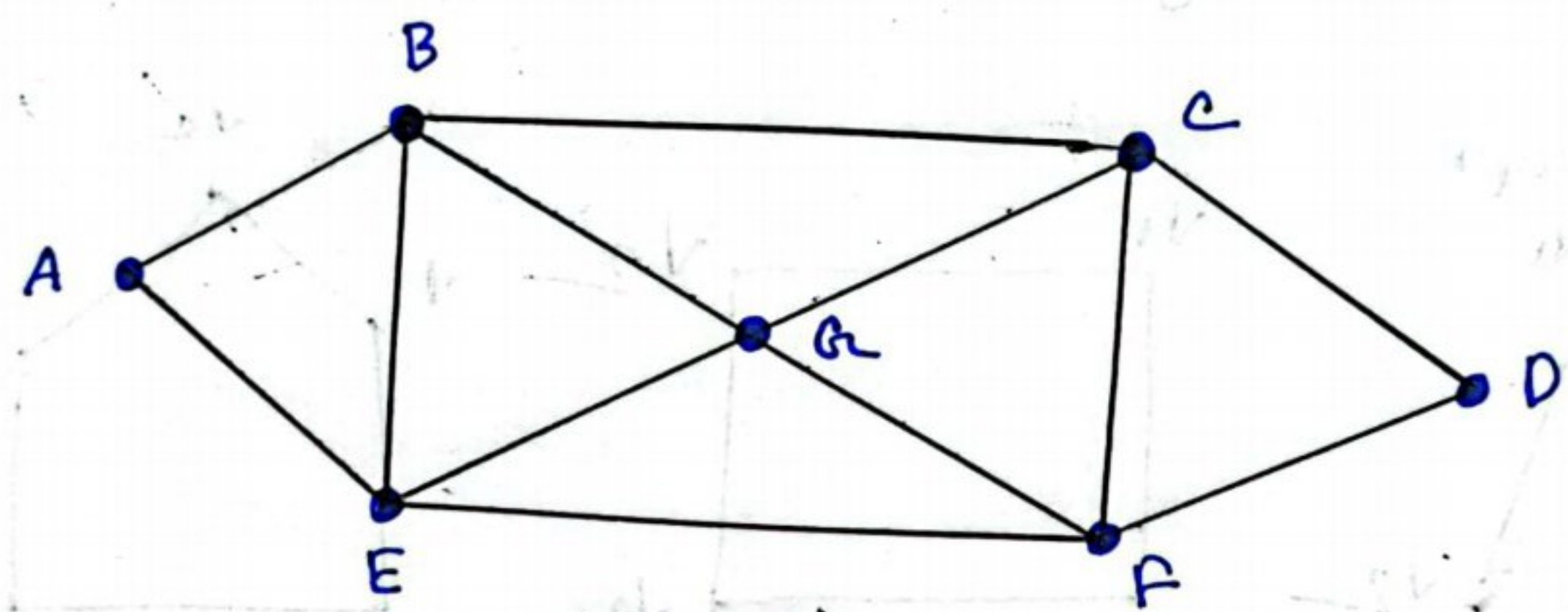
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Traversable

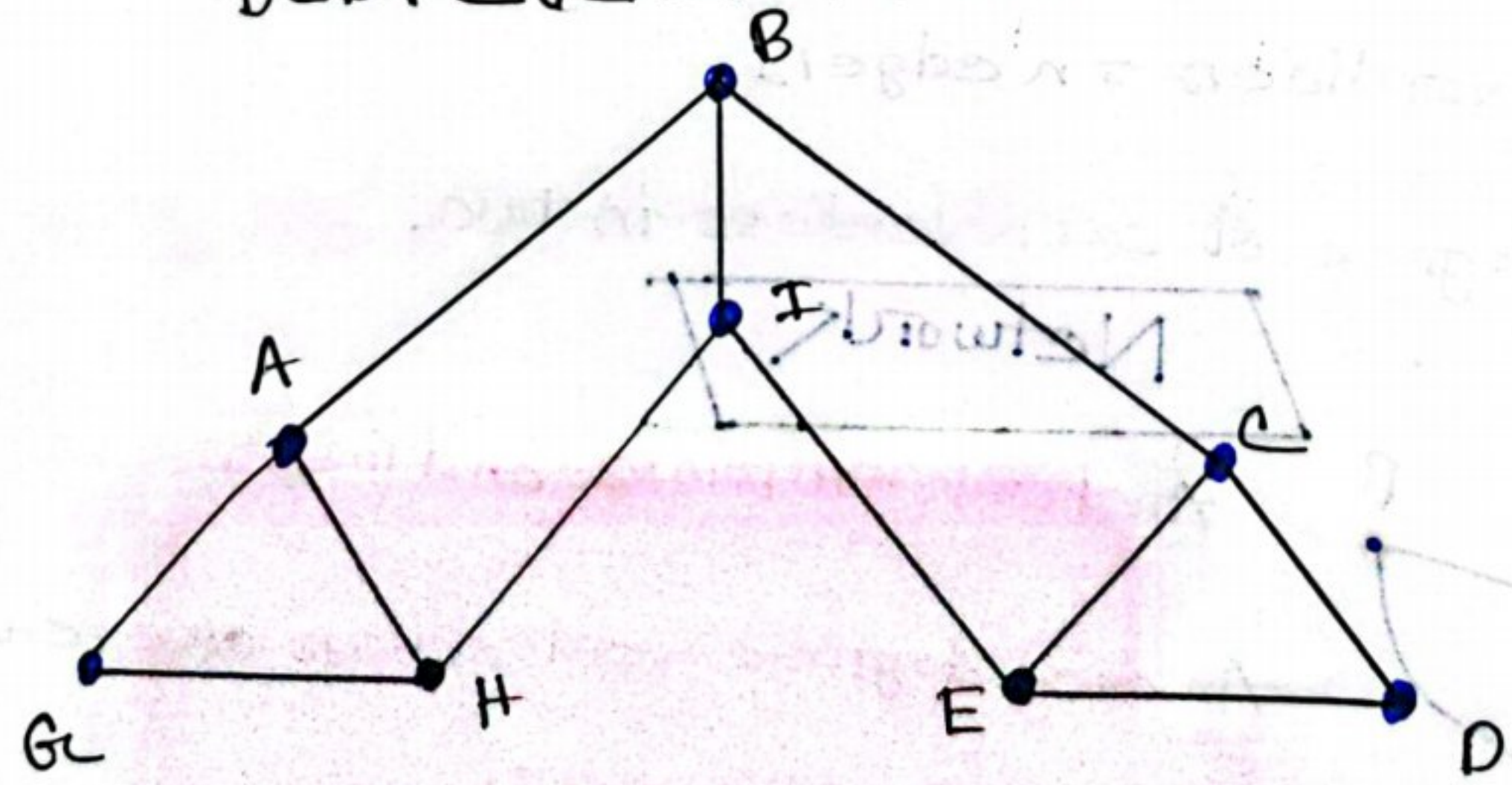


A network is traversable if you can trace the shape without lifting your pen and without going over a side/edge more than once.



(fig : 1)

BCDFCGBAEFGLED



(fig : 2)

(NOT Traversable)

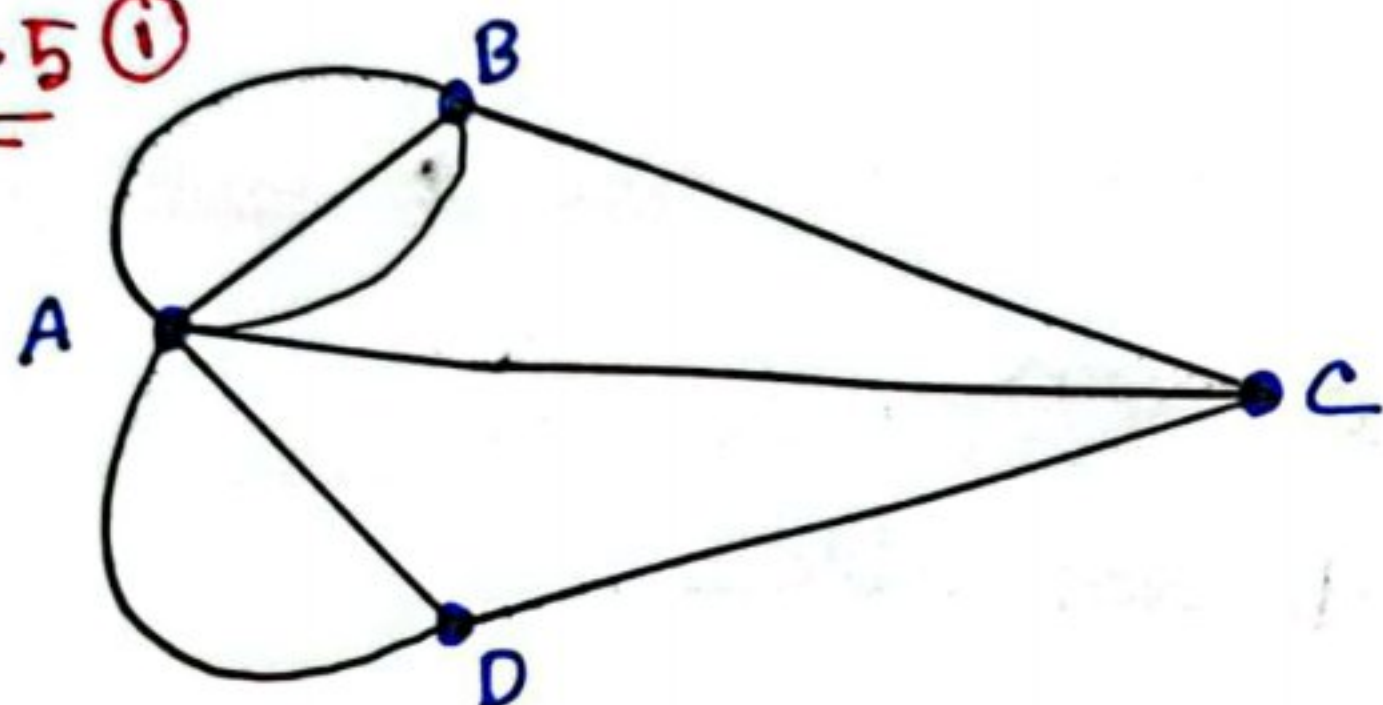
for a shape to be traversable

① All vertices are of even degree. [Rule-1]

② exactly two vertices are of odd order and rest are even

2 odd 2 even (Rule-2)

Q-5 ①



A even
B even
C odd
D odd.

Ans: Traversable

(Rule-2)

②

from fig - 1, we can say (previous page)

A, B, C, D, E, F = even degree

So, fig - 1 is traversable. (Rule-1)

③ from fig - 2, we can say

A odd

E even

3 odd 6 even not in rule 1 and

B odd

F even

2 So not traversable

C even

G odd

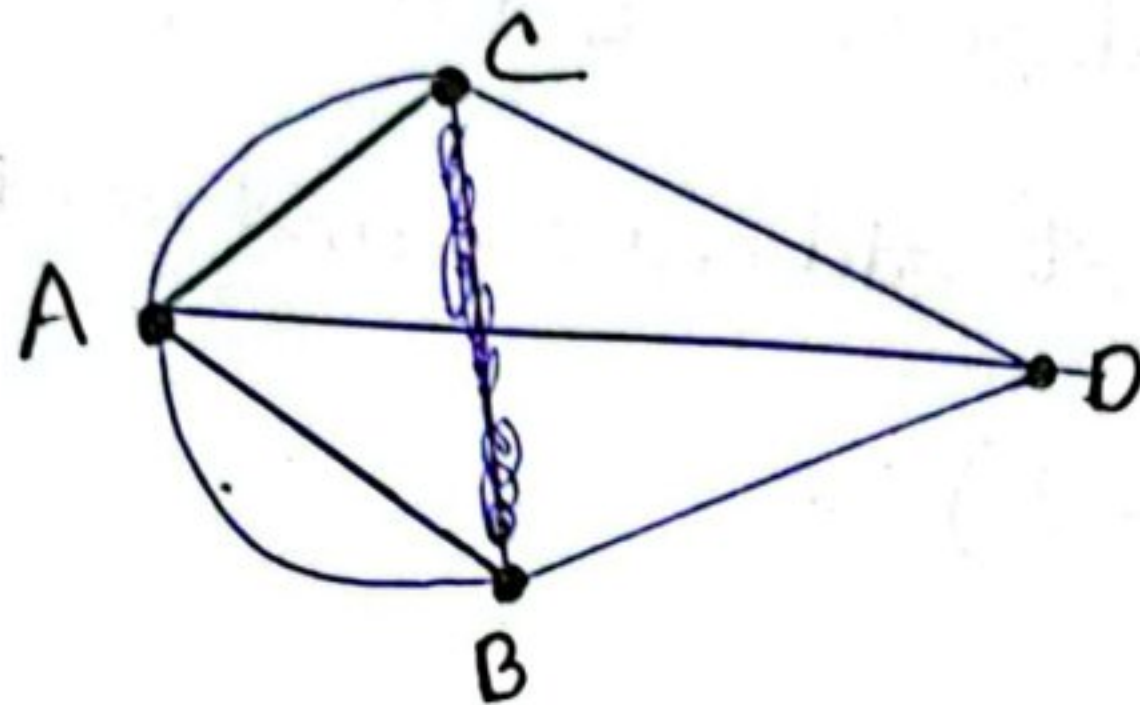
D odd

H odd

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Question: Can you cross over the each bridge exactly

once,

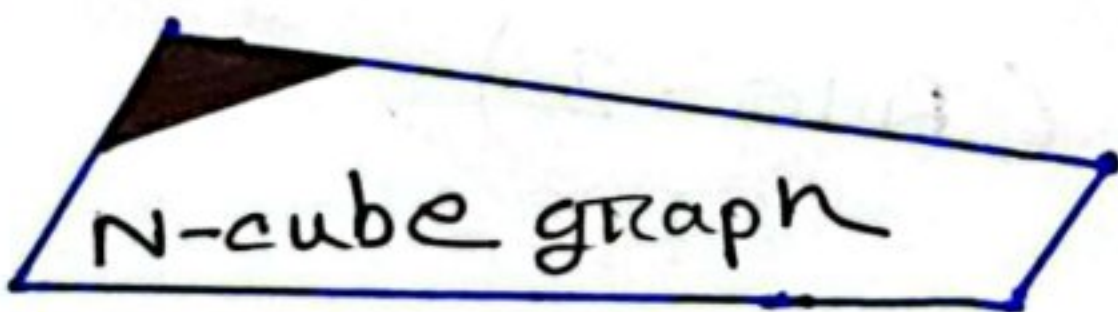


Here,

A odd
B odd
C odd
D odd

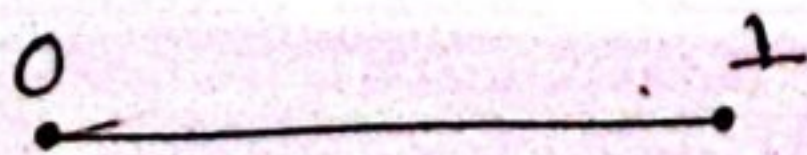
4 odd & even

Not traversable

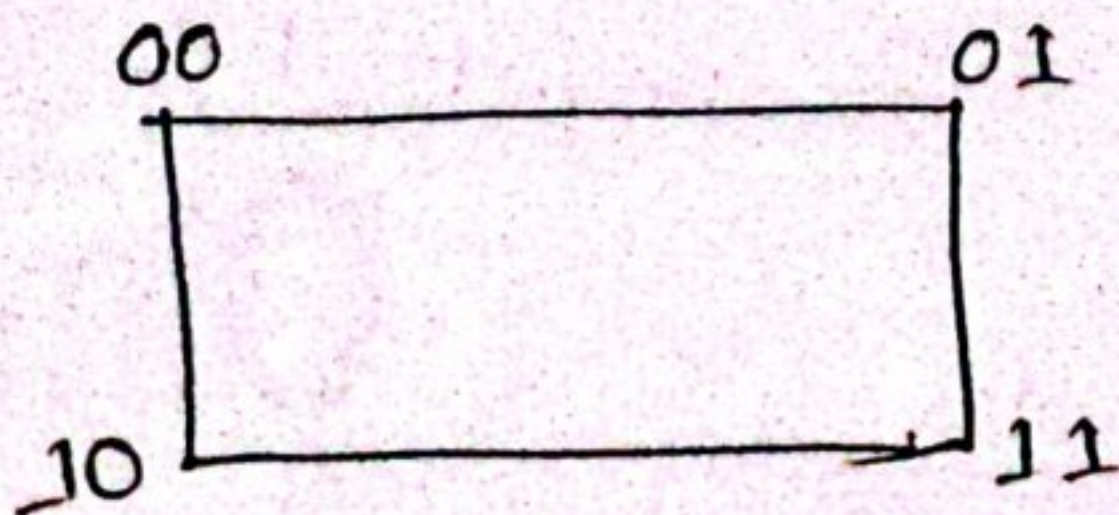


The N-cube denoted by G_n , is the graph that has vertices representing the 2^n bit strings of length n .

G_1 , $n=1$, no of vertex, $2^n = 2^1 = 2$



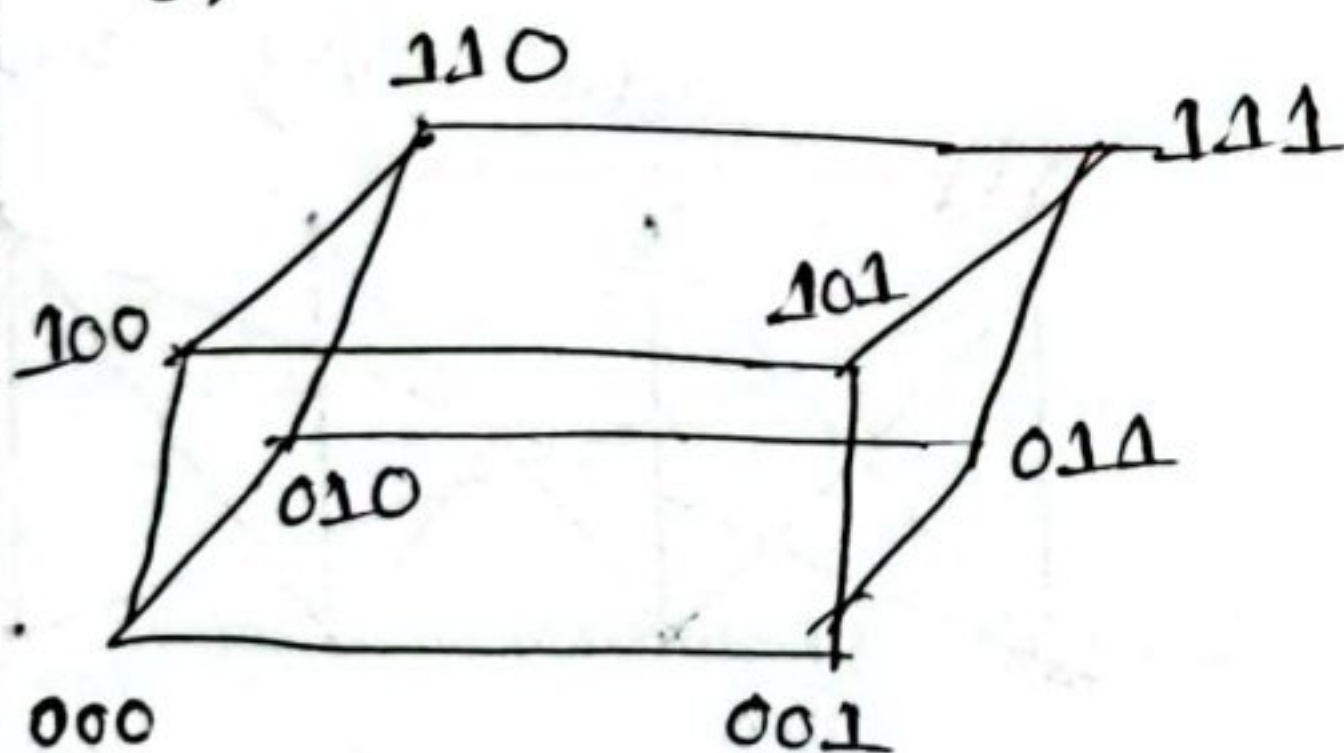
G_2 , $n=2$, no of vertex, $2^n = 2^2 = 4$



Note:

- (i) Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.
- (ii) The Q_n has 2^n vertices and $n \cdot 2^{n-1}$ edges, and is regular of degree n .

$Q_3, n=3$, no. of vertices = 8 (2^3).



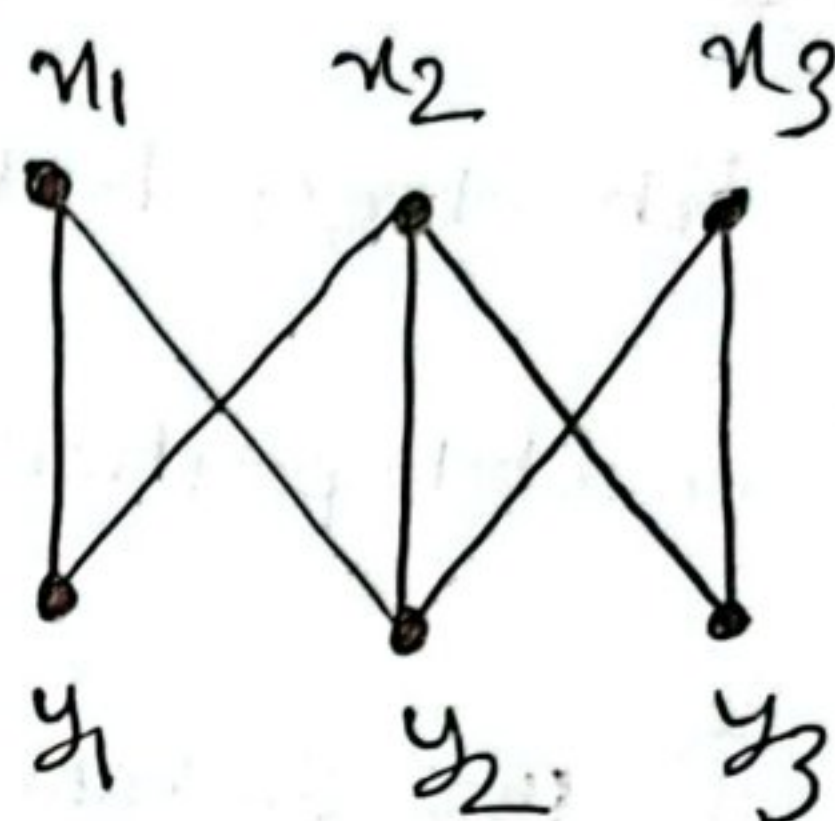
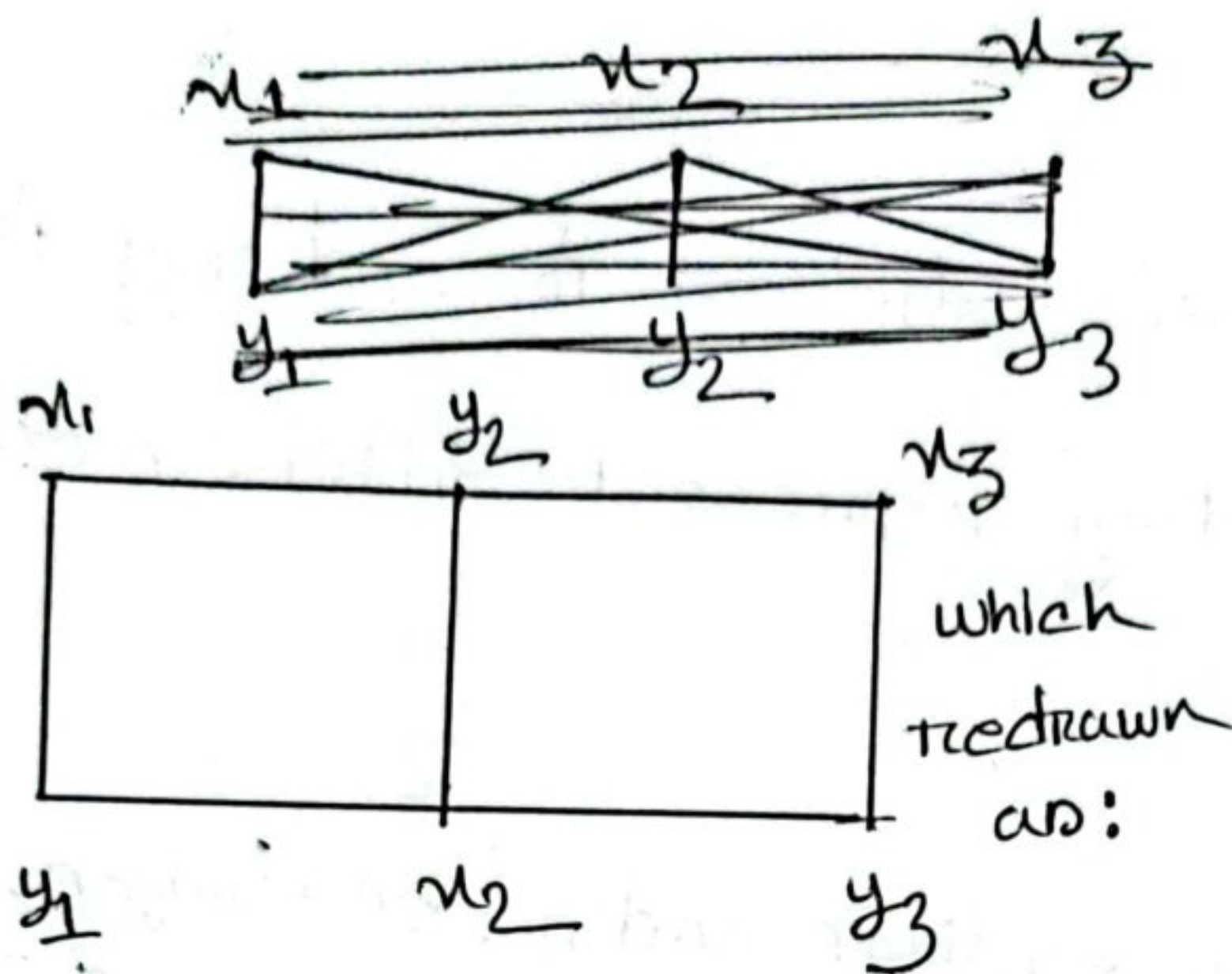
Bipartite graph

A graph G is said to be bipartite graph if its vertex set V can be partitioned into two sets, say V_1 and V_2 , such that no two vertices in the same partition can be adjacent.

Here, the pair (V_1, V_2) is called the bipartition of G .

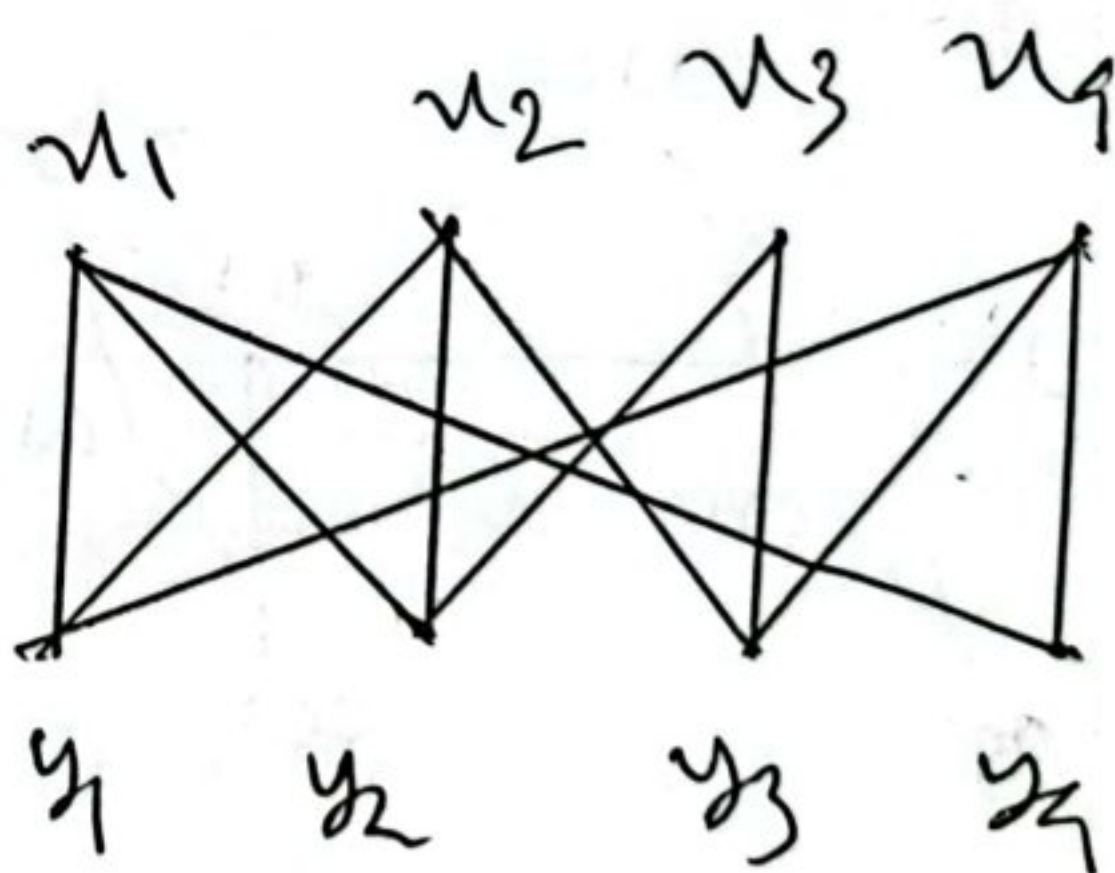
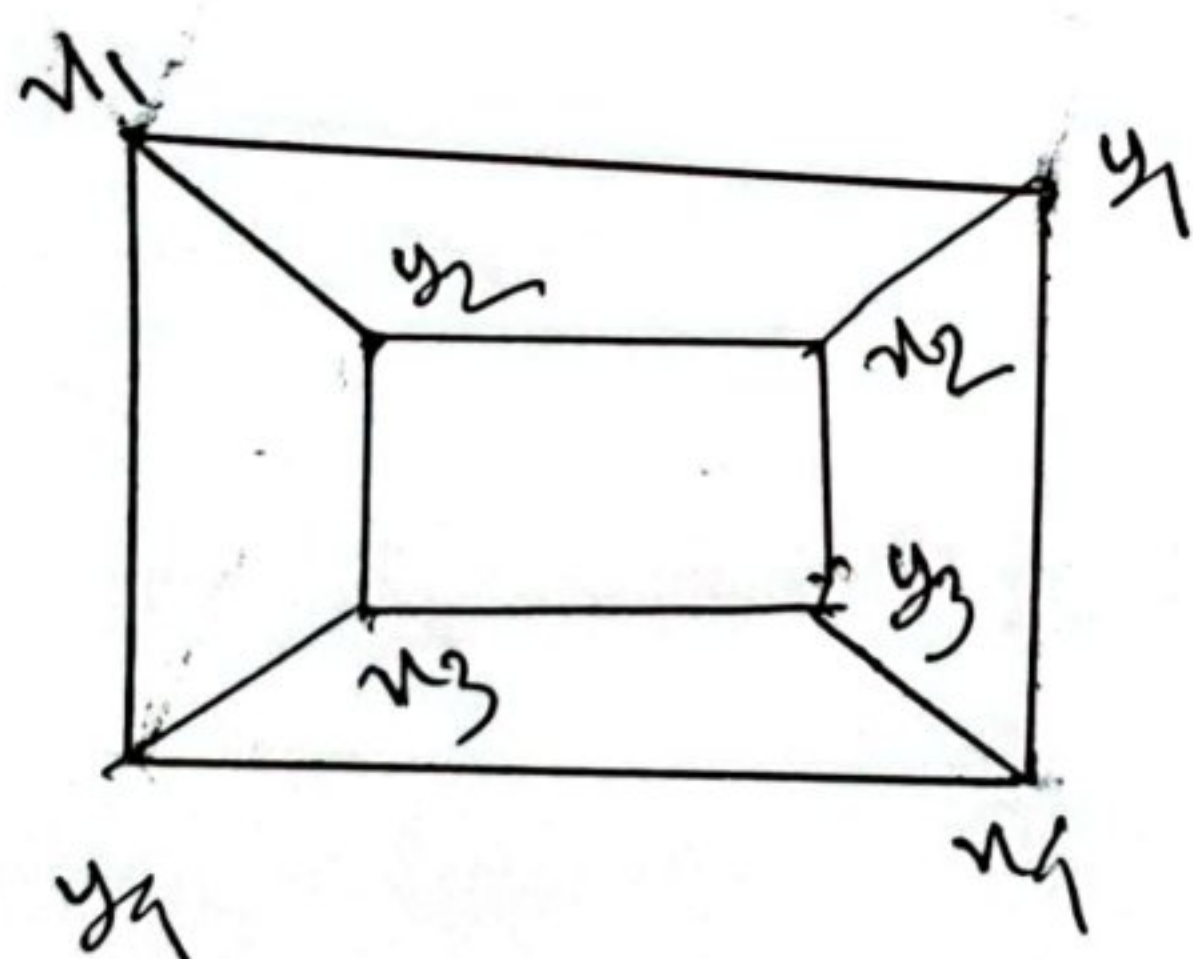
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Ex:



$$V_1 = \{u_1, u_2, u_3\}$$

$$V_2 = \{y_1, y_2, y_3\}$$

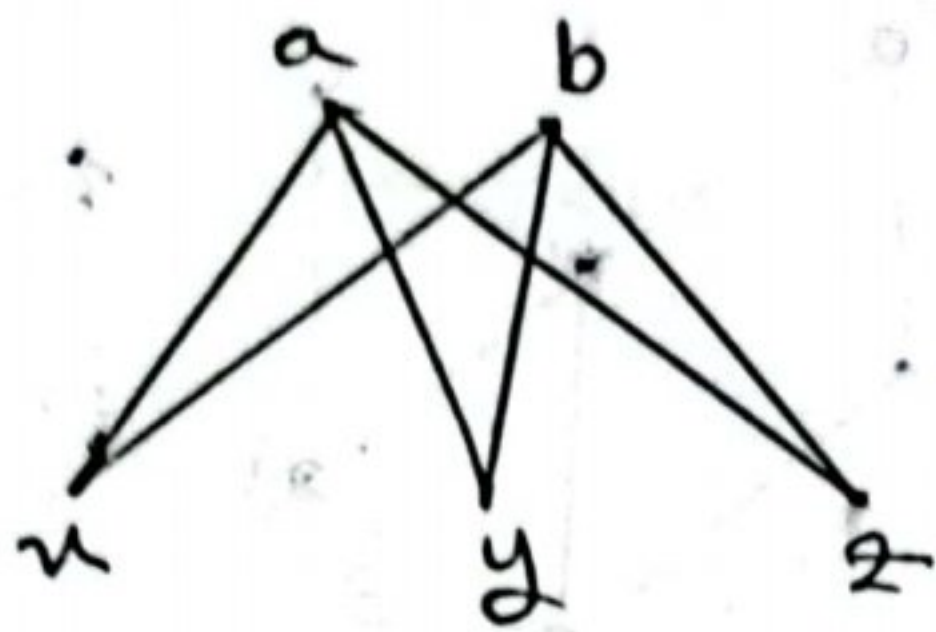


Complete Bipartite graph

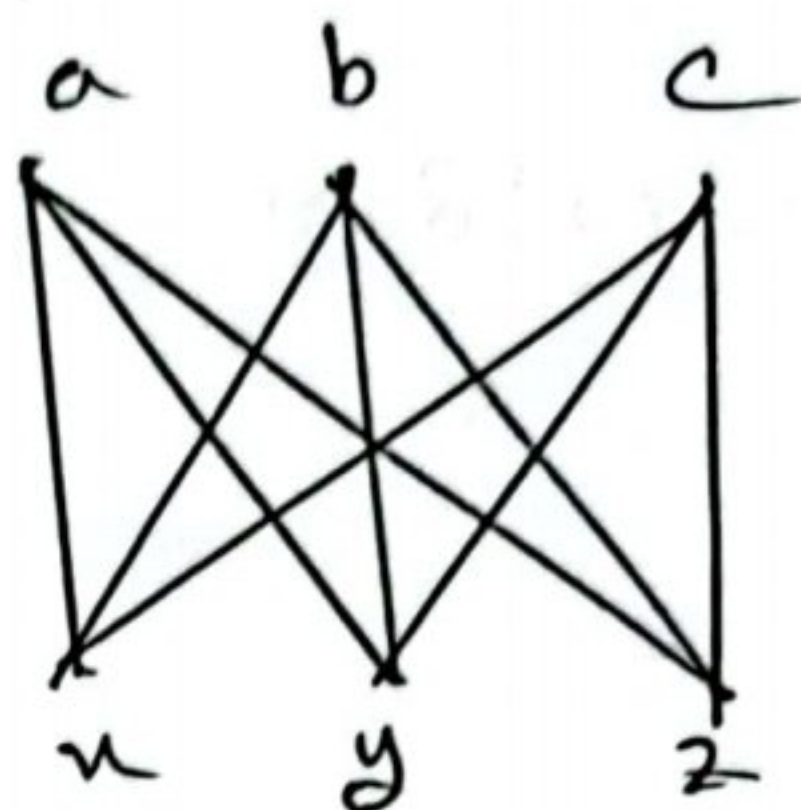
A complete bipartite graph $K_{m,n}$

is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between every pair of vertices if and only if one vertex in the pair is in the first subset and the other

vertex is in the second subset.



$K_{2,3}$



$K_{3,3}$

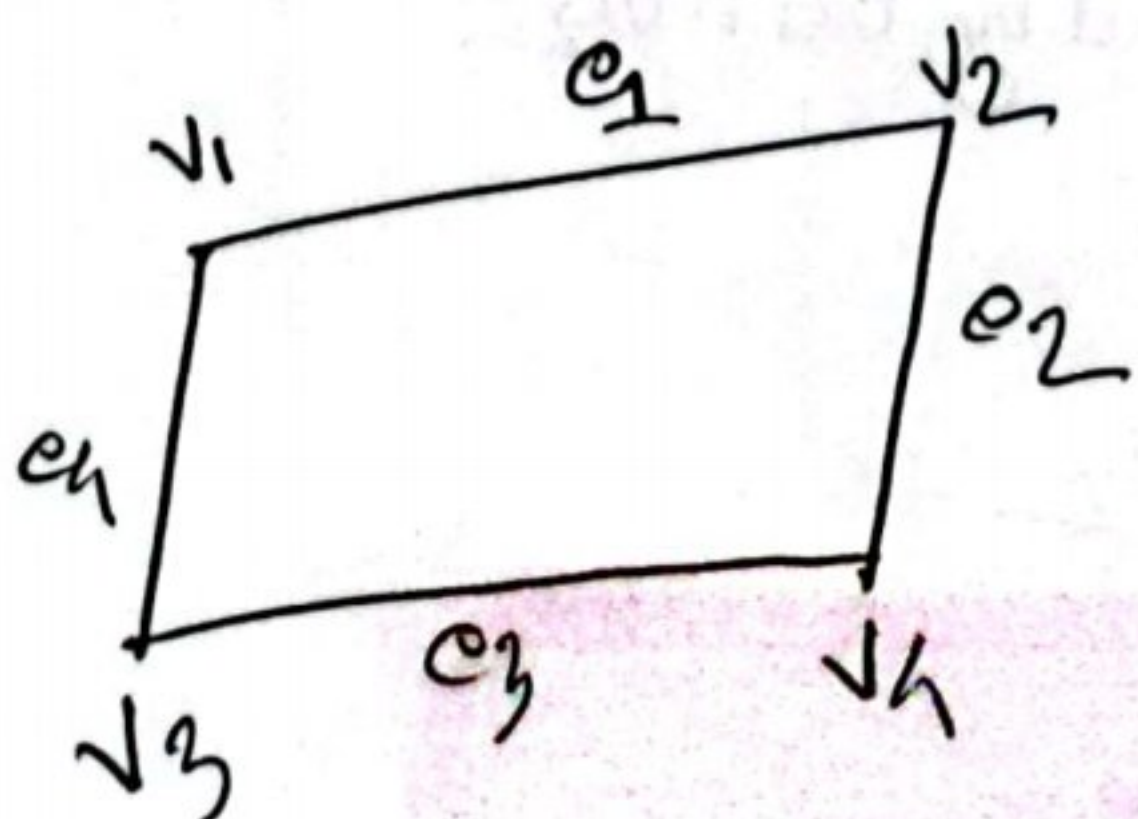
A complete bipartite graph is not a regular graph if $m \neq n$.

Subgraph of a graph

Given two graphs G and G_1 , we say that G_1 is a subgraph of G if the following condition hold:

- ① All the vertices and all the edges of G_1 are in G
- ② Each edge of G_1 has the same end vertices in G as in G_1

Ex:



$$G(V, E), \quad G_1(V_1, E_1)$$

$$V_1 \subseteq V, \quad E_1 \subseteq E$$

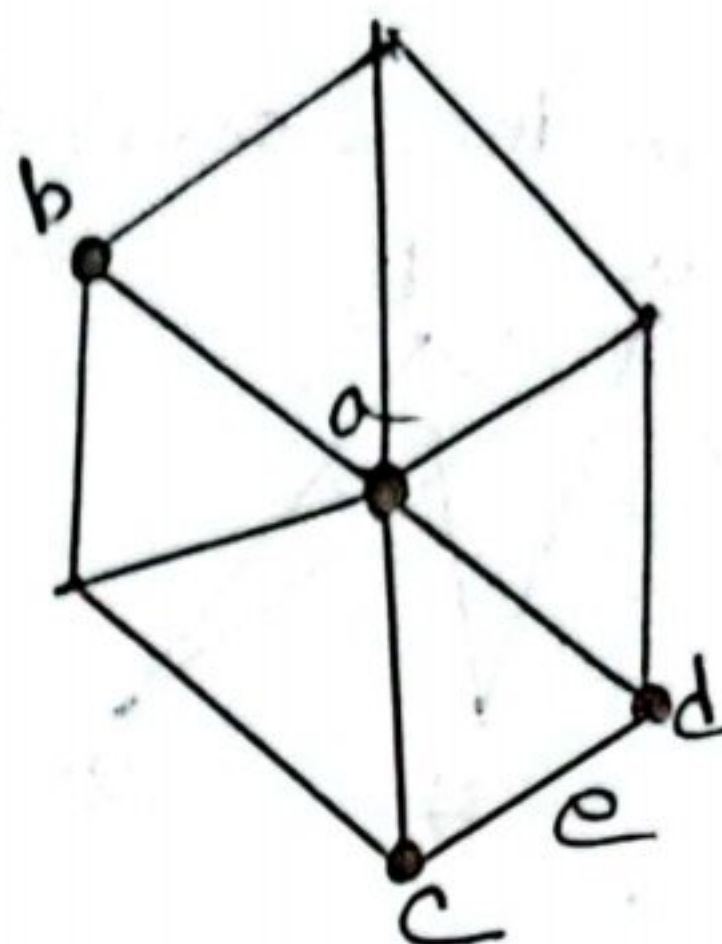
$$V_1 = \{v_1, v_2, v_3, v_4\} \subseteq V \{v_1, \dots, v_n\}$$

$$E_1 = \{e_1, e_2, e_3, e_4\} \subseteq E \{e_1, \dots, e_m\}$$

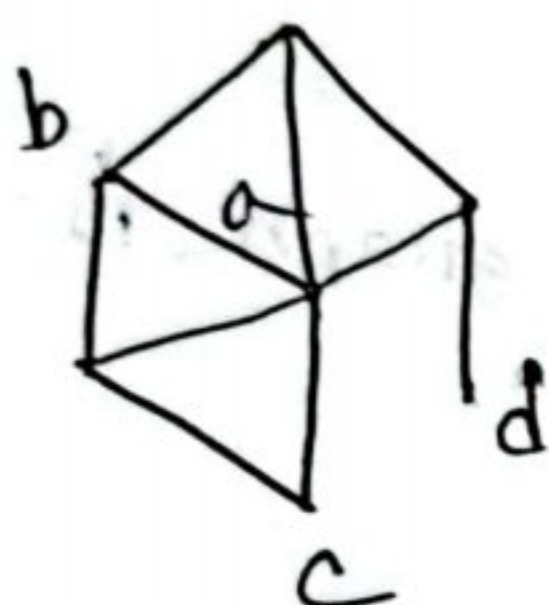
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Q-6 :- Draw subgraph of G .

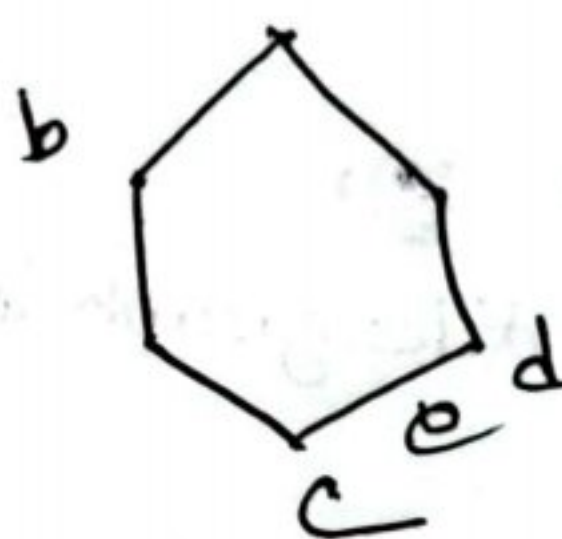
(i) $G-e$, (ii) $G-a$.



(i) $G-e$



(ii) $G-a$



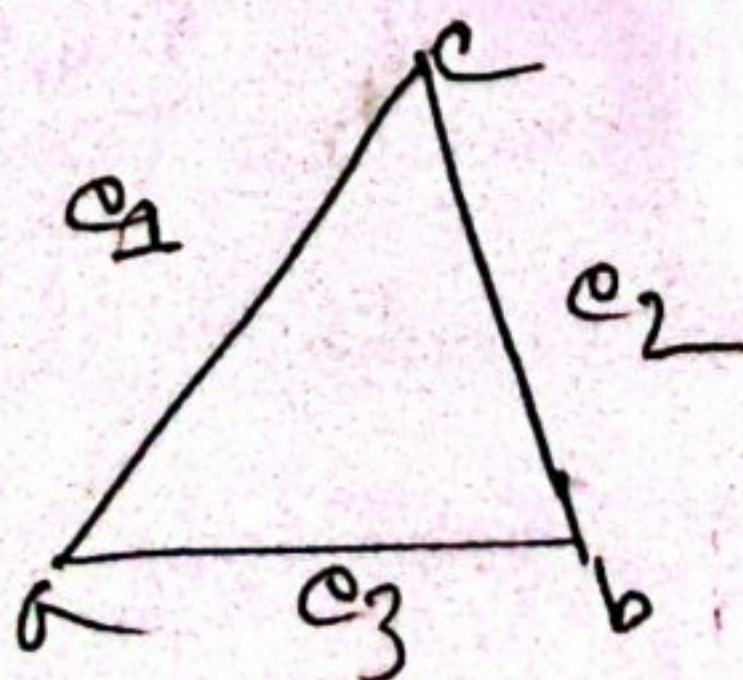
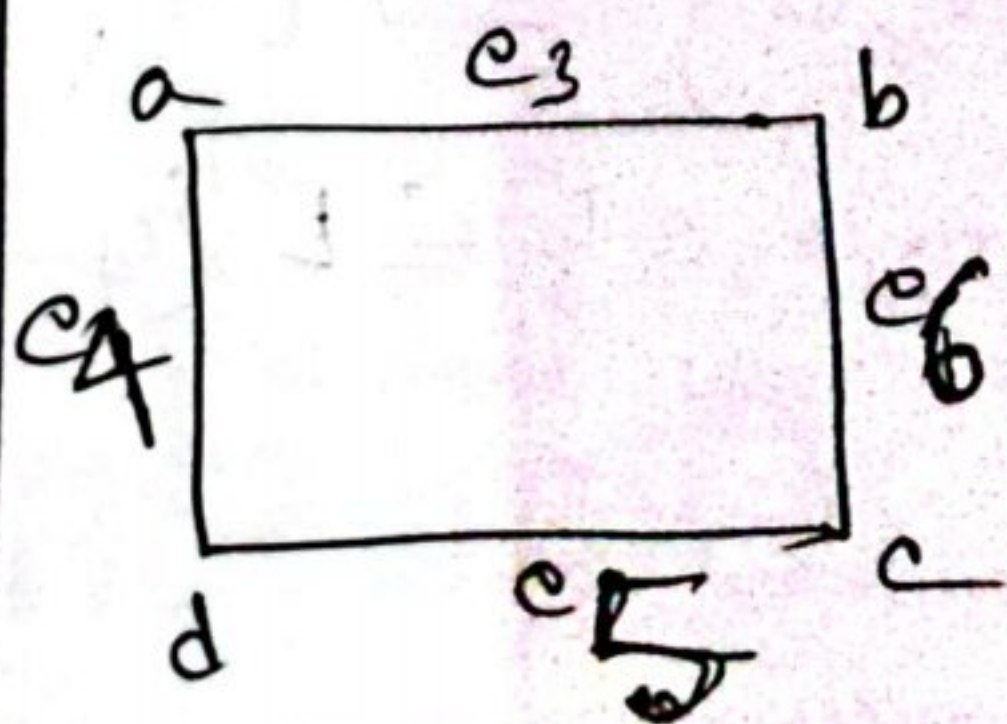
Operation of graph

Given two graph G_1 and G_2 with at least one vertex in common then their intersection will be a graph such that

$$V(G_1 \cap G_2) = V(G_1) \cap V(G_2)$$

$$E(G_1 \cap G_2) = E(G_1) \cap E(G_2)$$

Intersection of G_1 and G_2 is denoted by $G_1 \cap G_2$



$$V(G_1) = \{a, b, c, d\}$$

$$E(G_1) = \{e_3, e_4, e_5, e_6\}$$

$$V(G_2) = \{a, b, c\}$$

$$E(G_2) = \{e_1, e_2, e_3\}$$

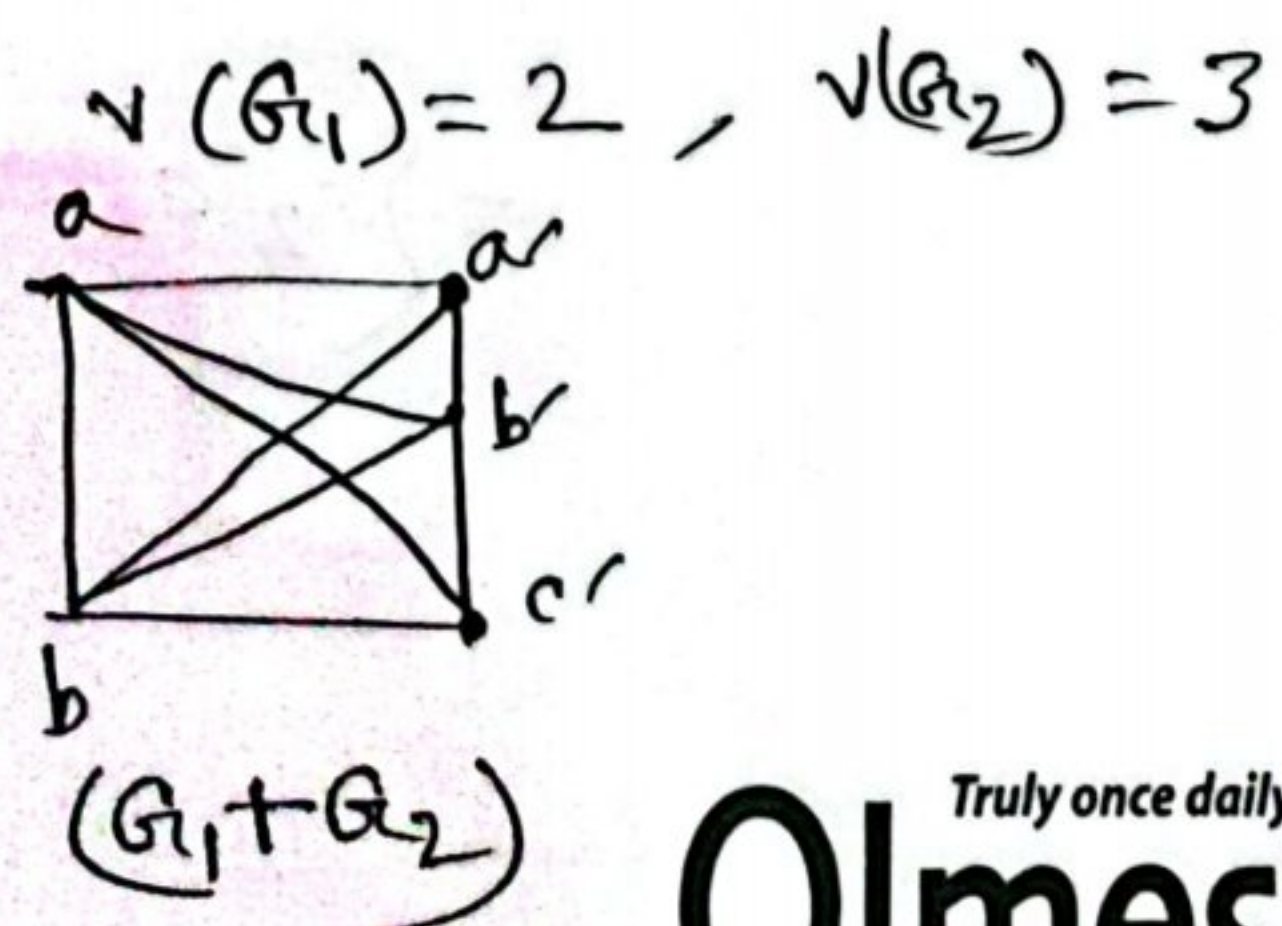
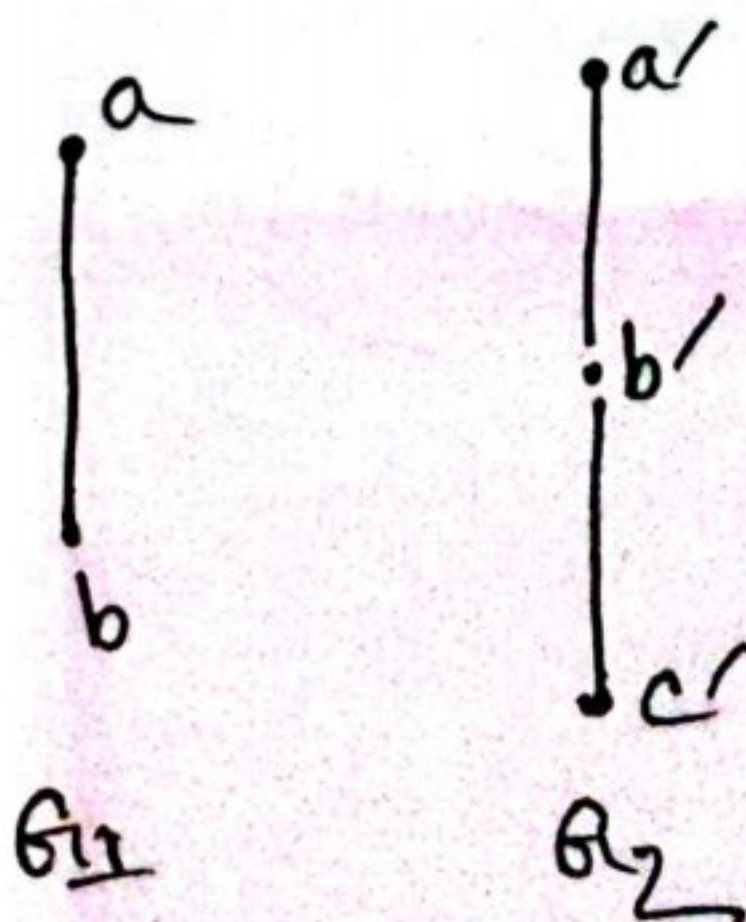
$$V(G_1 \cap G_2) = \{a, b\}$$

$$E(G_1 \cap G_2) = \{e_3\}$$



Sum of two graphs :-

Let the graphs G_1 and G_2 such that $V(G_1) \cap V(G_2) \neq \emptyset$ then the sum $G_1 + G_2$ is defined as the graph whose vertex set is $V(G_1) \cup V(G_2)$ and the edge set is consisting those edges, which are in G_1 and in G_2 and the edges obtained, by joining each vertex of G_1 and to each vertex of G_2 .

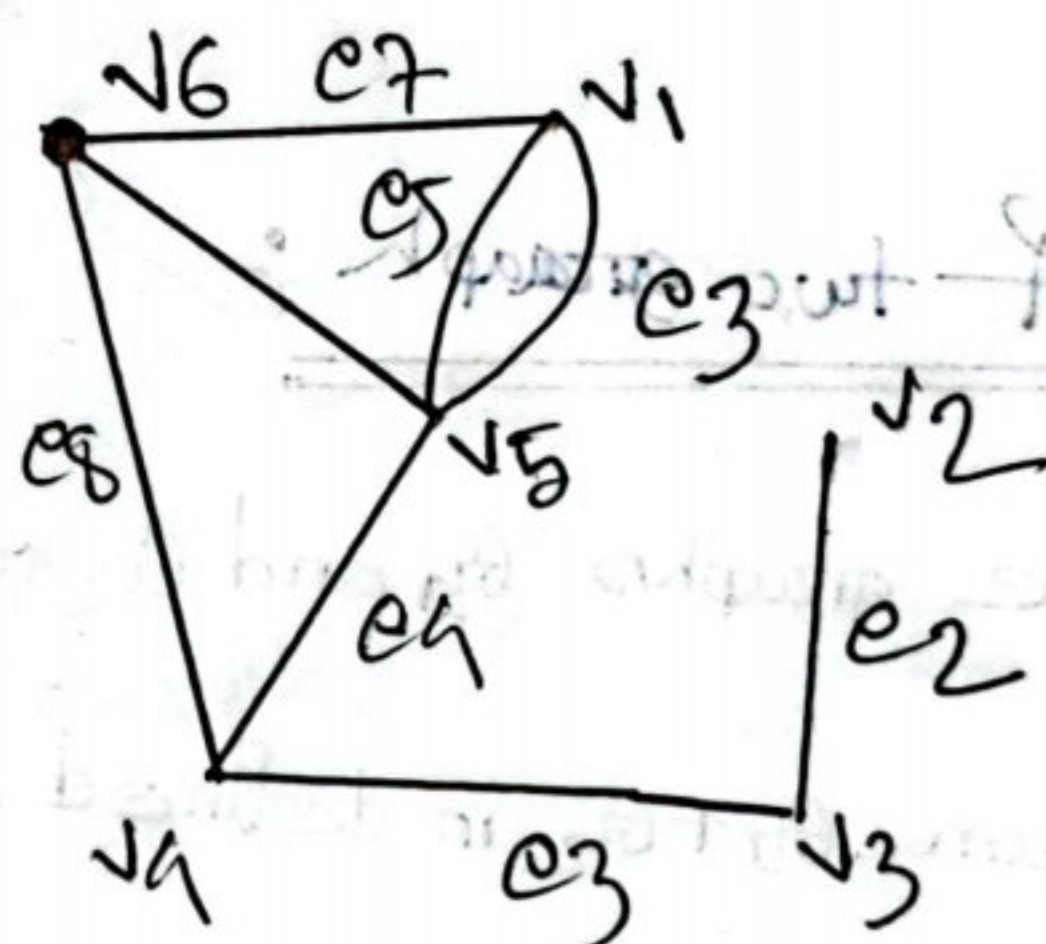
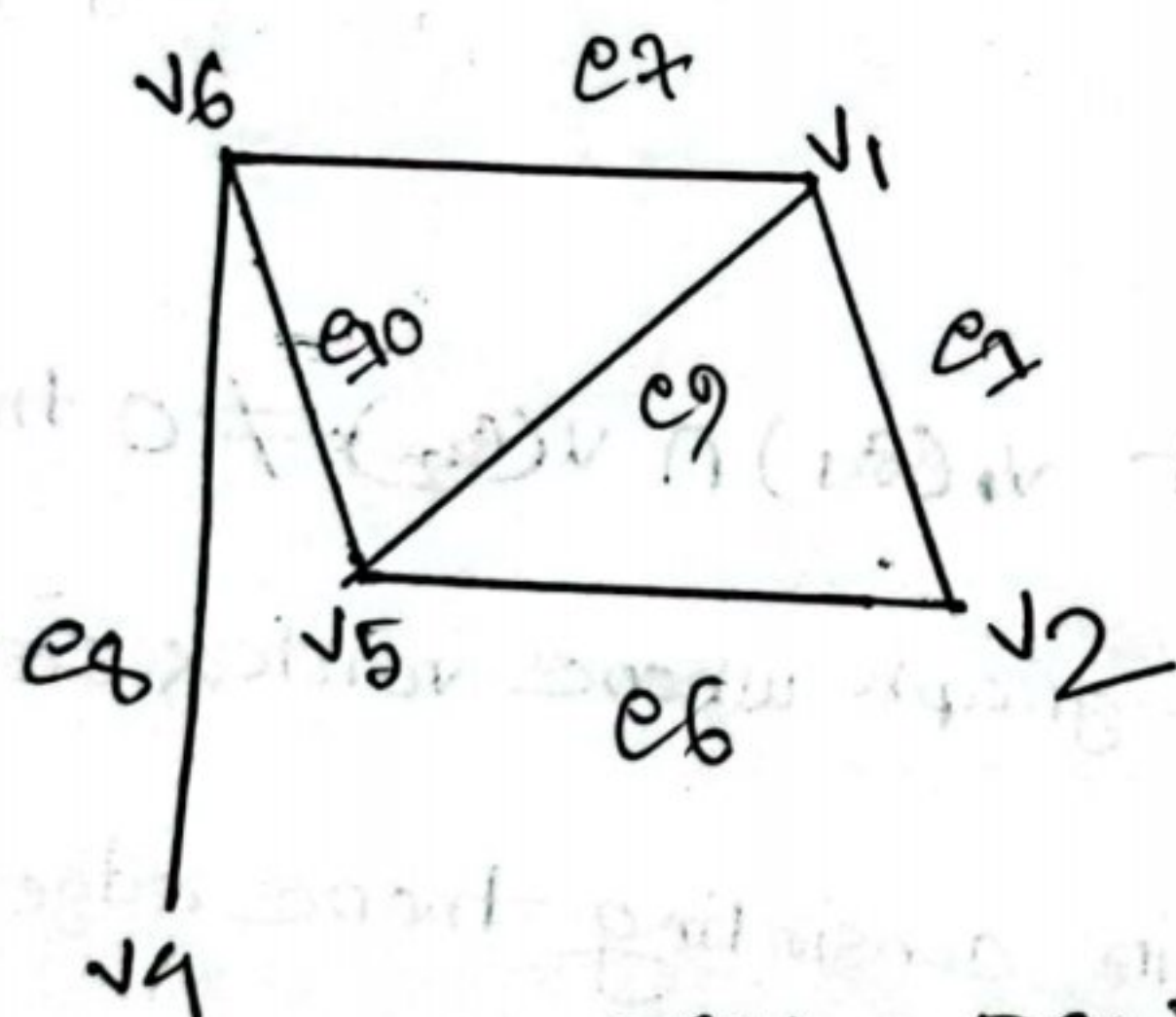


Ring Sum: Let $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ be two graphs then

the ring sum of G_1 and G_2 , denoted by $G_1 \oplus G_2$ is defined as the graph G such that

(i) $V(G) = V(G_1) \cup V(G_2)$

(ii) $E(G) = E(G_1) \cup E(G_2) - E(G_1) \cap E(G_2)$.



$$E(G) = E(G_1) \cup E(G_2) - E(G_1) \cap E(G_2)$$

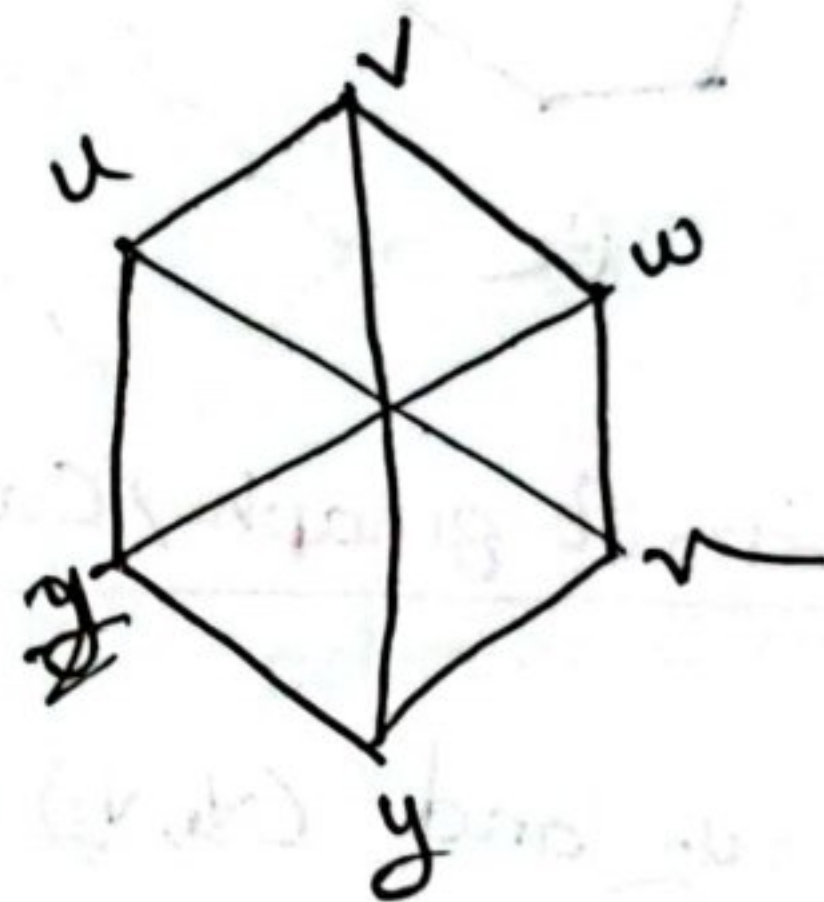
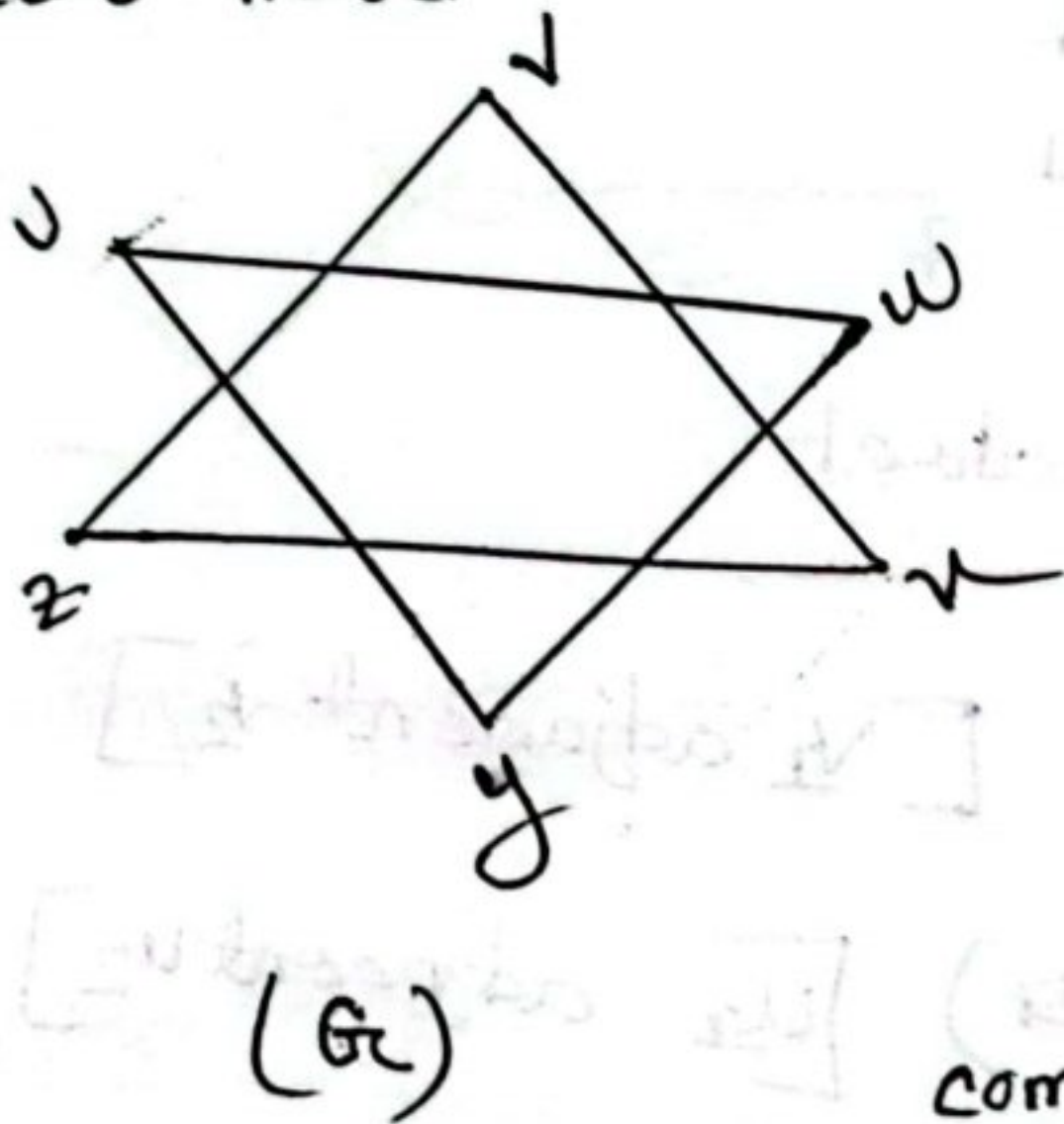
$$= \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10} -$$

$$\{e_1, e_9\}$$

$$= \{e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_{10}\}$$

Complement The complement G^c on (G) is defined

as a simple graph with the same vertex set as G and where two vertices u and v are adjacent only when they are not adjacent in G .



complement $\longrightarrow$

For a graph G and its complement G^c ,

- (i) $G \cup G^c = K_n$
- (ii) $V(G) = V(G^c)$
- (iii) $E(G) \cup E(G^c) =$

Selfcomplementary: A graph G is self-complementary if it

is isomorphic to its complement.



G



G^c

Products of graph / Cartesian product

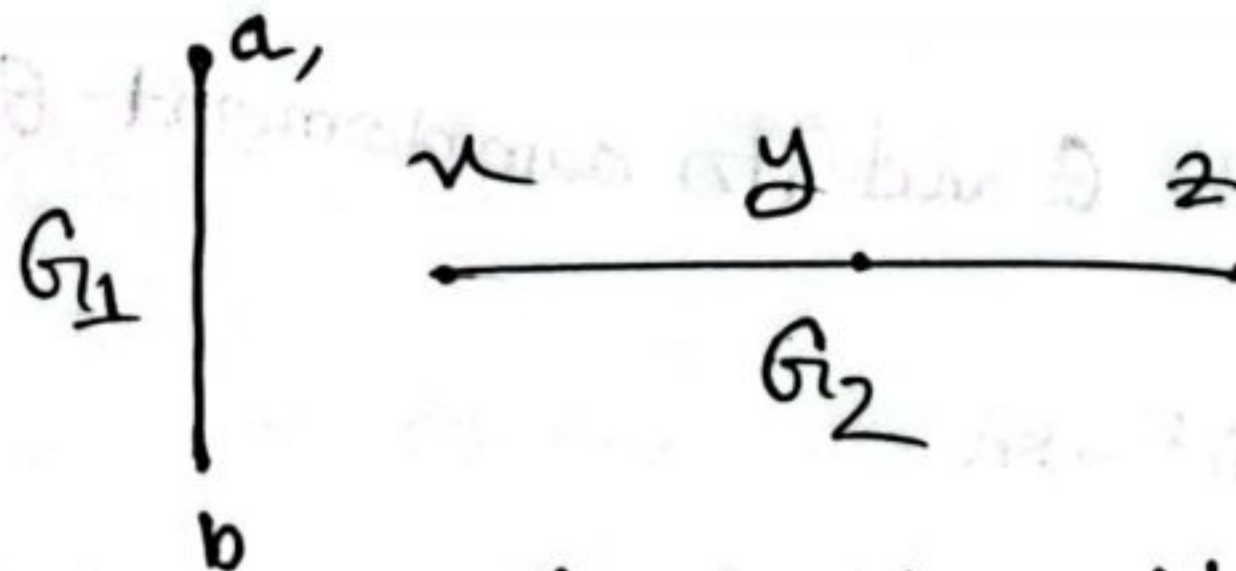
(i) $u_1 = u_2$ and $(v_1, v_2) \in E(G_2)$ [v_1 adjacent v_2]

(ii) $v_1 = v_2$ and $(u_1, u_2) \in E(G_1)$ [u_1 adjacent u_2]

Ex:

$$A = \{a, b\}$$

$$B = \{x, y, z\}$$



$$A \times B = \{(a, x), (a, y), (a, z), (b, x), (b, y), (b, z)\}$$

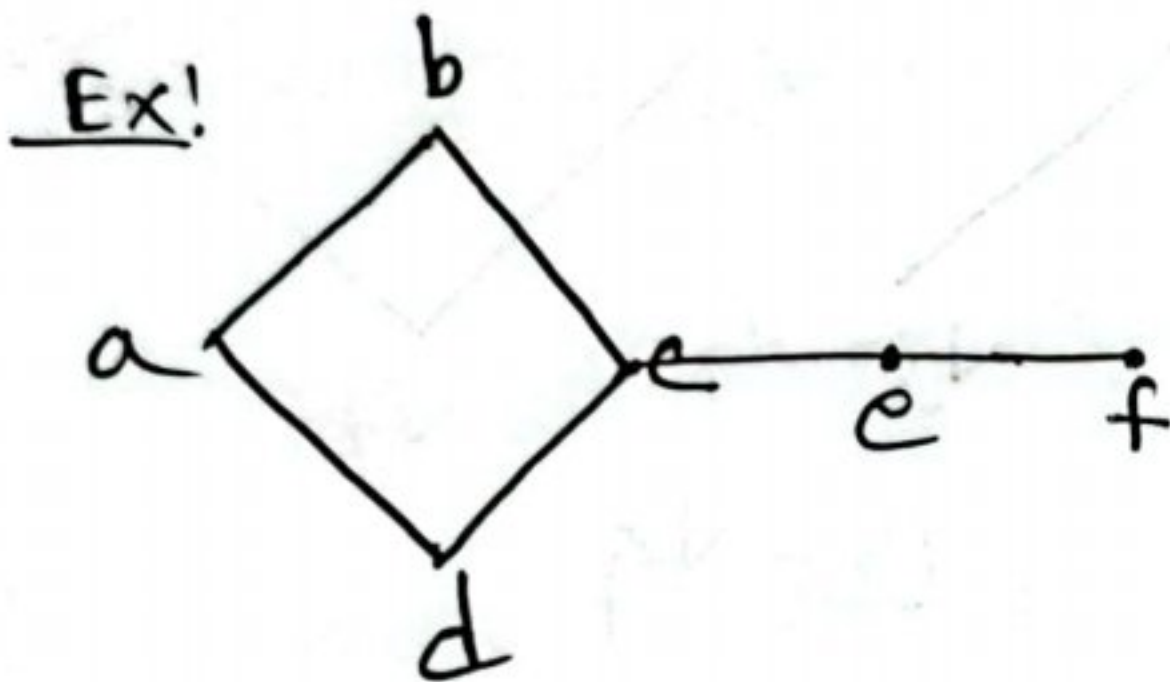
and

$$|V(G_1)| = 2, |V(G_2)| = 3$$

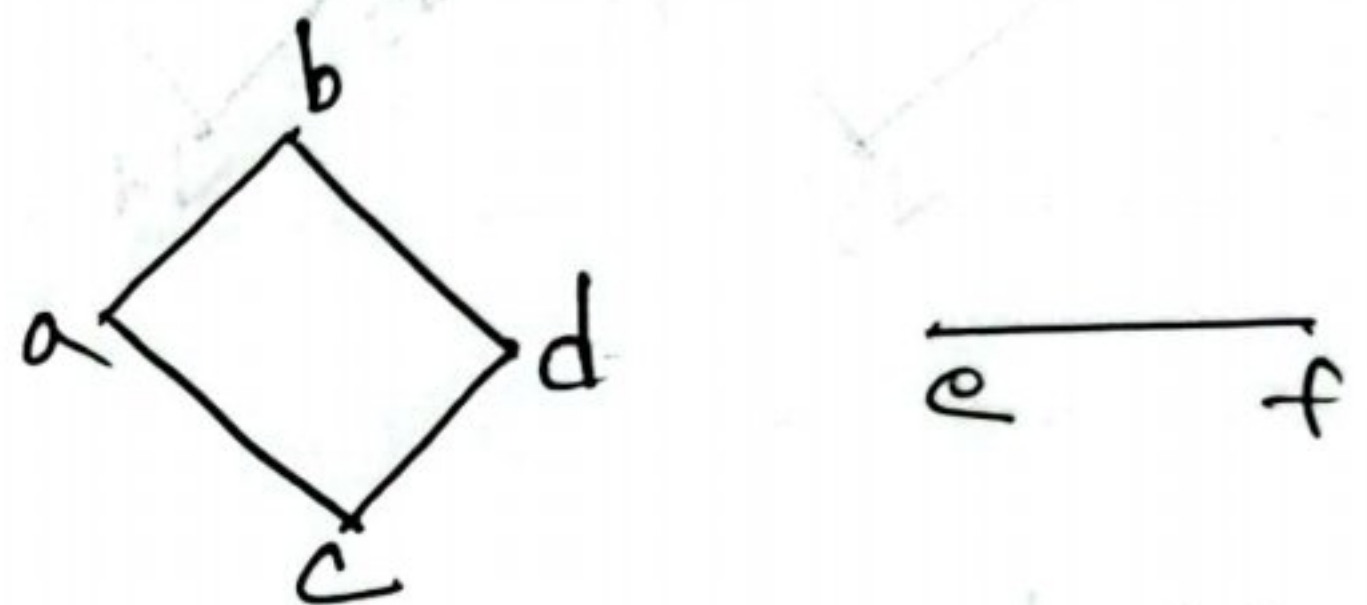
$$|E(G_1)| = 1, |E(G_2)| = 2$$

Connected graphs

Two vertices u and v are said to be connected if there exists a path between them.



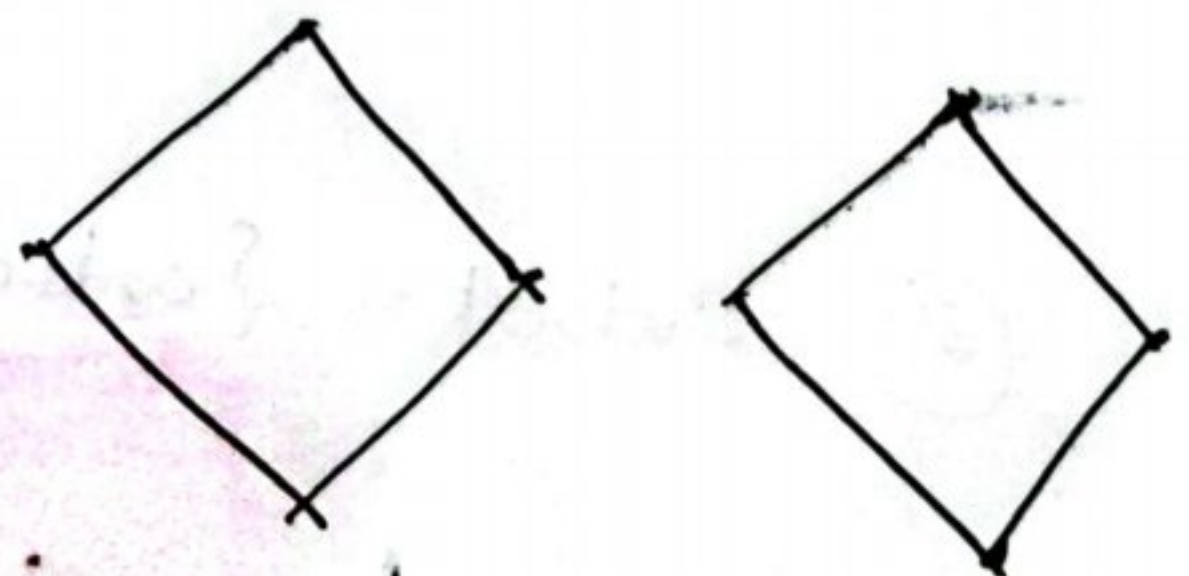
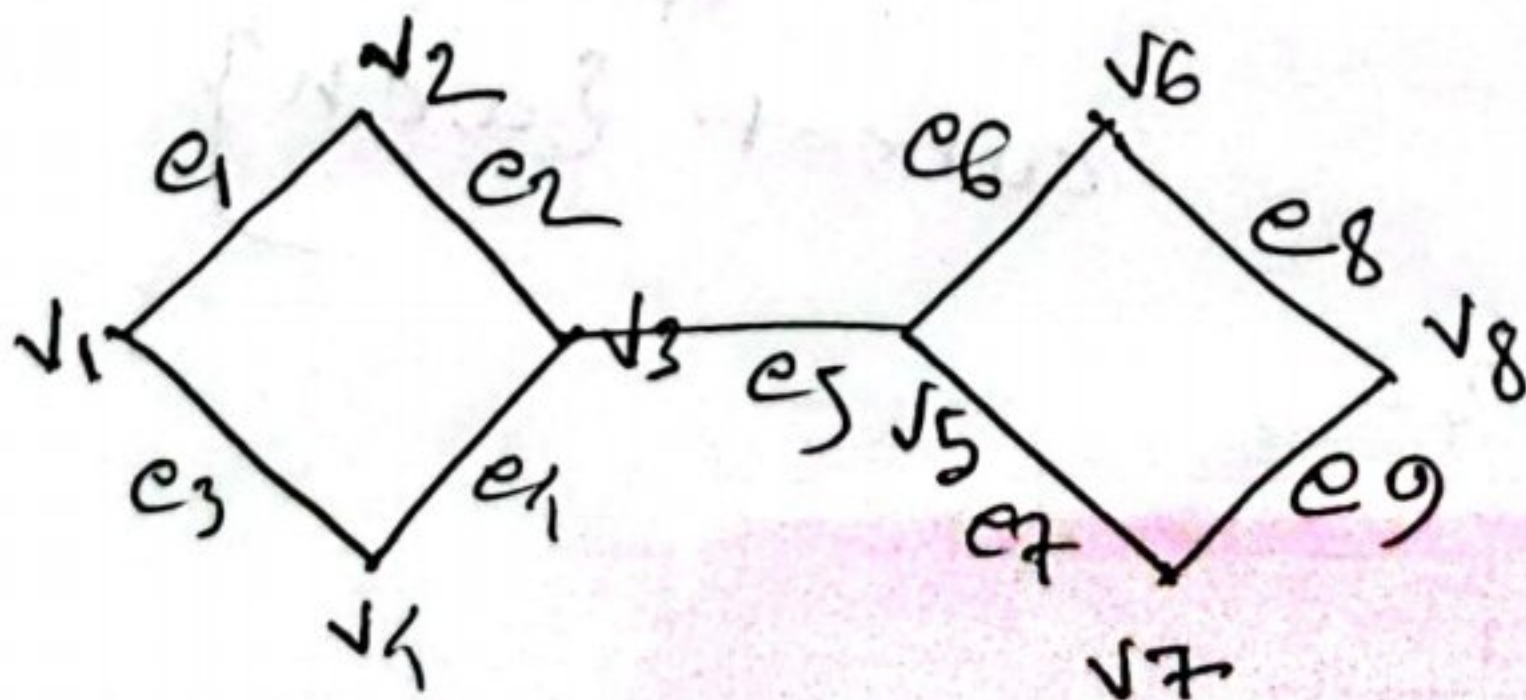
G_1 : connected



G_2 : not connected

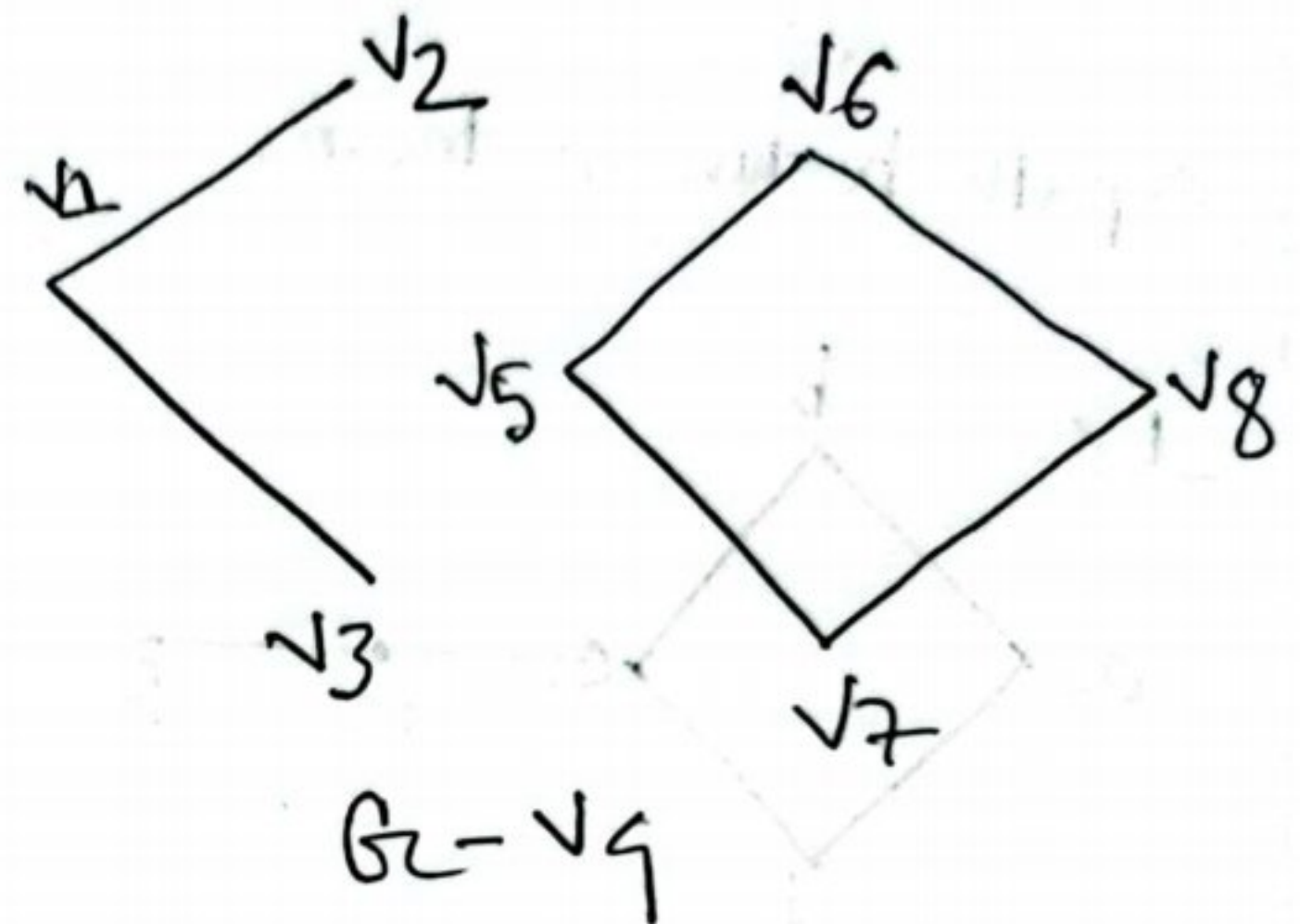
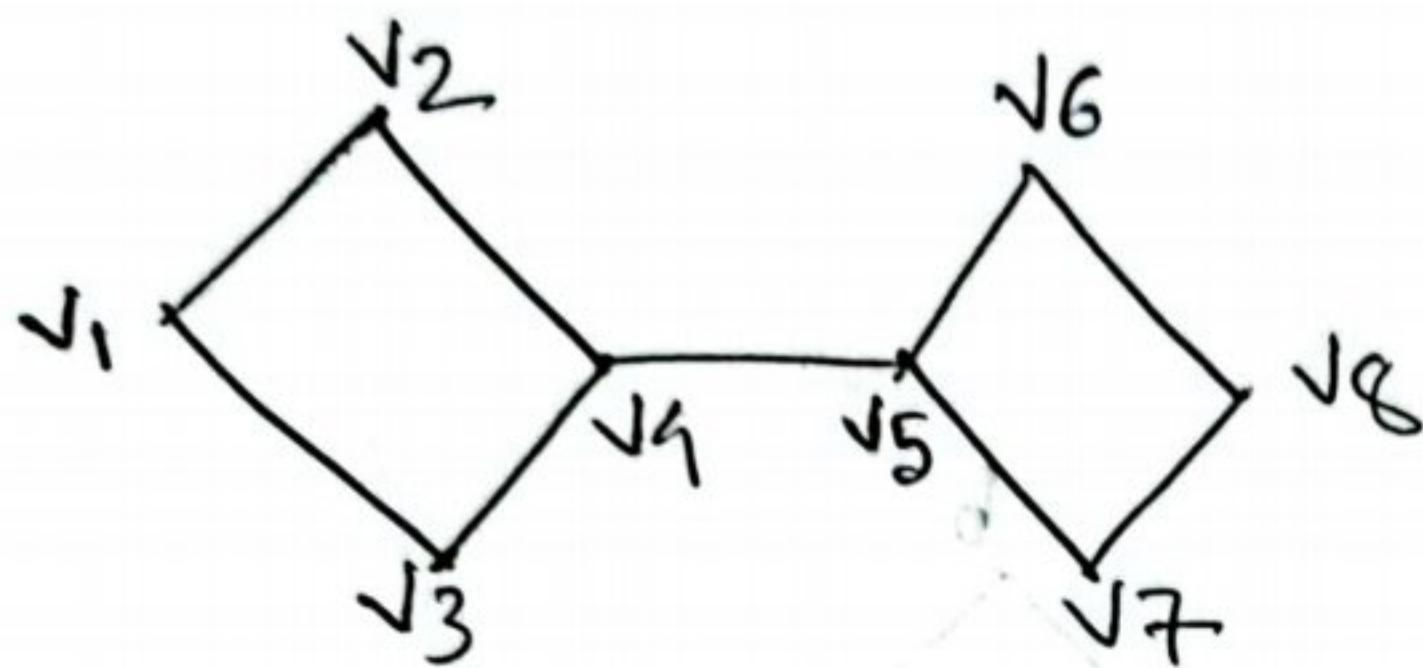
Cut edge

An edge e of a graph G is said to be a cut-edge or a bridge of G if $G - e$ is disconnected.



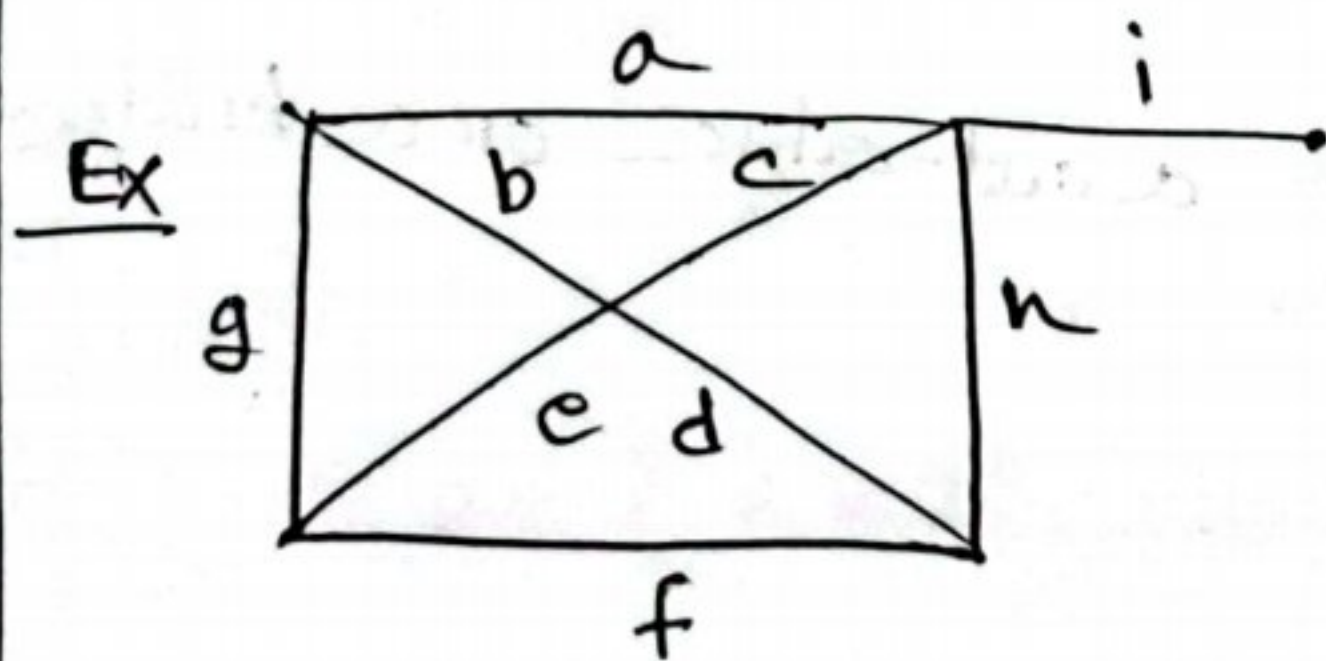
$G - e_5$

Cut-Vertex A vertex v of a graph G is said to be a cut vertex of G if $G-v$ is disconnected.

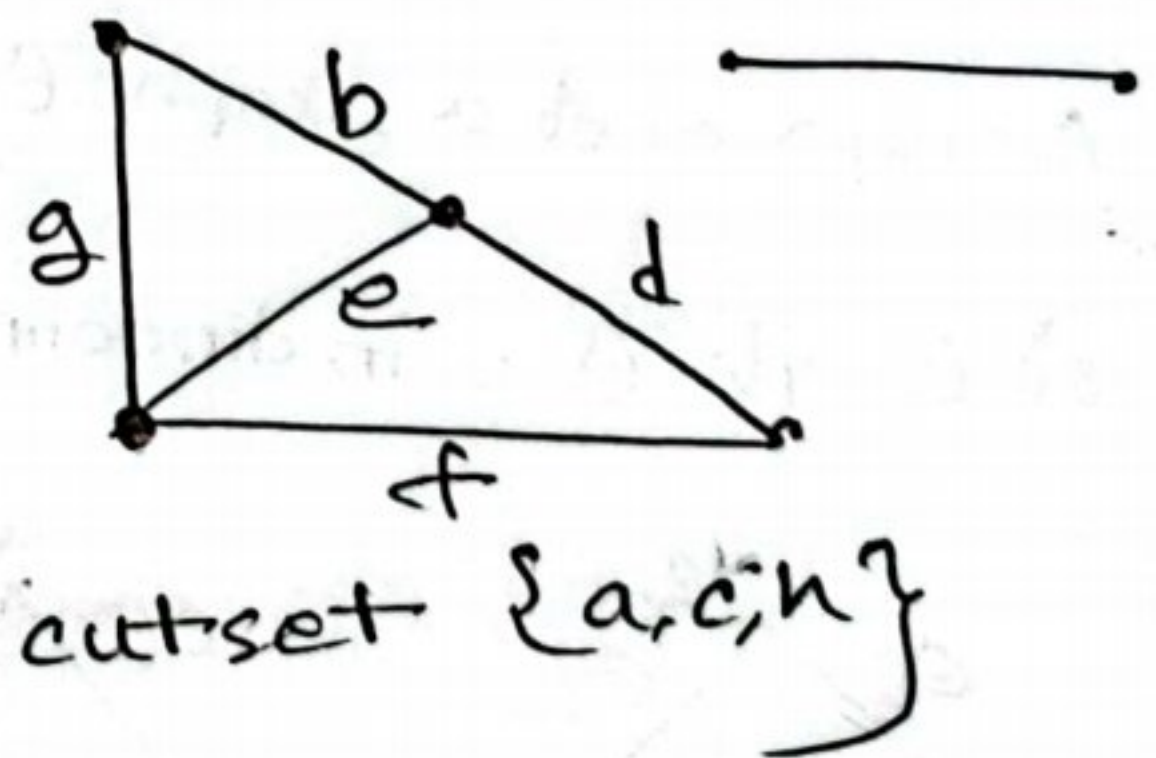


Cutset

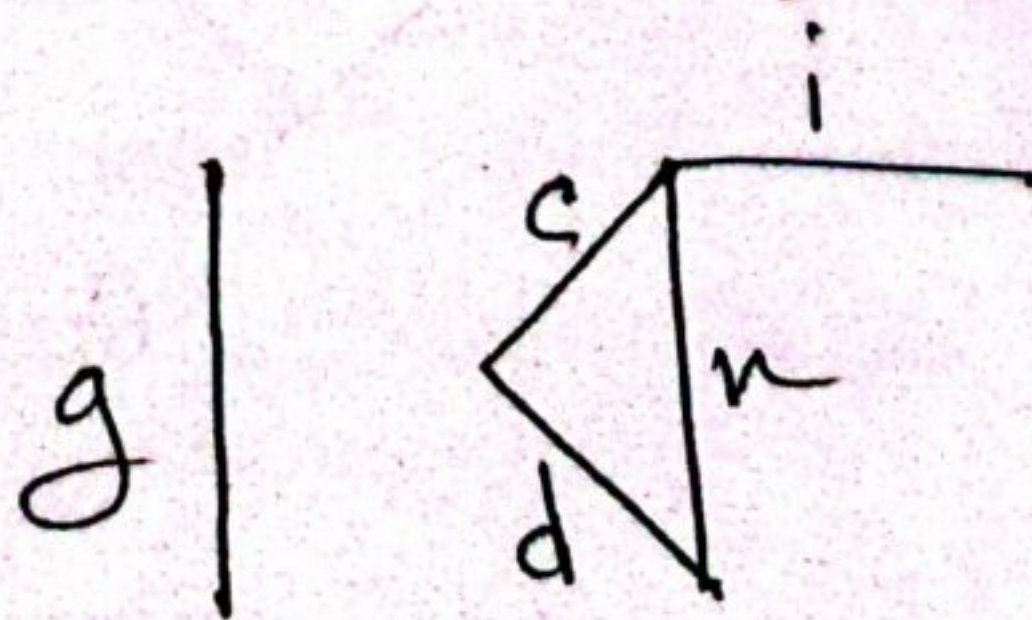
The set of all minimum number of edges of G whose removal disconnects a graph G is called a cutset of G .



(i)



(2) Cutset = $\{a, b, e, f\}$

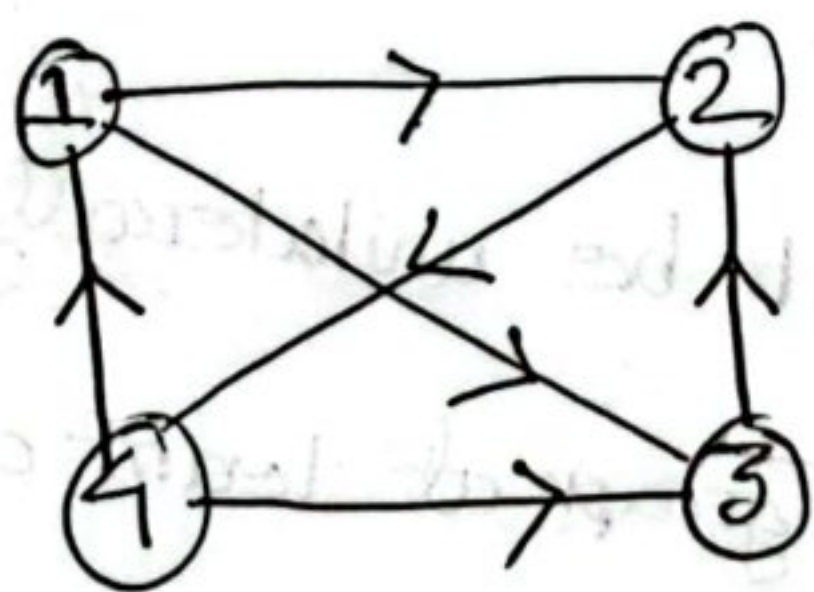


Connectedness in Direct graph

- (i) Strongly connected (ii) weakly connected (iii) Unilaterally

(i) Strongly connected:

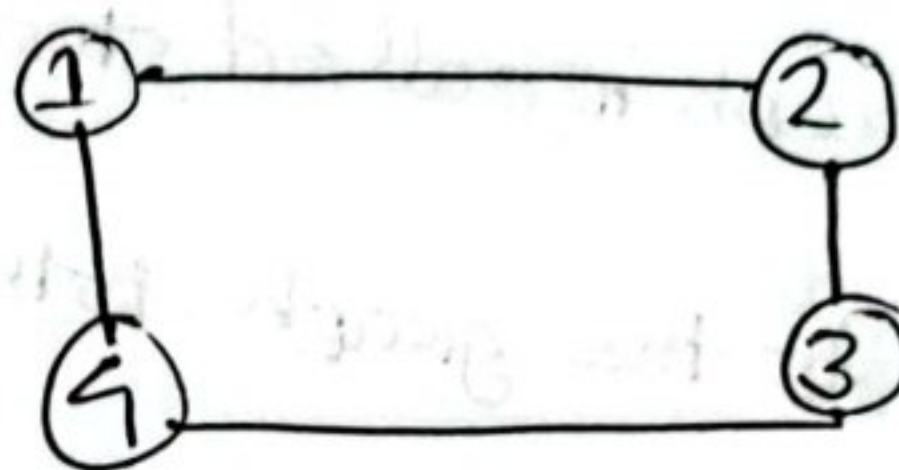
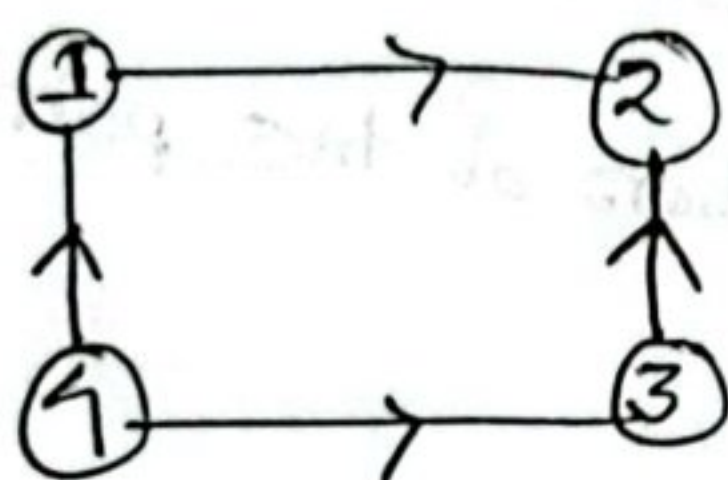
A directed graph is called strongly connected if for any pair of vertices of the graph both the vertices of the pair are reachable from one another.



<u>pair of vertices</u>	<u>forward path</u>	<u>Reverse Path</u>	<u>Backward Path</u>
(1,2)	1-2	(2,1)	2-4-1
(1,3)	1-3	(3,1)	3-2-4-1
(1,4)	1-2-4	(4,1)	4-1
(2,3)	2-4-3	(3,2)	3-2
(2,4)	2-4	(4,2)	4-1-2
(3,4)	3-2-4	(4,3)	4-3

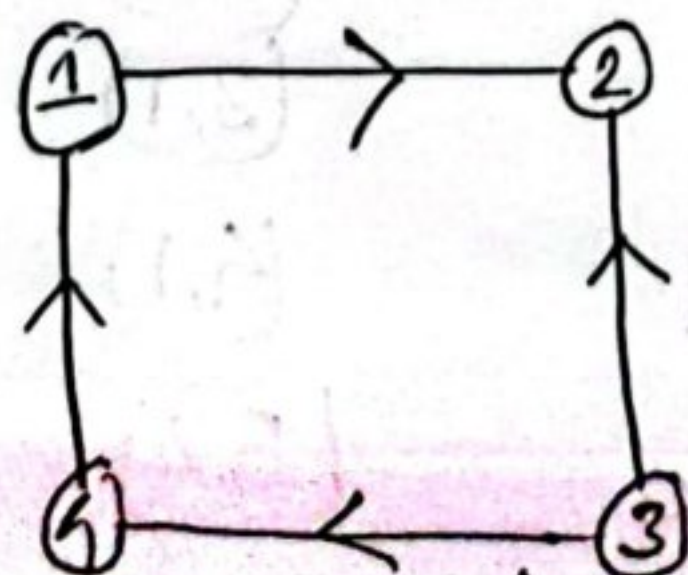
2. Connected or weakly connected

A directed graph is called connected or weakly connected if it is connected as an undirected graph in which each directed edge is converted to an undirected graph.



3. Unilaterally connected

A simple directed graph is said to be unilaterally connected if for any pair vertices of the graph at least one of the vertices of the pair is reachable from other vertex.

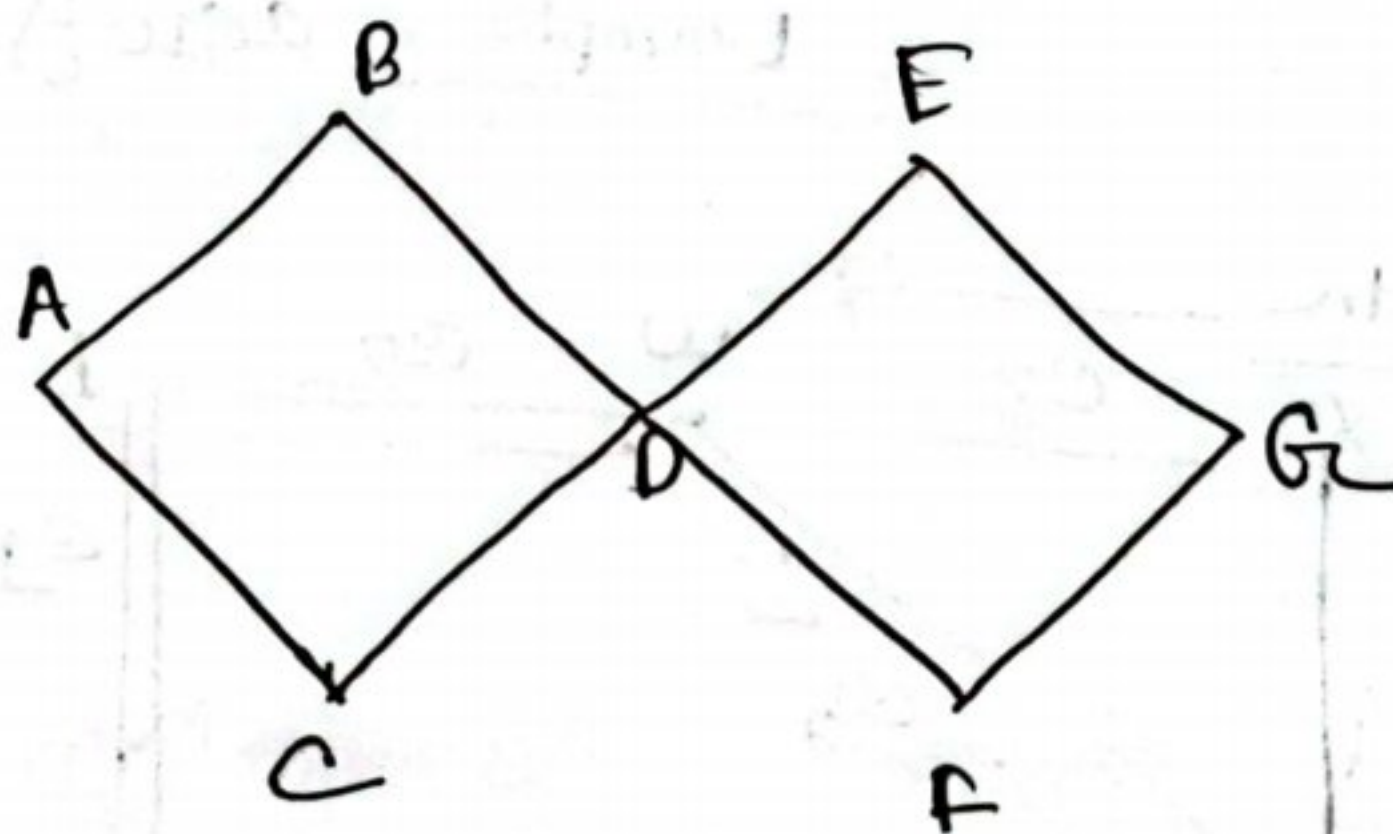
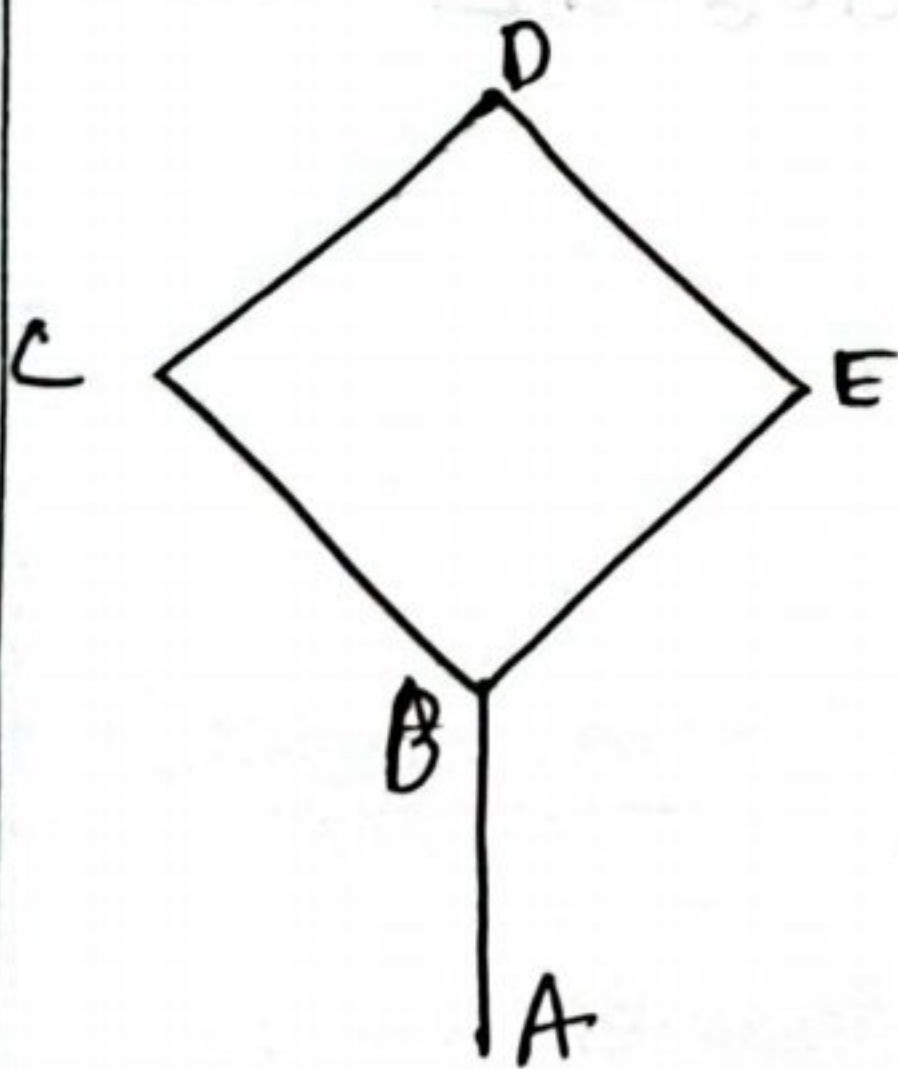


(pair)	forward	Backward
(1, 2)	1-2	X
(1, 3)	X	3-4-1
(1, 4)	X	4-1
(2, 3)	X	3-2
(2, 4)	X	4-1-2
(3, 4)	3-4	X

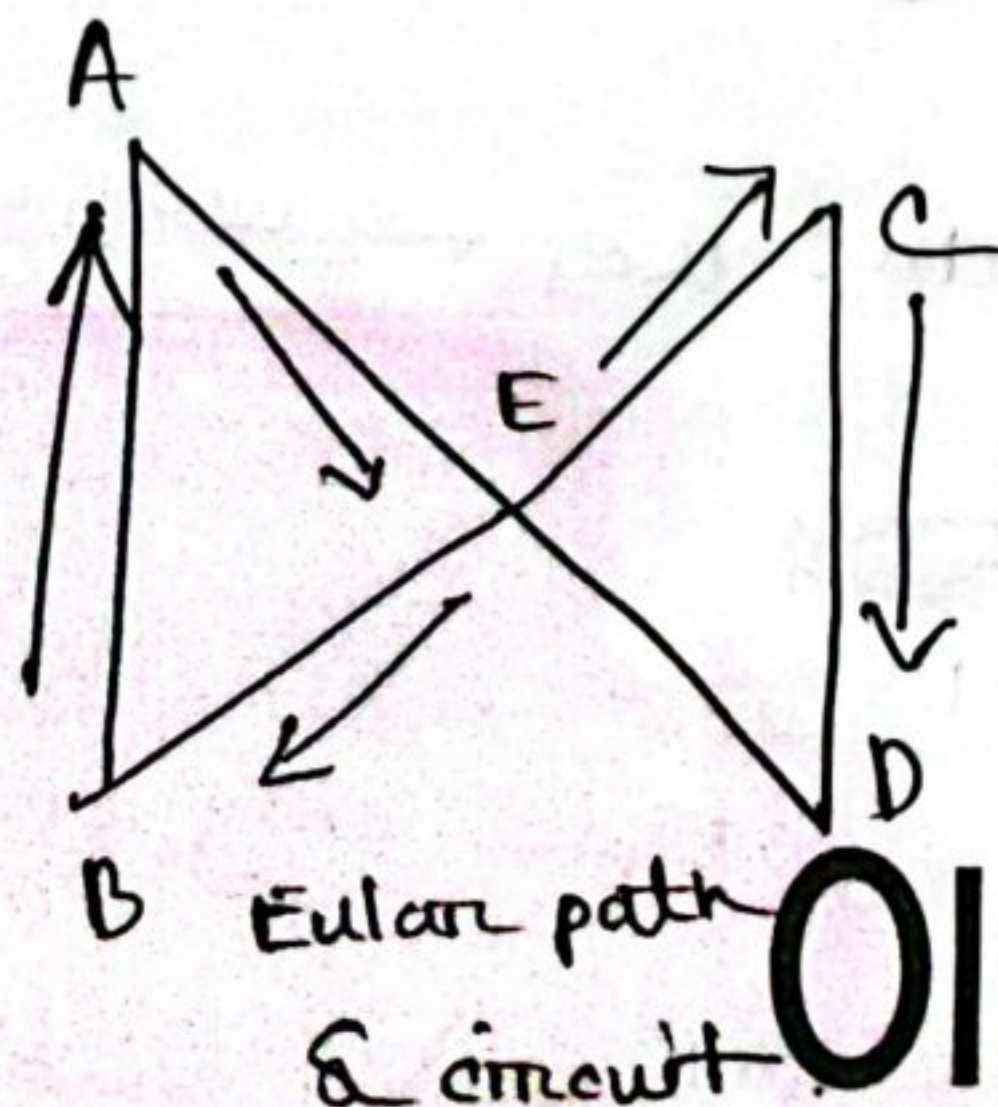
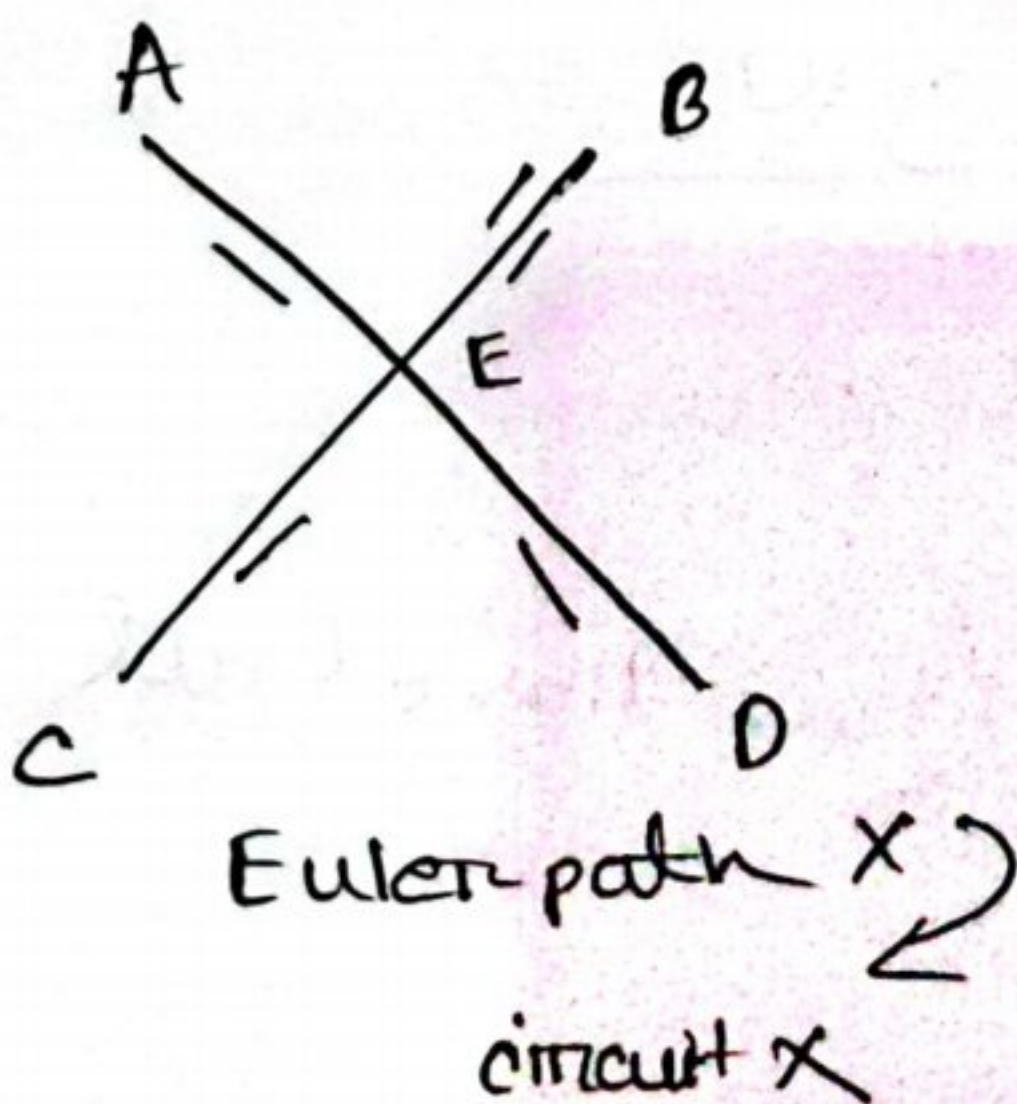
Eulerian graph

Euler Path: A path in a graph is called Euler path if it includes every edge exactly once. Since the path contains every edge exactly once, it is also called euler trail.

Euler circuit: An euler path that is circuit is called Euler circuit. A graph which has a Eulerian circuit is called an eulerian graph.

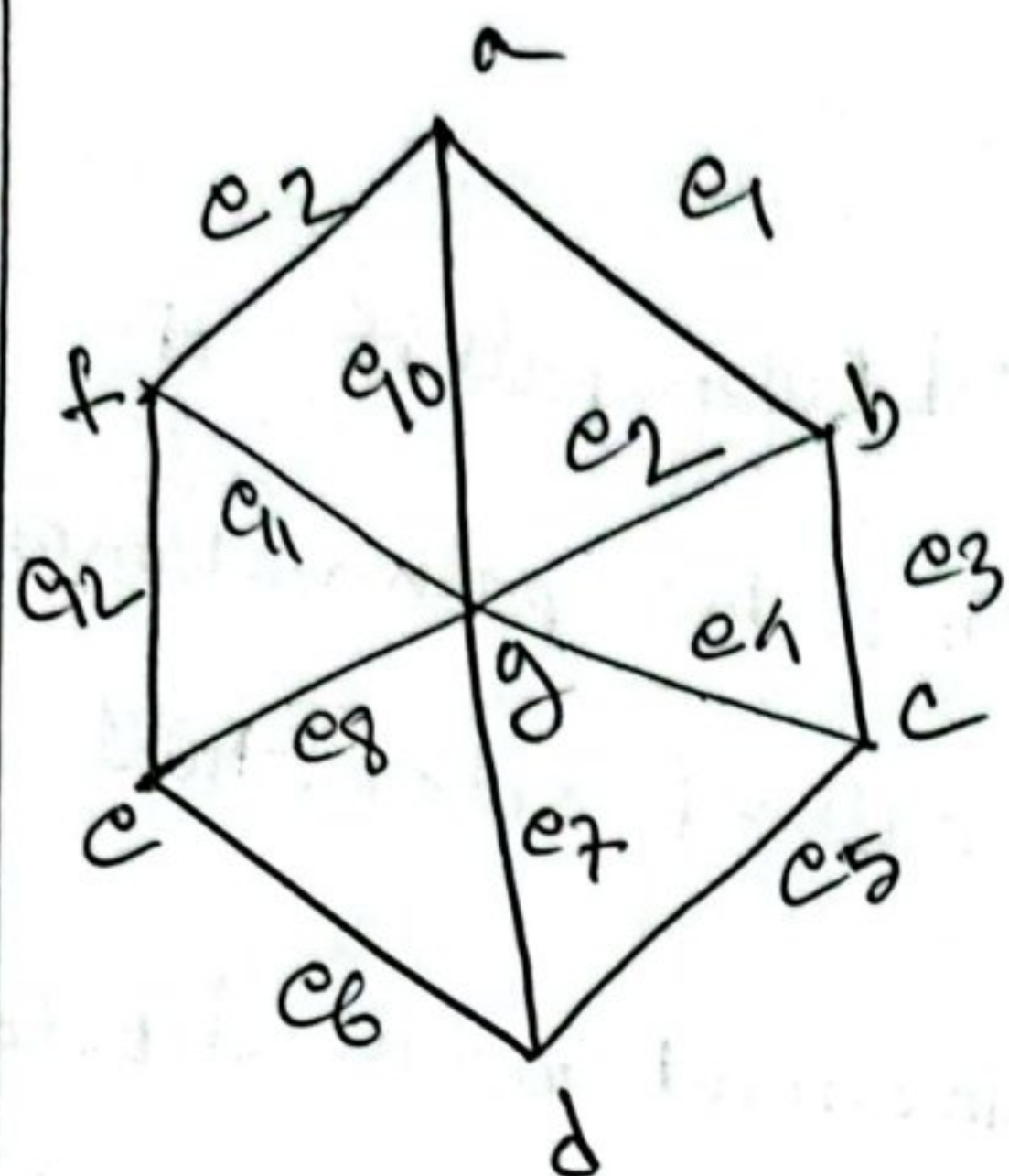


Euler path & circuit → Eulerian graph



Truly once daily

Olmesan[®]



Trail: Edge can't be repeated

Circuit: A closed trail is called circuit.

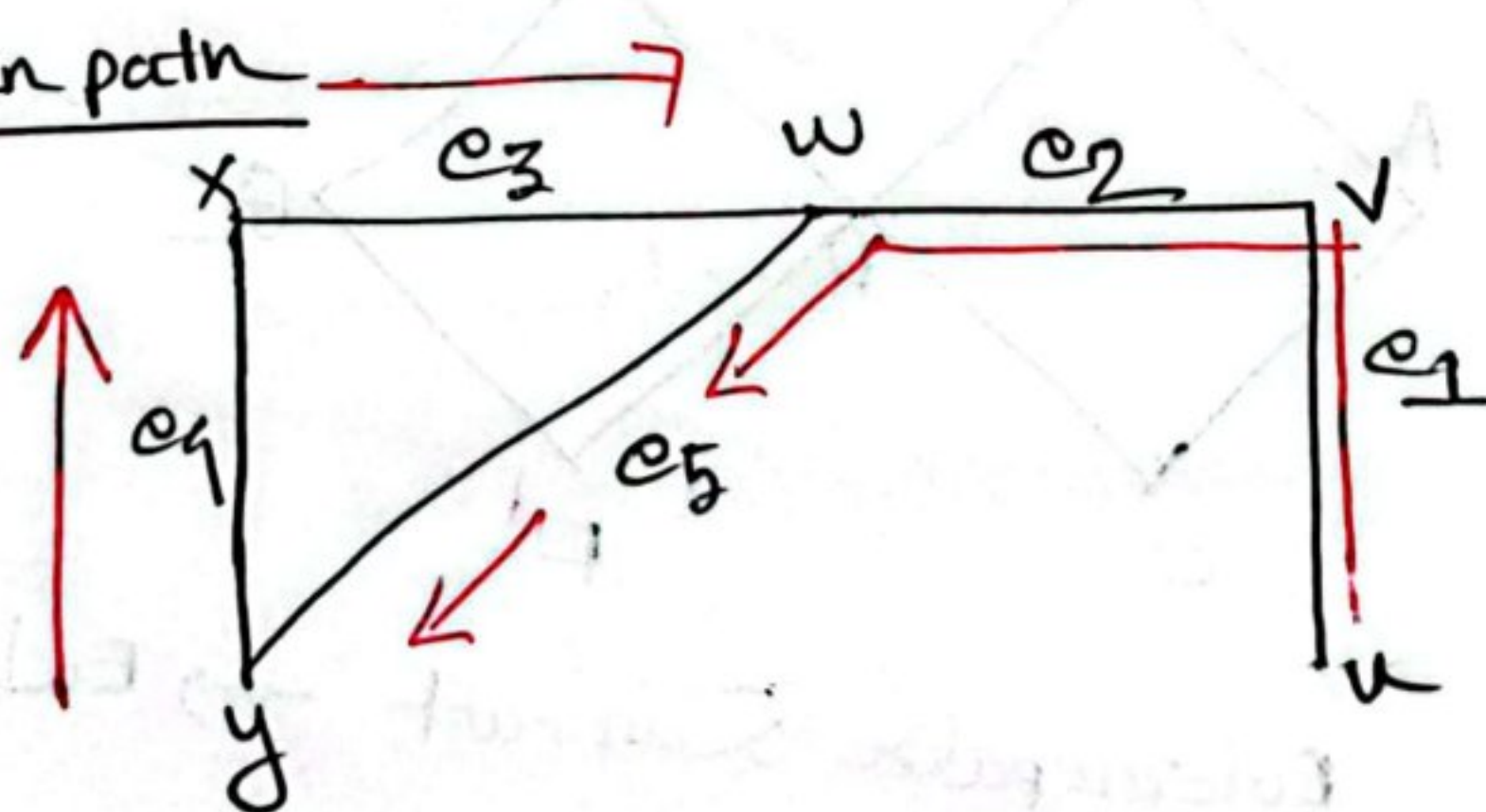
Example: ~~a e2 f e11 g e2 b e3 a~~

Path: Vertices can't be repeated

Cycle: closed path is called cycle

Example: a e10 g e7 d e6 e4 f e2 a

Eulerian path



path: $u e_1 v e_2 w e_5 y e_4 x e_3 w$

~~Edge~~

Note:

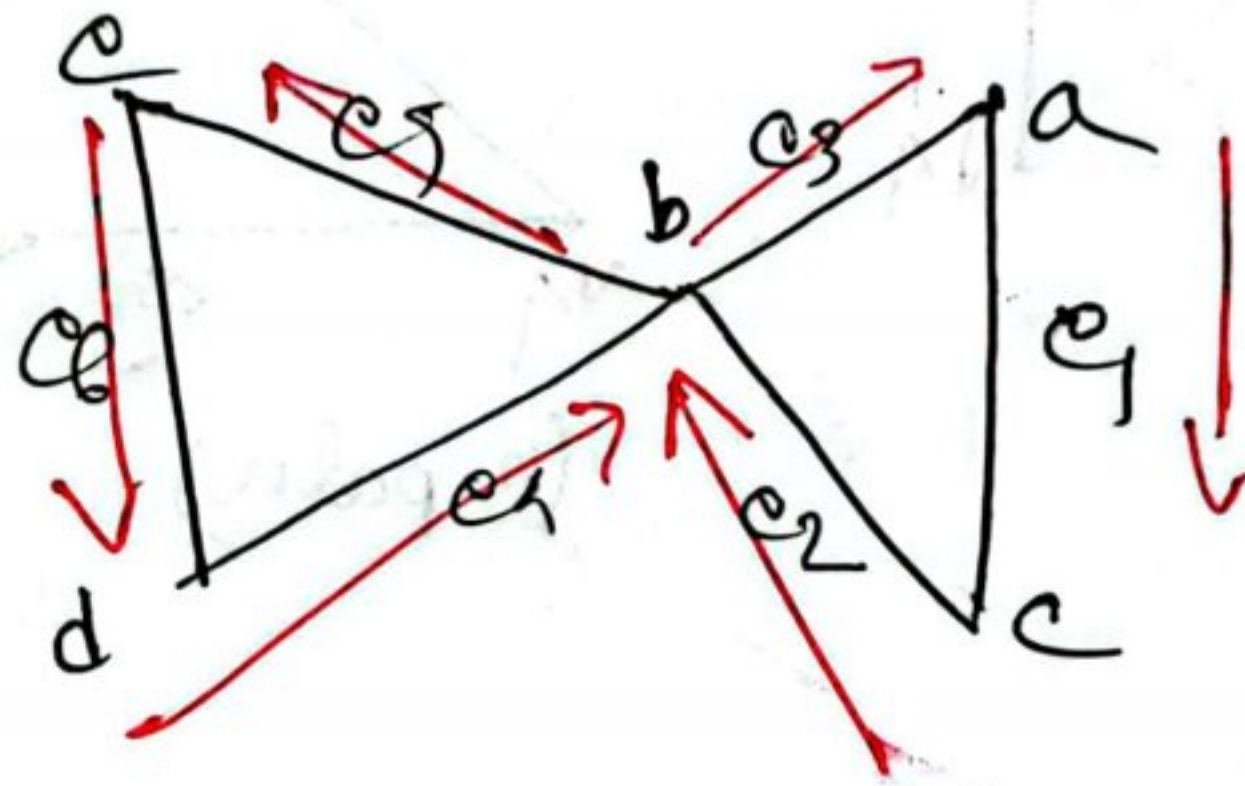
Edge only once (has travel all edges).

Eulerian Path Circuit

if it traverses each edge in the graph once and has to be closed

ଆରମ୍ଭ ଓ ଶେଷ ସମାନ ଥାଏ ଏବଂ ସବୁ ଇଡ୍ଜେସ୍ କଭର୍ଡ୍ -

eulerian graph.



b e₃ a e₁ c e₂ d e₆ c e₅ b e₄ a

Hamiltonian Path

Hamiltonian path: A hamiltonian path is a simple path that

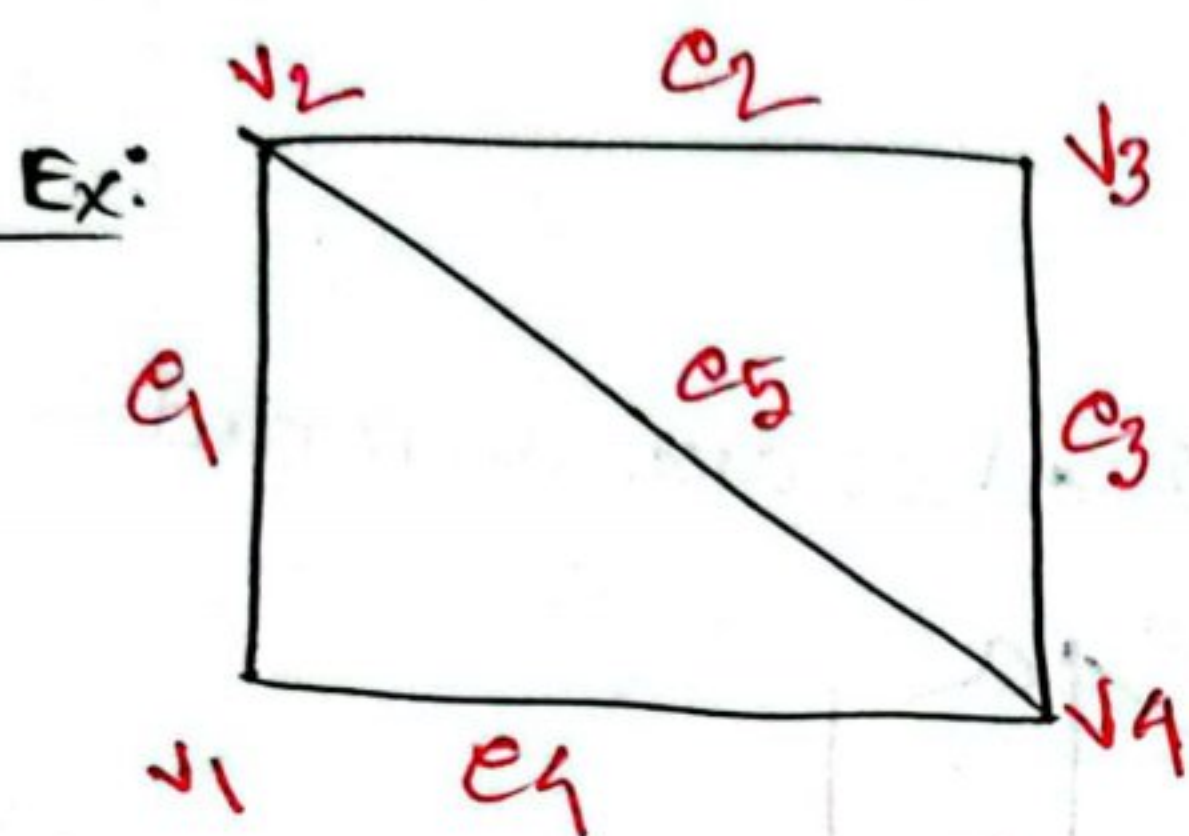
contains all vertices of G where the end points may be distinct.

Hamiltonian Circuit: A circuit in a graph that contains each

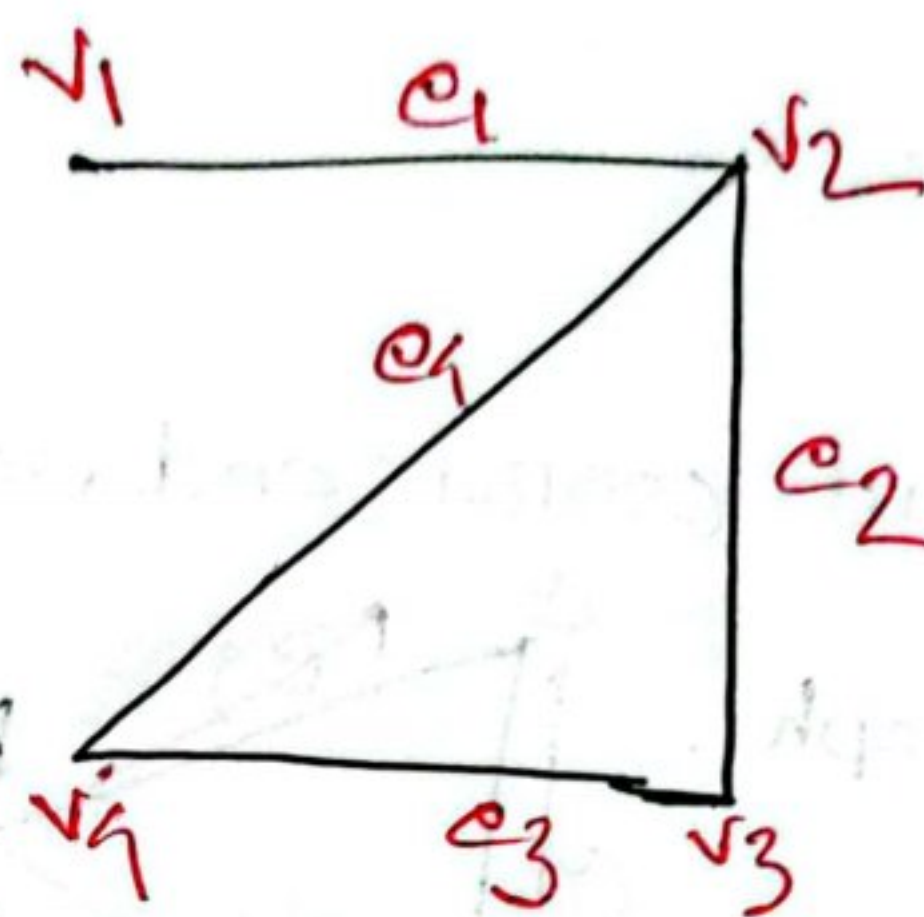
vertex in G exactly once, except for the starting and ending

vertex that appears twice known as hamiltonian circuit.

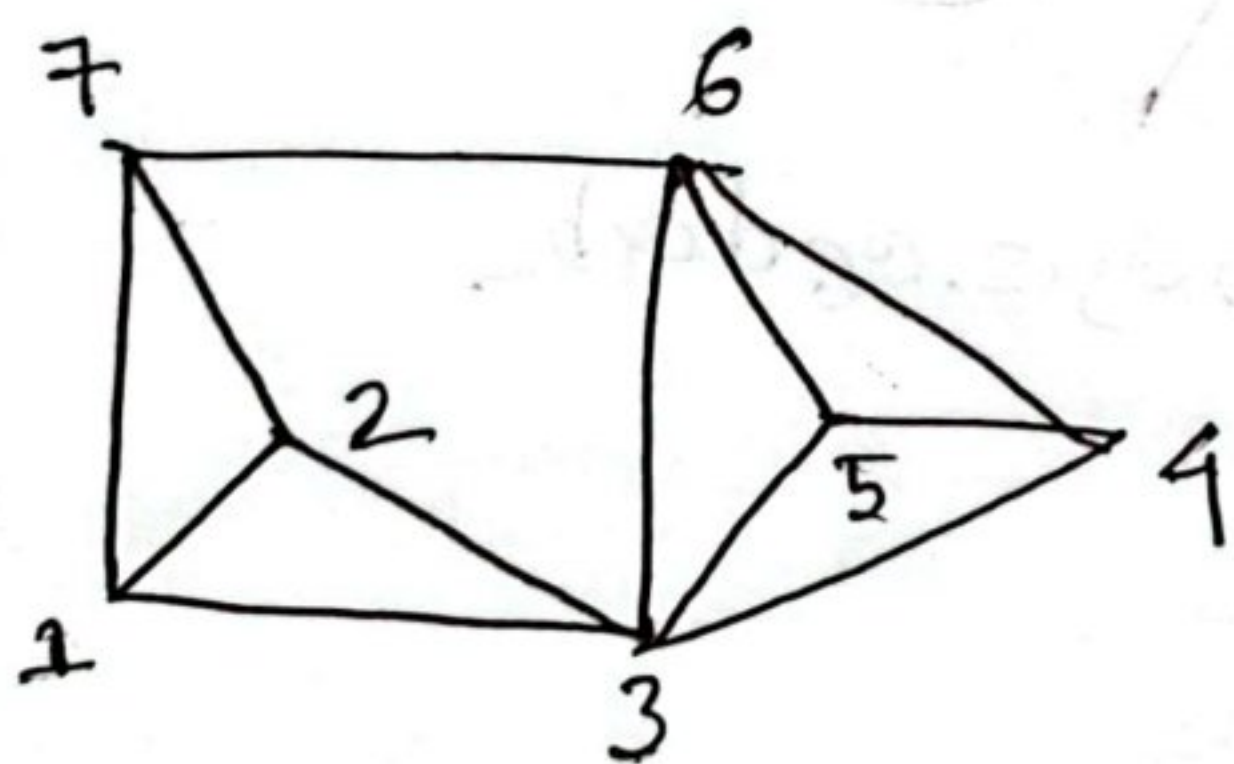
A graph G is called a hamiltonian graph if it contains a hamintontan circuit.



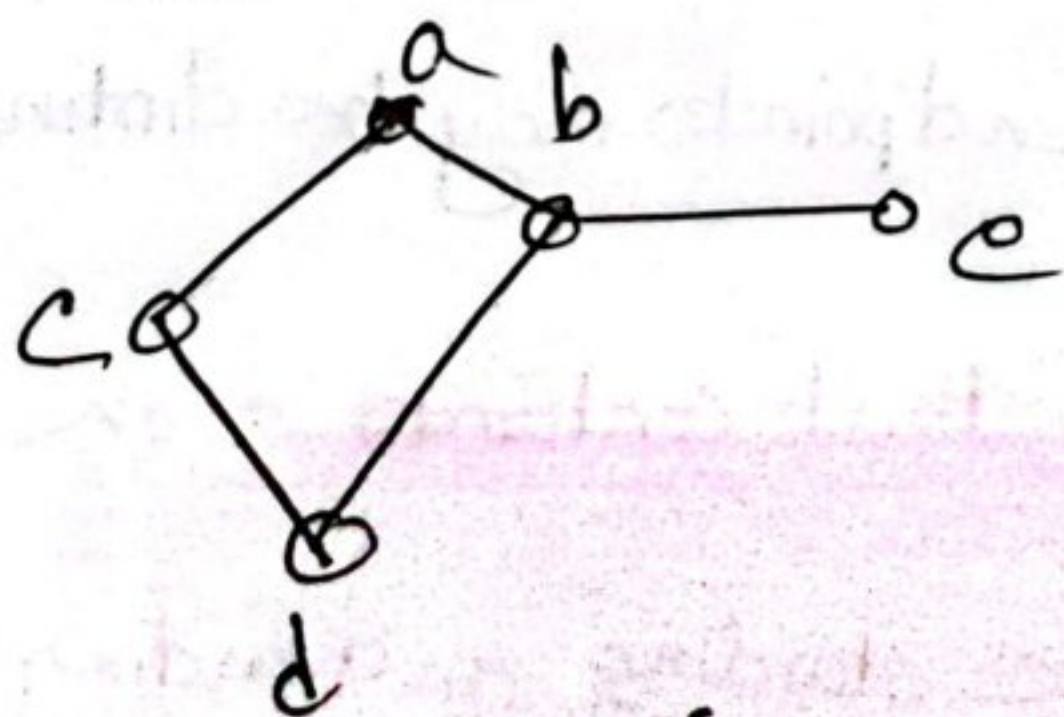
(h. circuit)



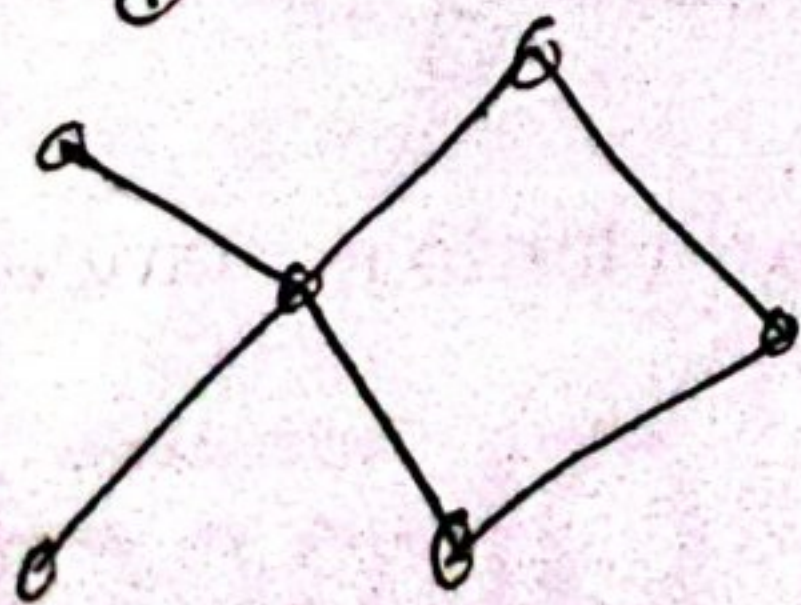
(h. path)



(72135467) h. circuit + has a h. path $\rightarrow$ h. graph



has h. path but no circuit $\rightarrow$ not h. graph

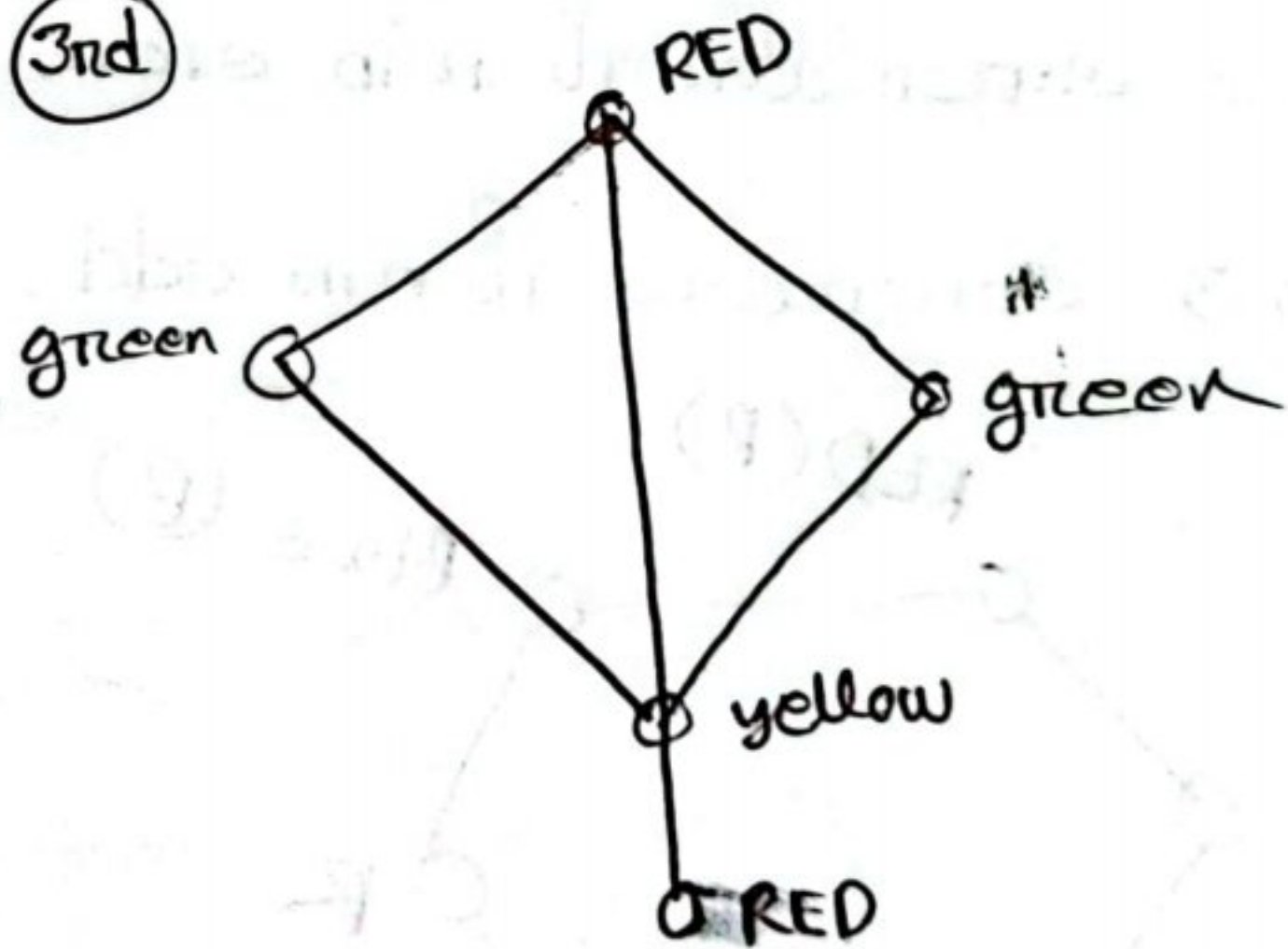
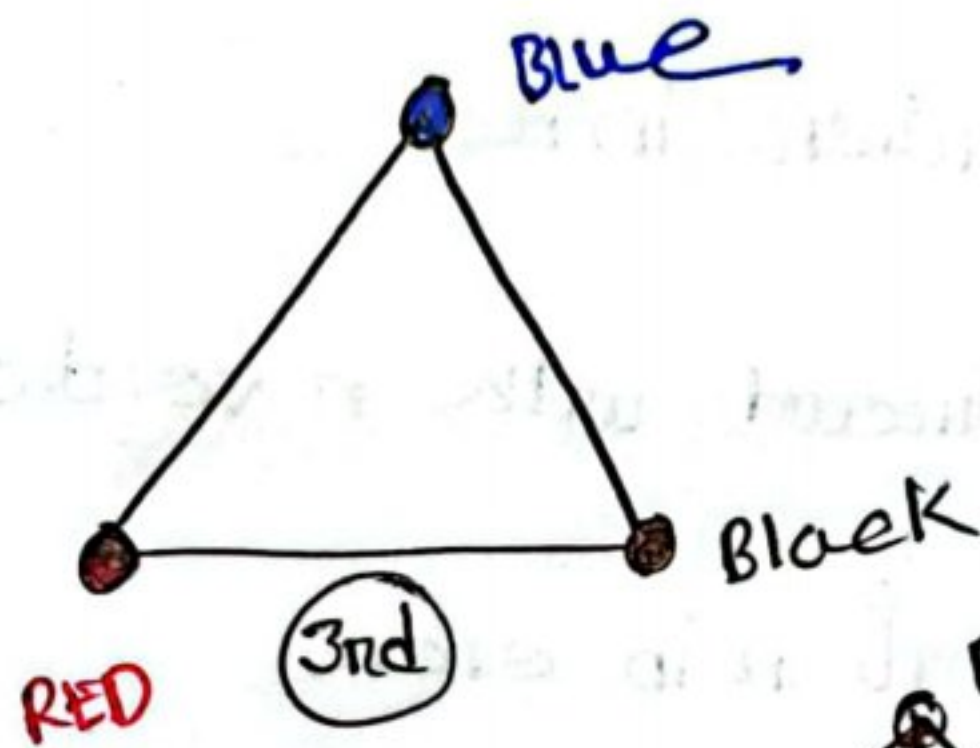
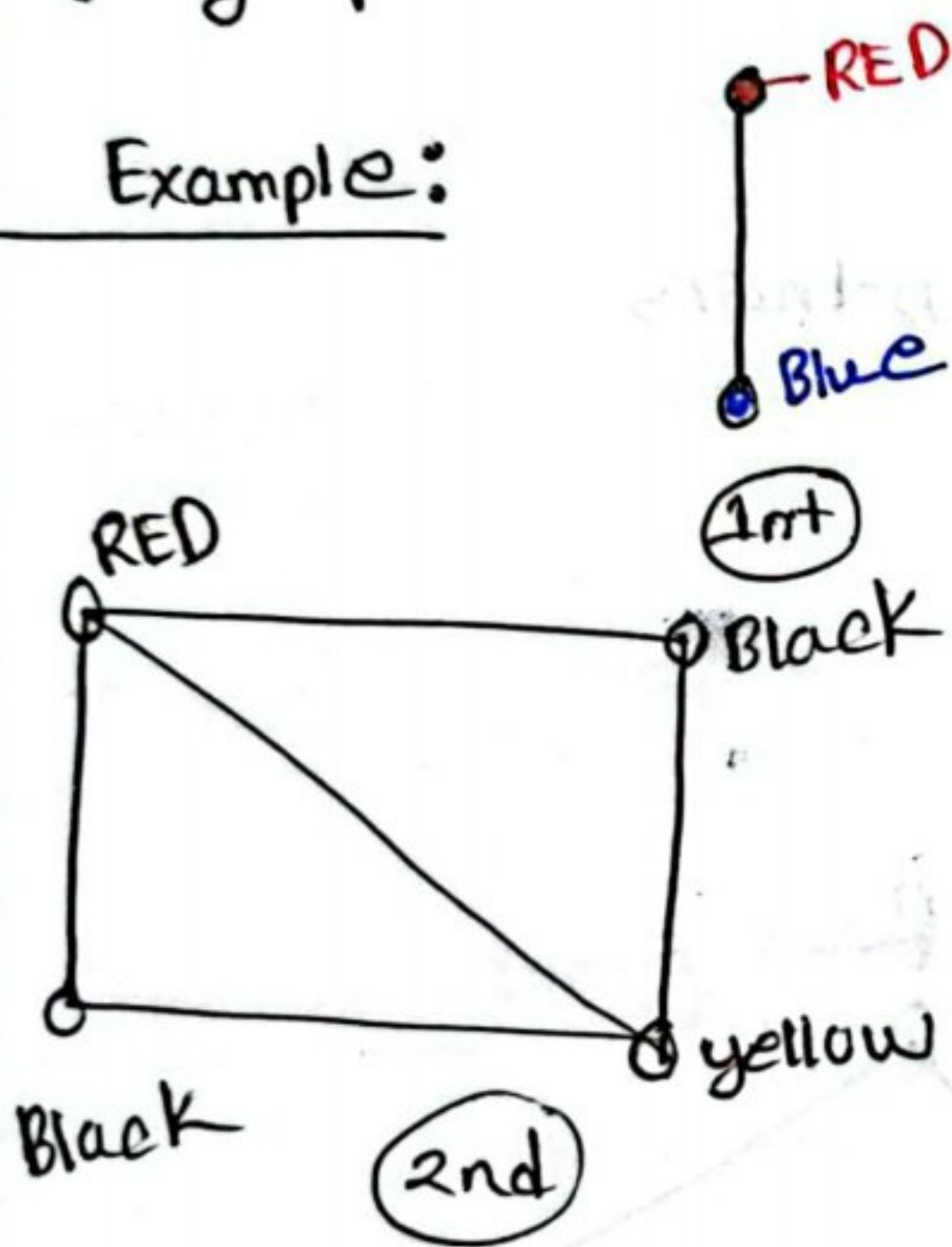


no hamiltonian path

graph coloring

Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called coloring of graph

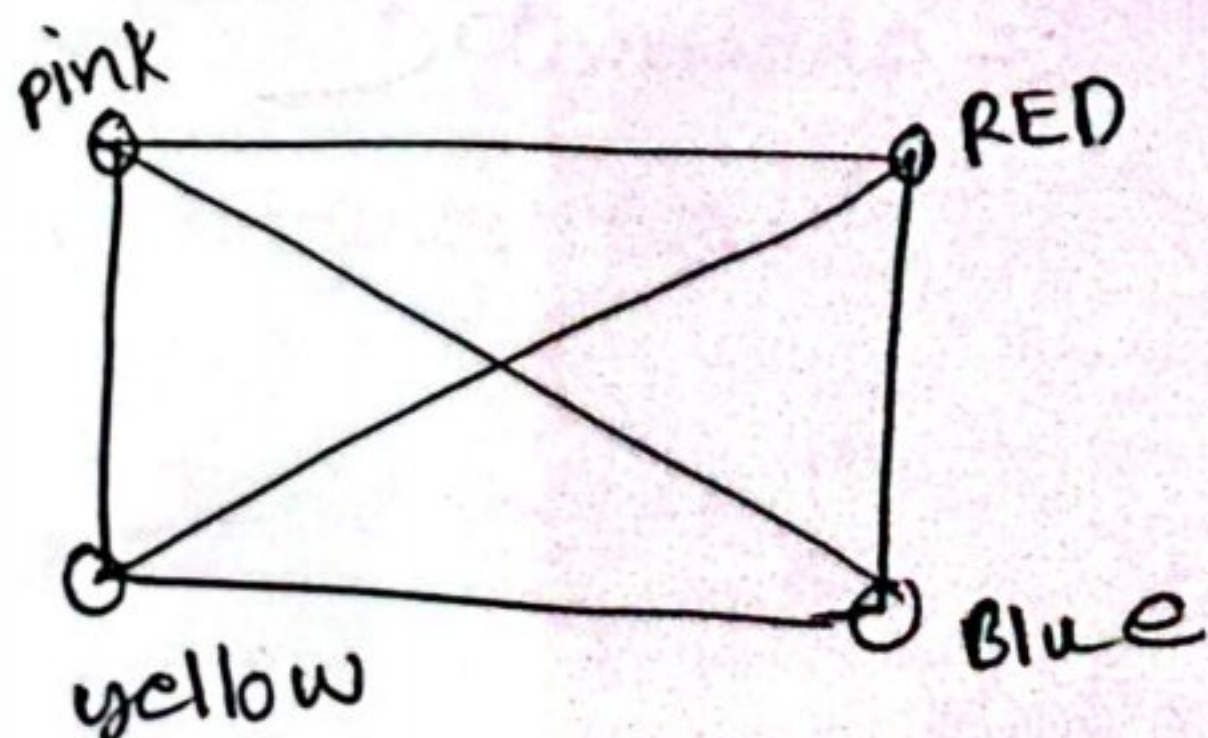
Example:



CHROMATIC NUMBER

1st 2-C number

2nd 3-C number



4-C number

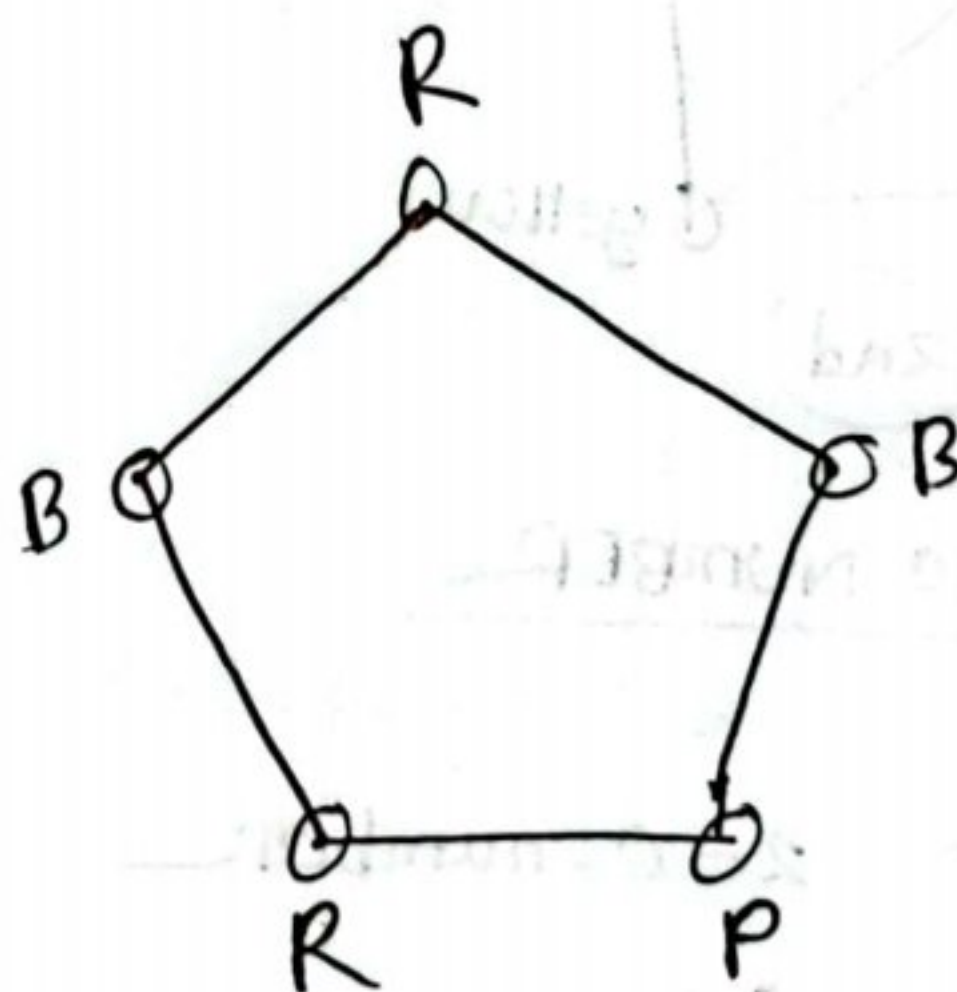
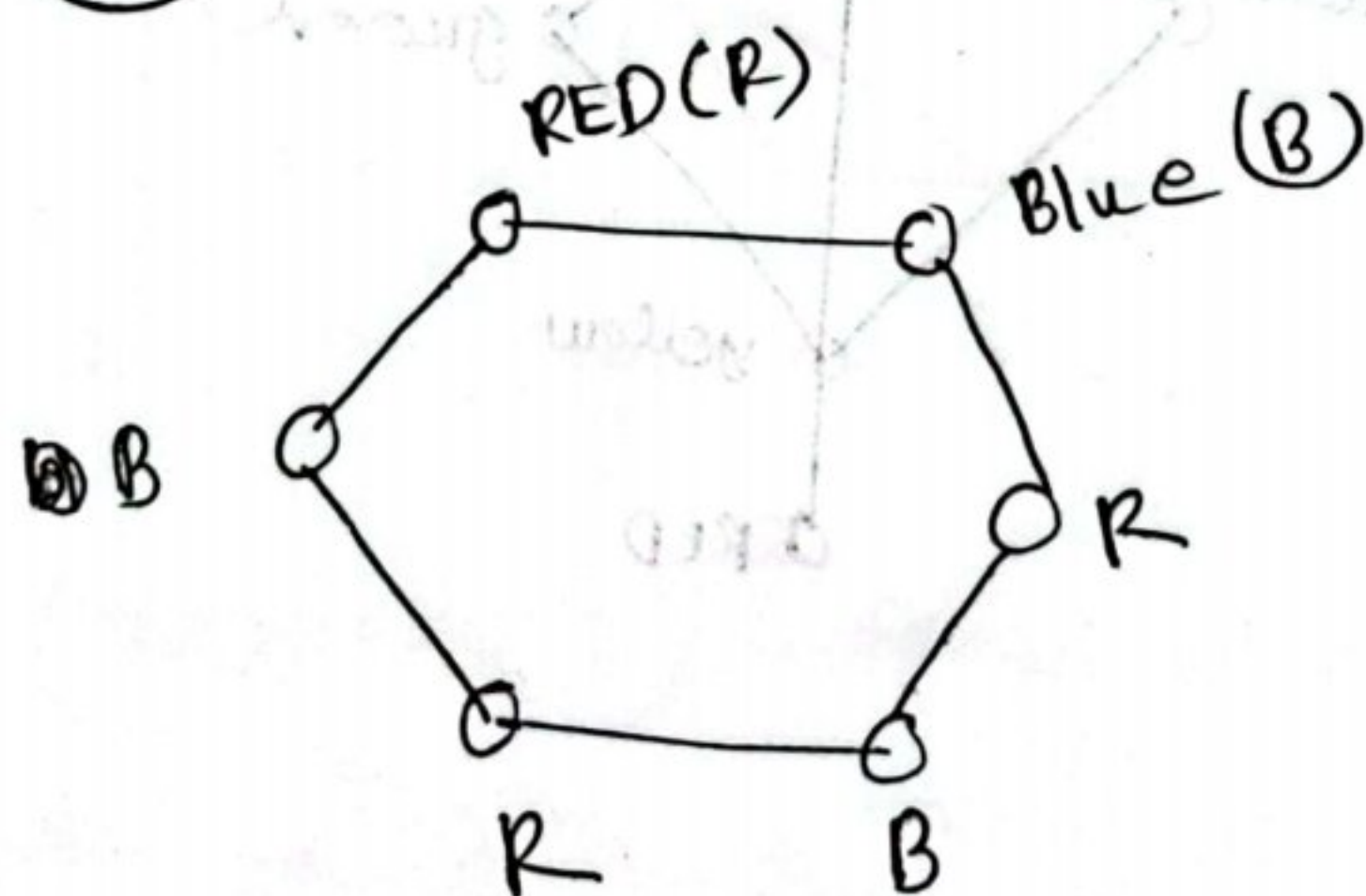
Note

The chromatic number of graph G is denoted by $\chi(G)$. If $\chi(G) = k$, then the graph is called k -chromatic. Chromatic number of null graph is 1. Chromatic number of complete graph K_n of n vertices is n .

*. If a graph is circuit with n vertices then,

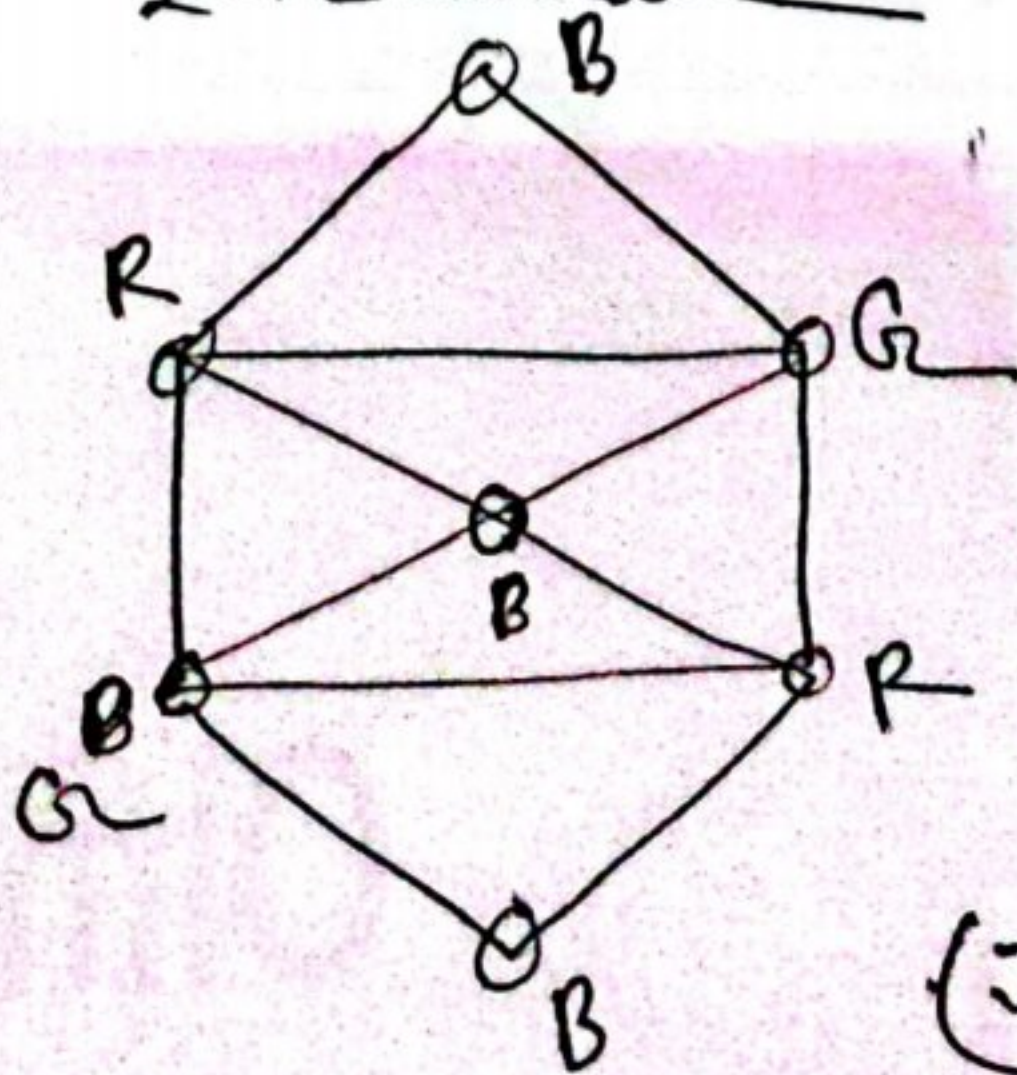
(i) 2-chromatic if n is even.

(ii) 3-chromatic if n is odd.



2-chromatic

3-chromatic



(3-c)

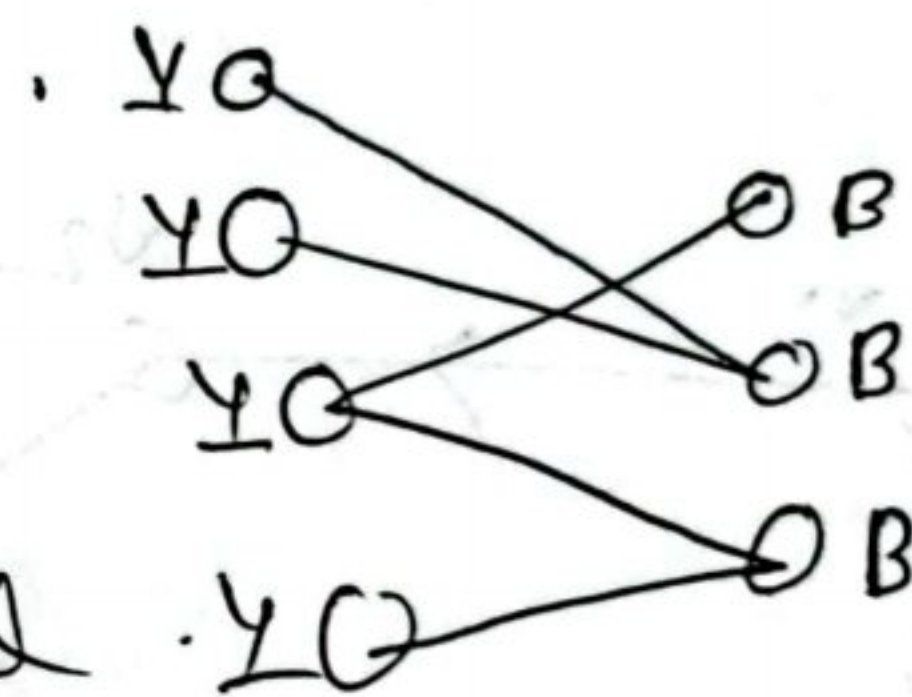
Planer graph cycle graph are also planer

A planer graph is a graph that can be drawn in a plane with such that none of its edges cross each other.

[chromatic number = ~~number of~~ (≤ 4)]

complete graph (n -vertices = n chromatic)

Bipartite graph χ number = 2



Trees Every tree is a bipartite graph. A tree is a special

type of connected graph there are χ num = 2
no circuits

Representation of graph

1) Adjacency matrix

2) Incidence matrix

3) Adjacency list

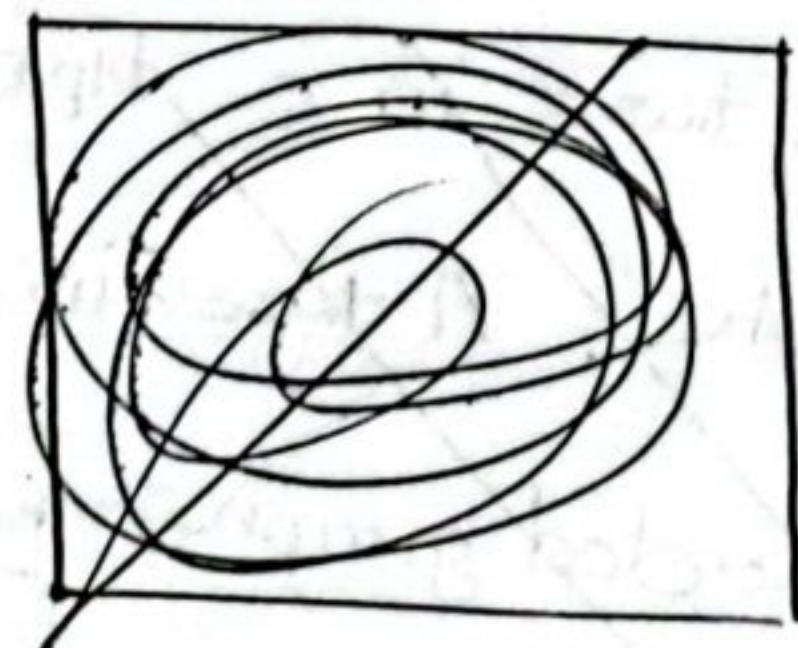
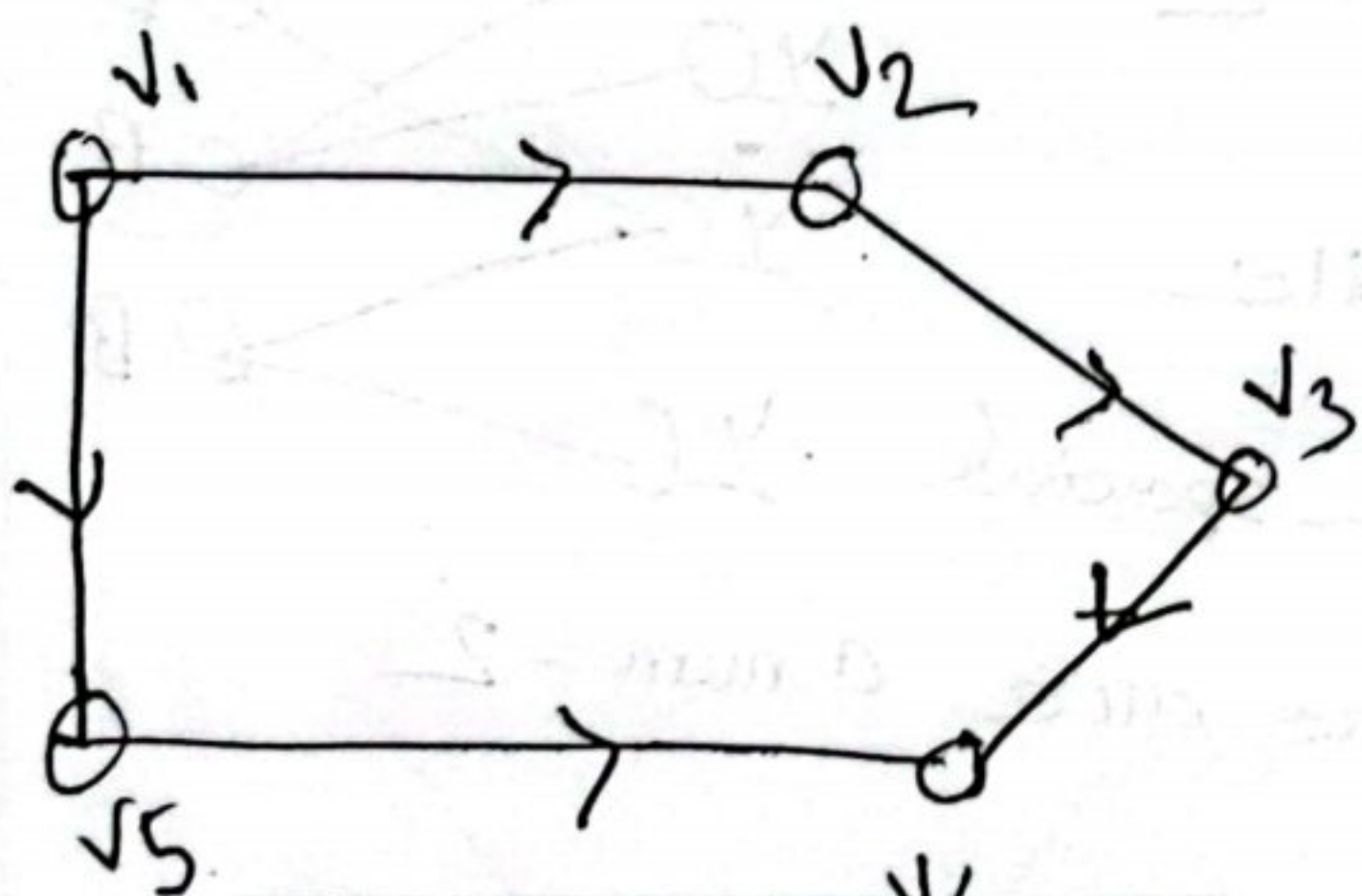
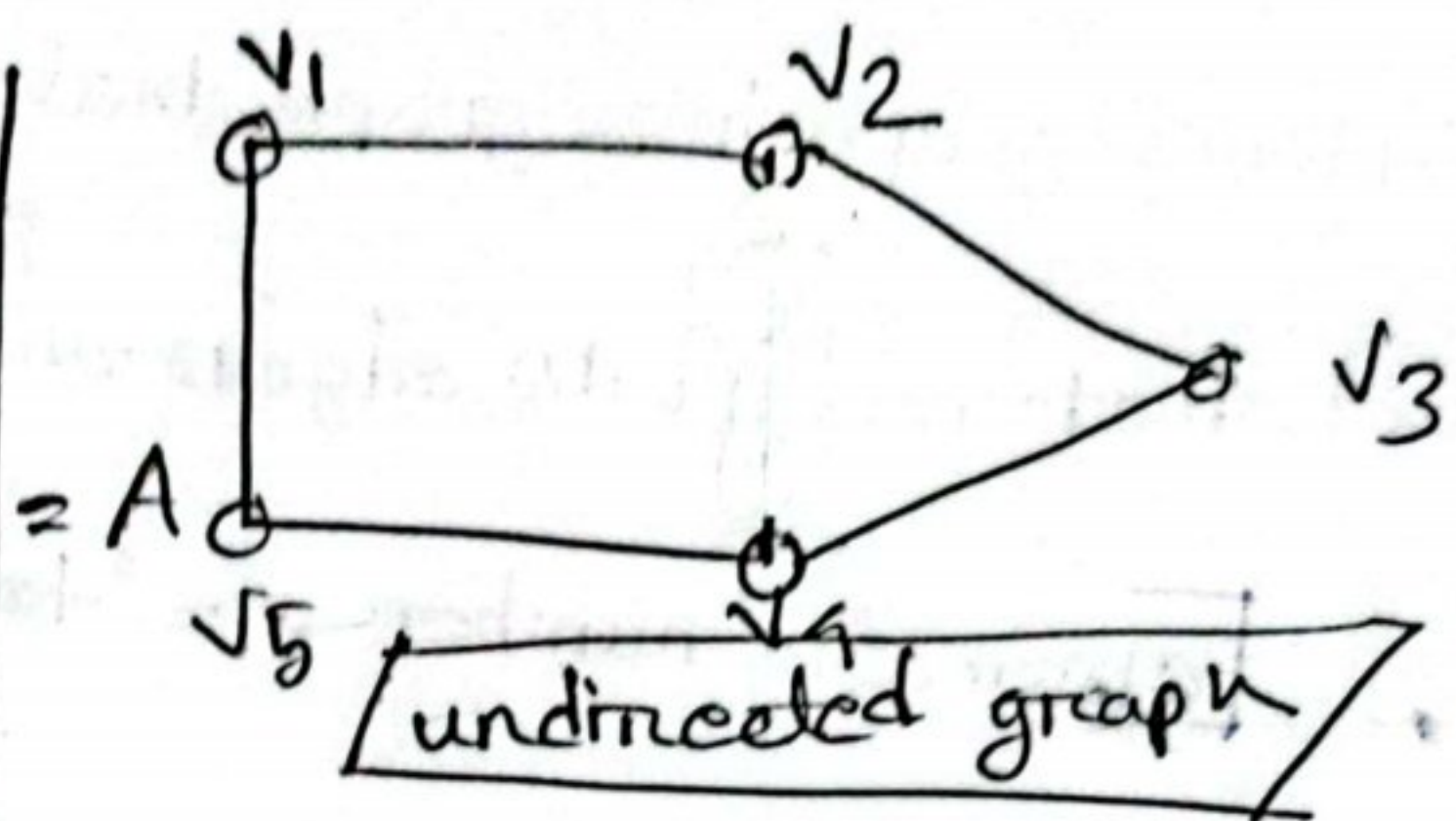
} matrix repre...

Adjacency

(i) vertices connected with another vertices = 1 .

v_1 v_2 v_3 v_4 v_5 not connected = 0 . self loop = 1 .

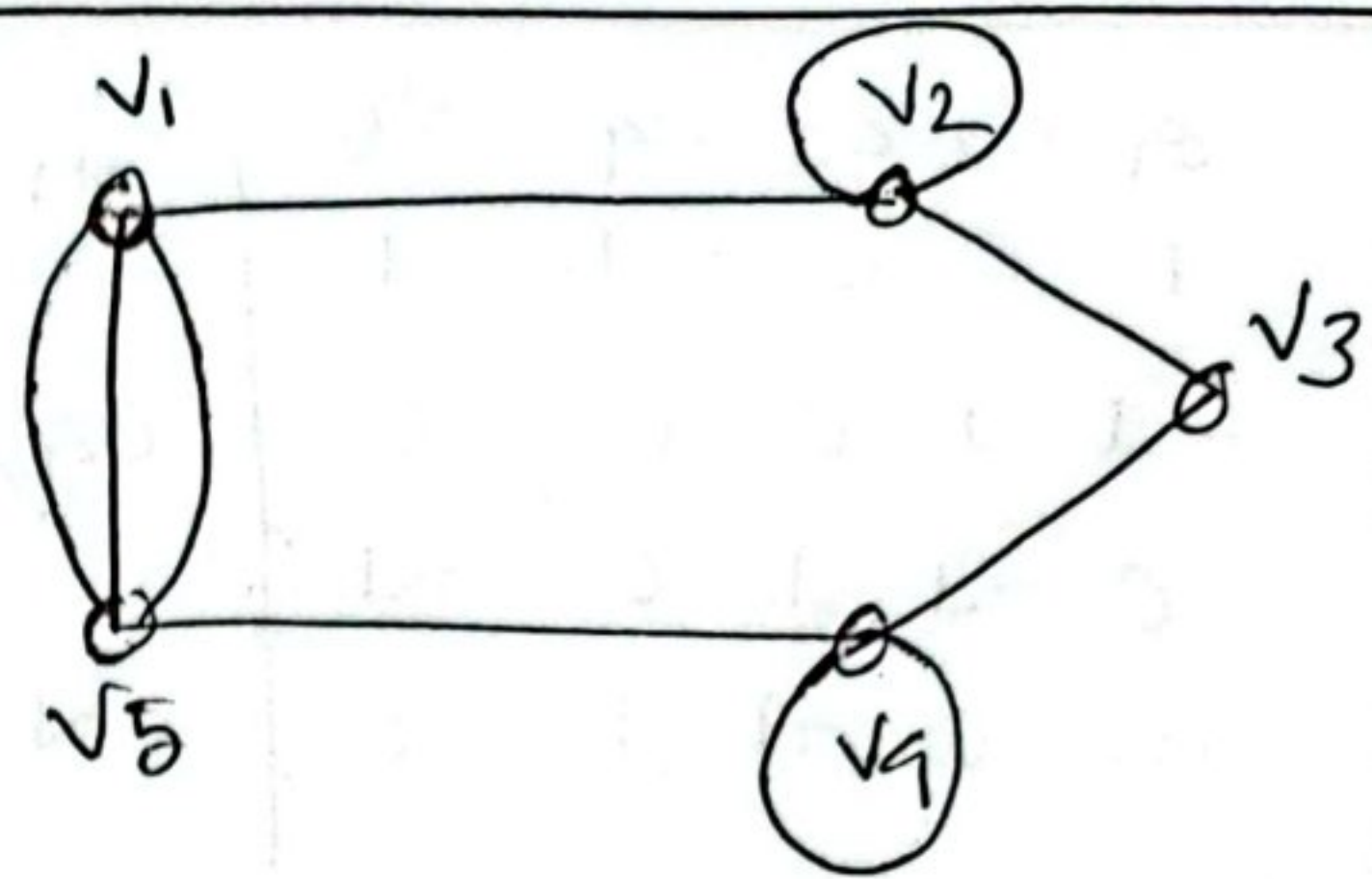
v_1	0	1	0	0	1
v_2	1	0	1	0	0
v_3	0	1	0	1	0
v_4	0	0	1	0	1
v_5	1	0	0	1	0



directed graph

	v_1	v_2	v_3	v_4	v_5
v_1	0	1	0	0	1
v_2	0	0	1	0	0
v_3	0	0	0	1	0
v_4	0	0	0	0	0
v_5	0	0	0	1	0

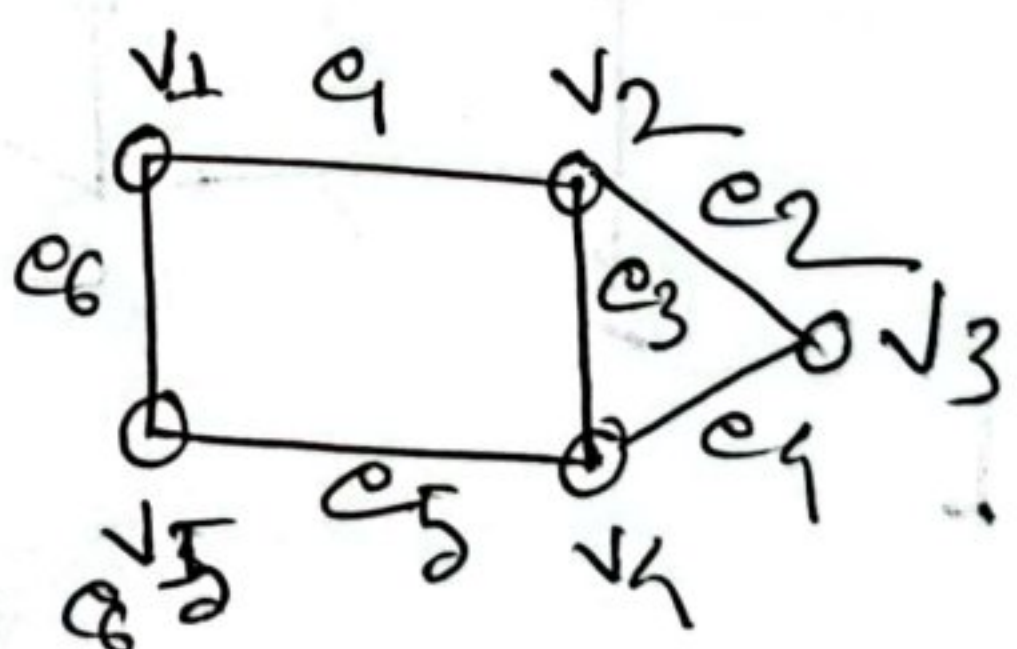
$[\begin{matrix} v_{in} = 0 \\ v_{out} = 1 \end{matrix}]$

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 3 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$


(2) Incidence matrix

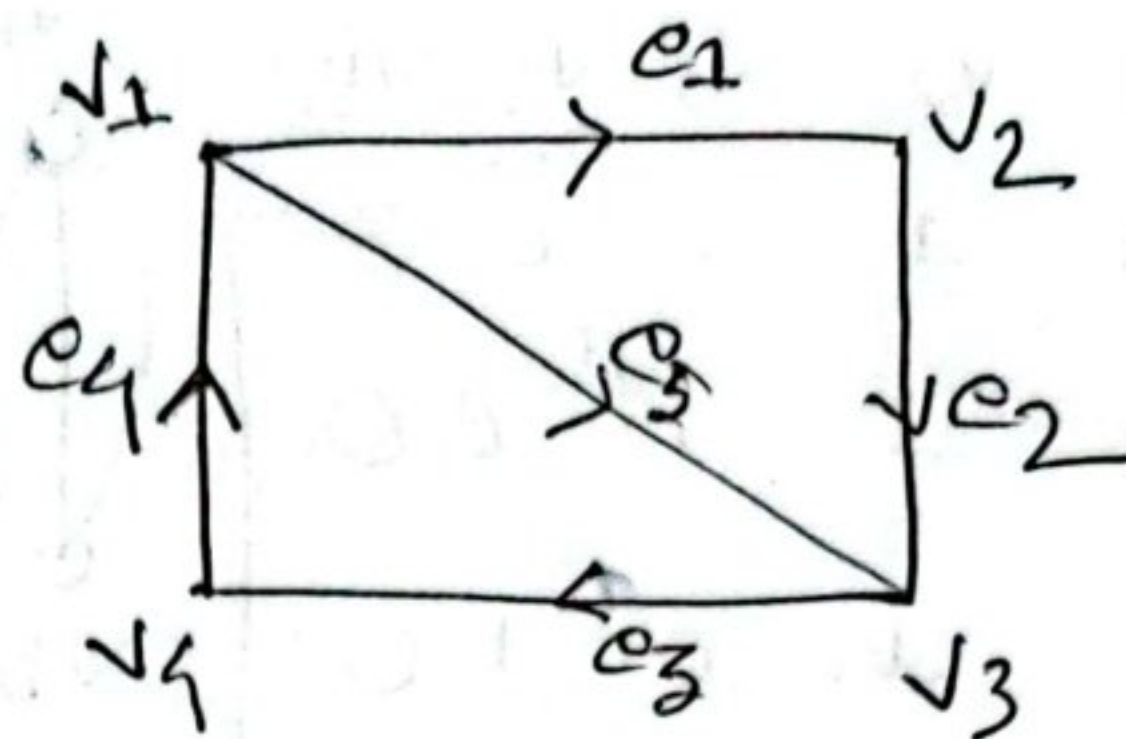
edges connected to vertices = 1
not connected = 0

* self loop = 2



	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	0	0	0	0	1
v_2	1	1	1	0	0	0
v_3	0	1	0	1	0	0
v_4	0	0	1	1	1	0
v_5	0	0	0	0	1	1

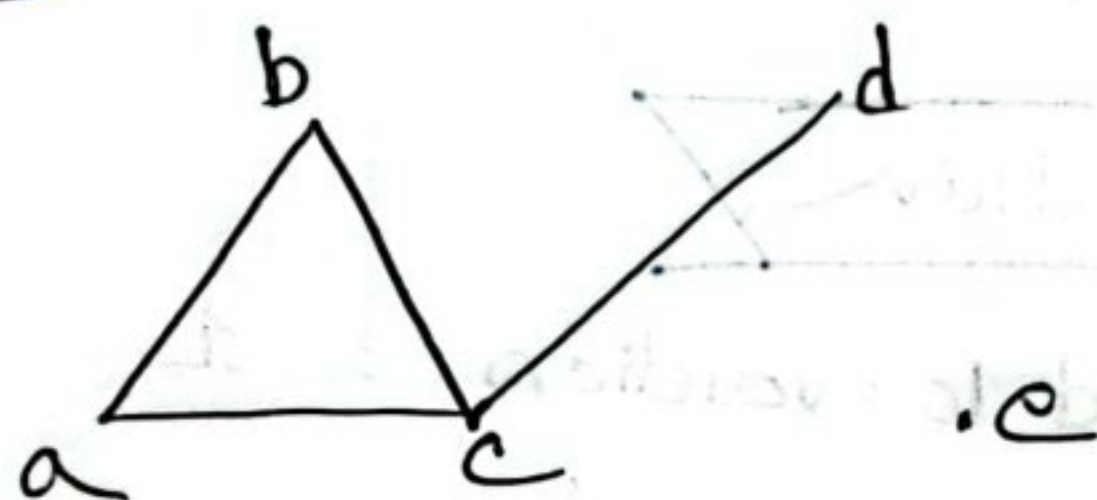
	e_1	e_2	e_3	e_4	e_5
v_1	1	0	0	-1	1
v_2	-1	1	0	0	0
v_3	0	-1	1	0	-1
v_4	0	0	-1	1	0



$v_{in} = (-1)$
 $v_{out} = 1$

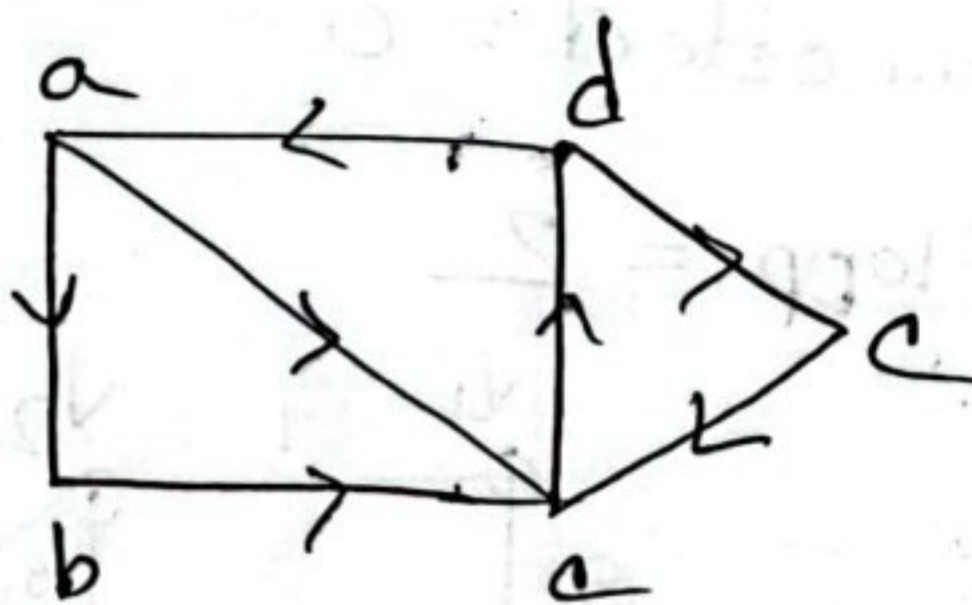
3. Adjacency list

undirected graph



Vertex	Adjacency list
a	b, c
b	a, c
c	a, b, d
d	c
e	c

Directed graph



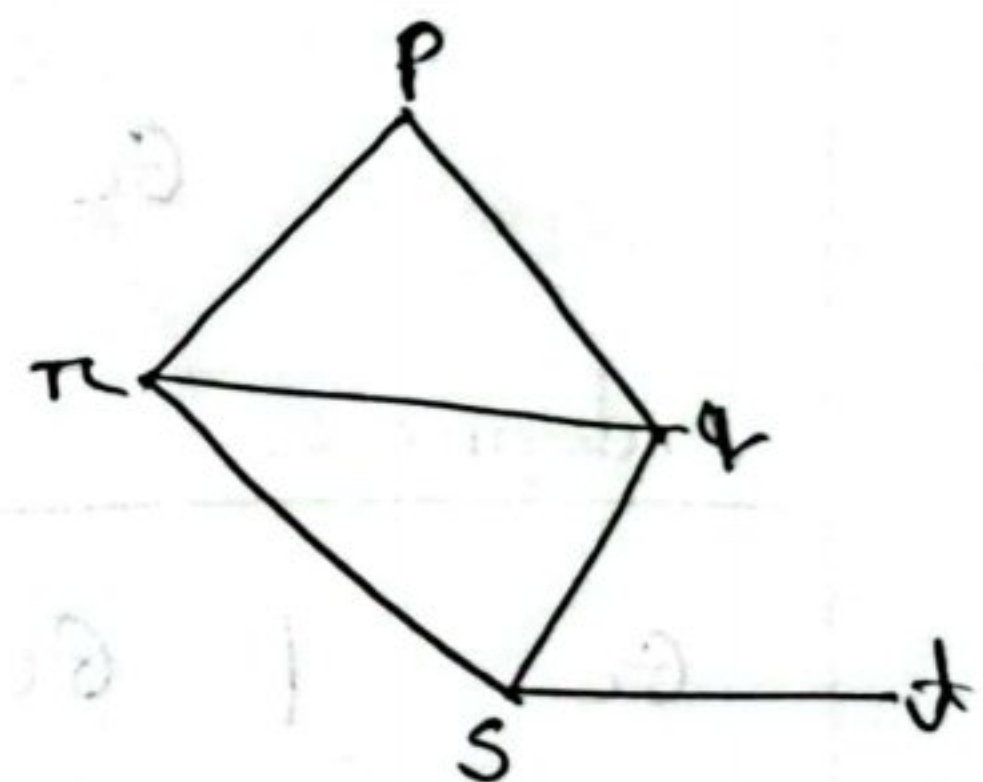
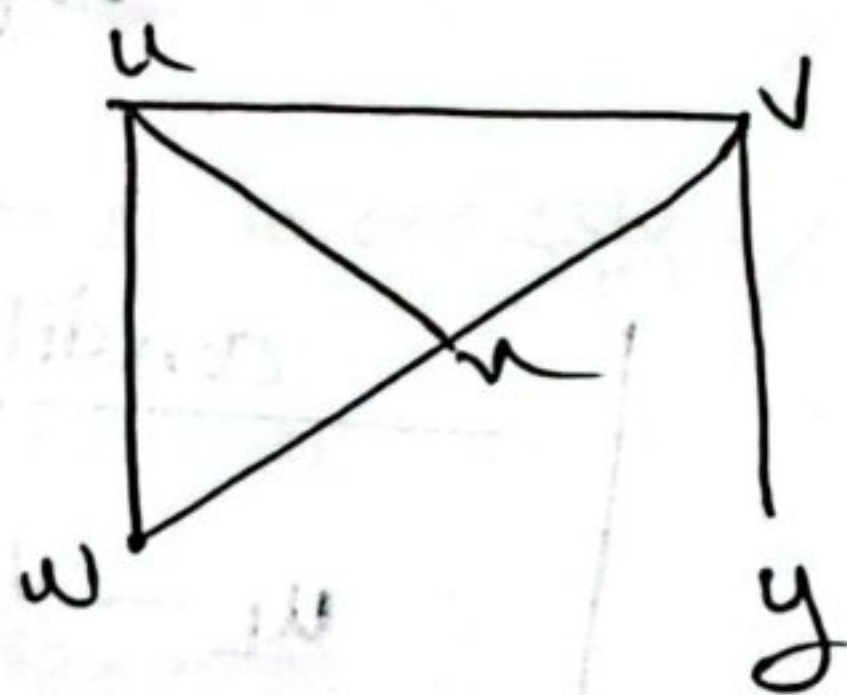
Vertex	Adjacency list
a	b, c
b	c
c	d
d	a, e
e	c

Isomorphism of graph

- (i) Vertices equal
- (ii) Edges equal
- (iii) An equal number of vertices with given degree
- (iv) one-to-one corresponding degree

$u = 3$
 $v = 3$
 $w = 2$
 $x = 3$
 $y = 1$

$p = 2$
 $q = 3$
 $r = 3$
 $s = 3$
 $t = 1$



i, ii, iii satisfied

$y - t$
 $w - p$
 $v - s$
 $u - q$
 $x - r$

edge theory

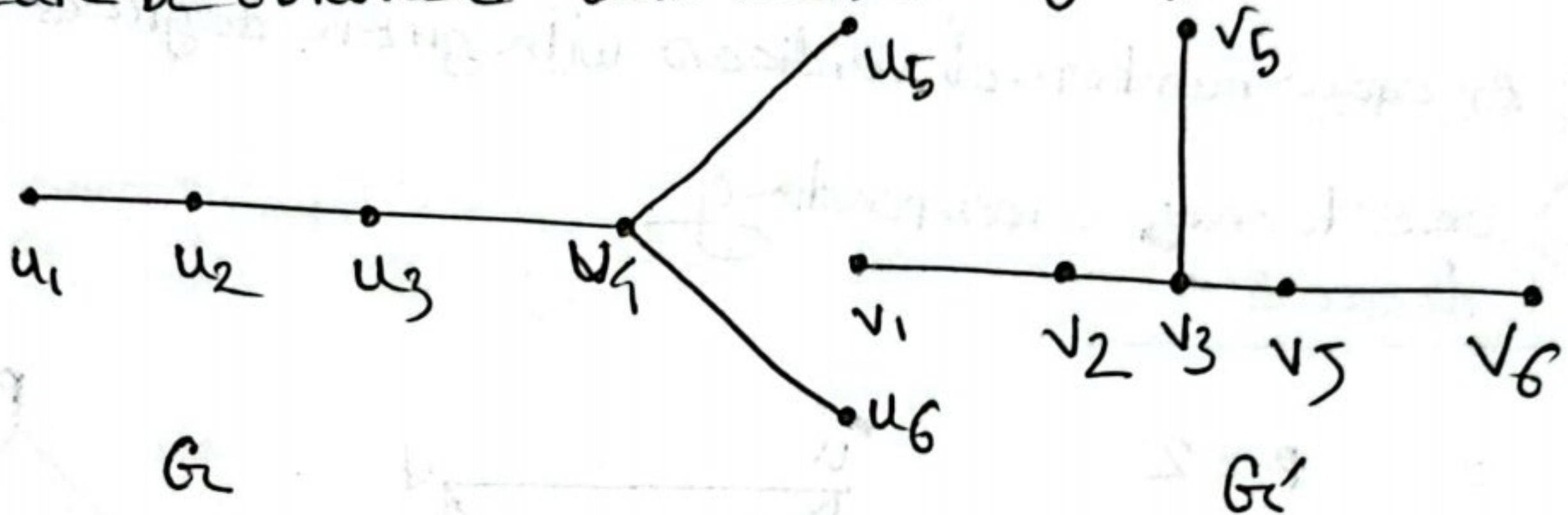
$y \leftrightarrow w$ no edge, $t \leftrightarrow p$ no edge
 $w \leftrightarrow v$ no "", $p \leftrightarrow s$ no edge
 $v \leftrightarrow u$ edge, $s \leftrightarrow q$ edge.
 $u \leftrightarrow x$ edge, $q \leftrightarrow r$ edge.

Both are isomorphic

Homeomorphic graph

The graph G and G' are said to be homeomorphic if they can be obtained from the same graph

Ex 1



degree

condition - 1 (check)

G	G'
$u_1 = 1$	$v_1 = 1$
$u_2 = 2$	$v_2 = 2$
$u_3 = 2$	$v_3 = 3$
$u_4 = 3$	$v_4 = 1$
$u_5 = 1$	$v_5 = 2$
$u_6 = 1$	$v_6 = 1$

$$u_1 - v_1$$

$$u_2 - v_2 \rightarrow v_3 (3)$$

$$\xrightarrow{\text{(connected)}} (u_3 (2))$$

(not one-to-one correspondence)

$$u_1 \leftrightarrow v_6$$

$$u_2 \leftrightarrow v_5 \rightarrow v_3 (3)$$

$$\xrightarrow{\text{(connected)}} (u_3 (2))$$

(not satisfied)

So, it's not isomorphic

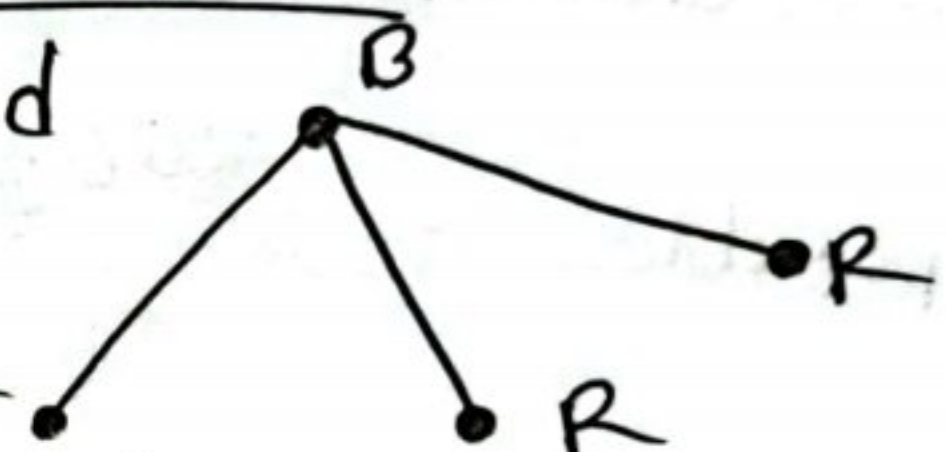
Chap- Tree

Trees: A Tree is a special type of connected graph in which there are no circuits. Every tree is a bipartite graph.
chromatic number of tree = 2

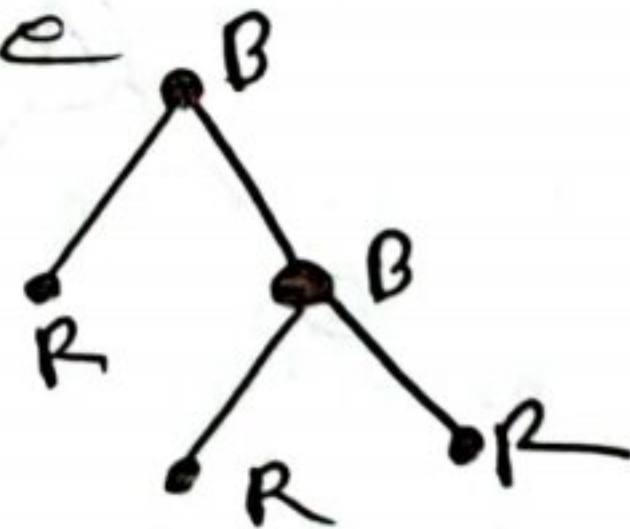
Binary tree: A binary tree is defined

as a tree in which there is exactly one vertex of degree two, and each of the remaining vertices is of degree one or three and vertex of degree two is set as a root.

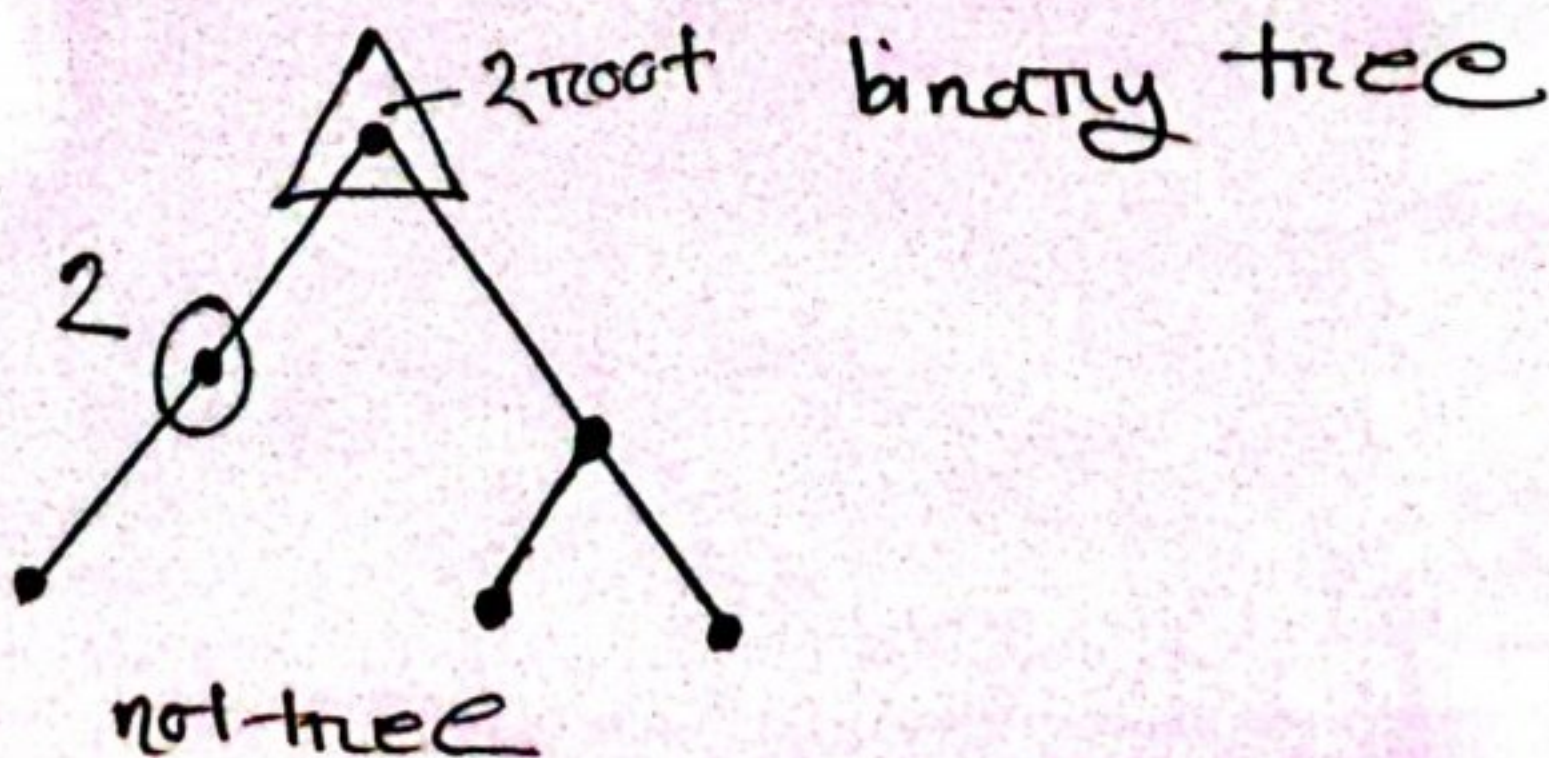
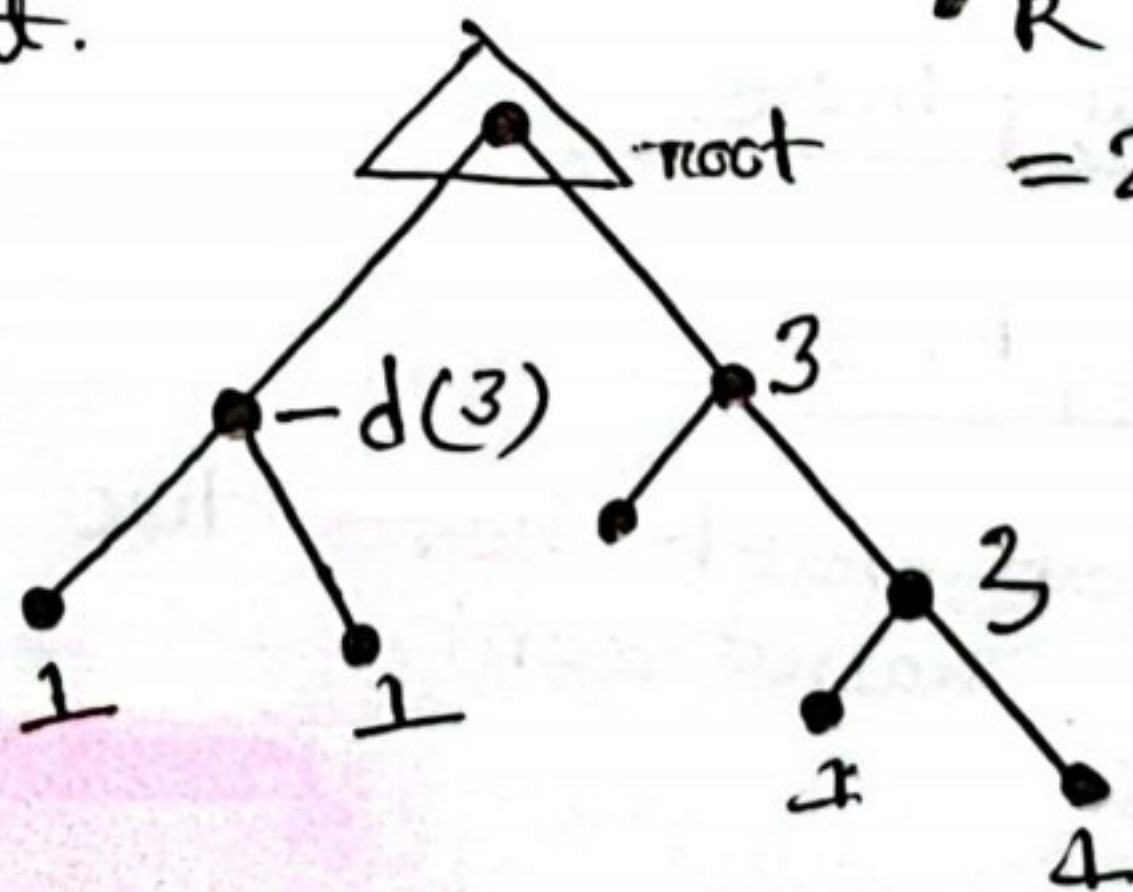
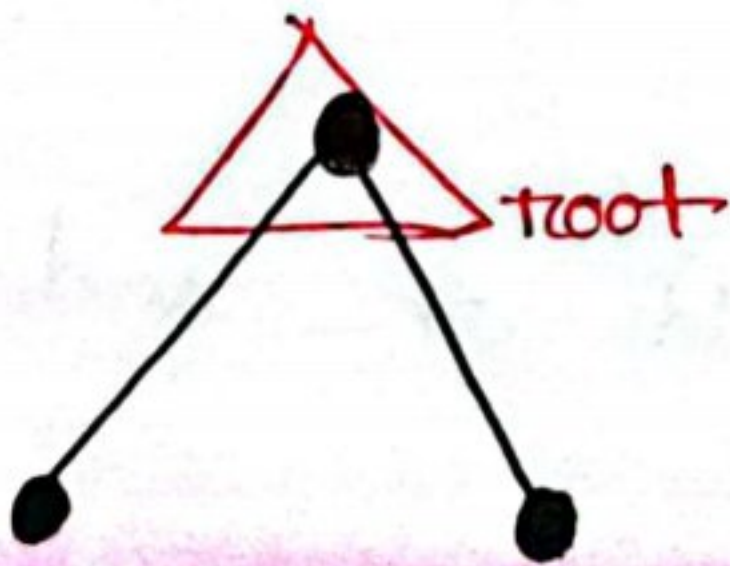
Example



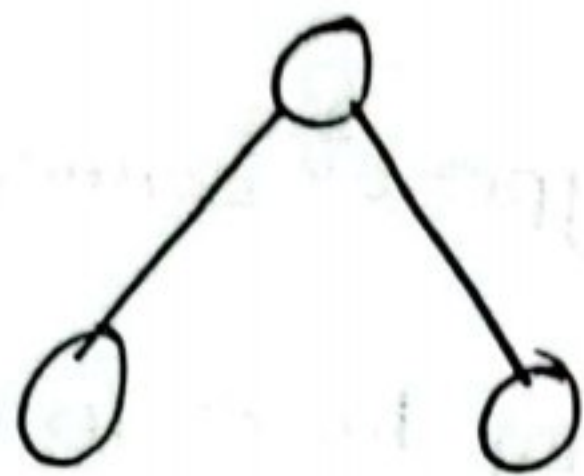
= 2



= 2



Unlabelled Binary tree if nodes are not assigned any level



Number of different binary trees = $\frac{2nC_n}{n+1}$

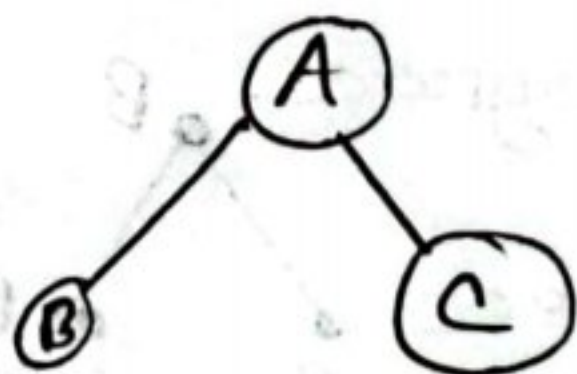
n unlabeled nodes

Ex:-1

n=3, unlabelled nodes, possible different binary trees

$$\text{possible} = \frac{2 \times 3C_3}{4} = 5$$

labelled tree



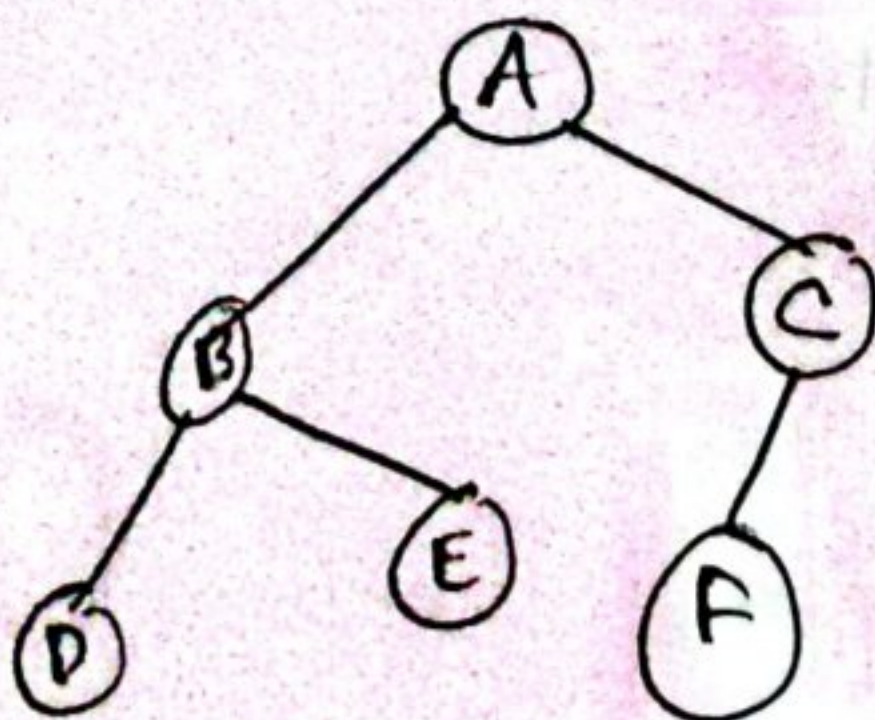
Number of diff. bi. tree = $\frac{2nC_n}{n+1} \times n!$

$$\text{if } n=3, = \frac{6C_3}{4} \times 3! = 30$$

Types of binary tree

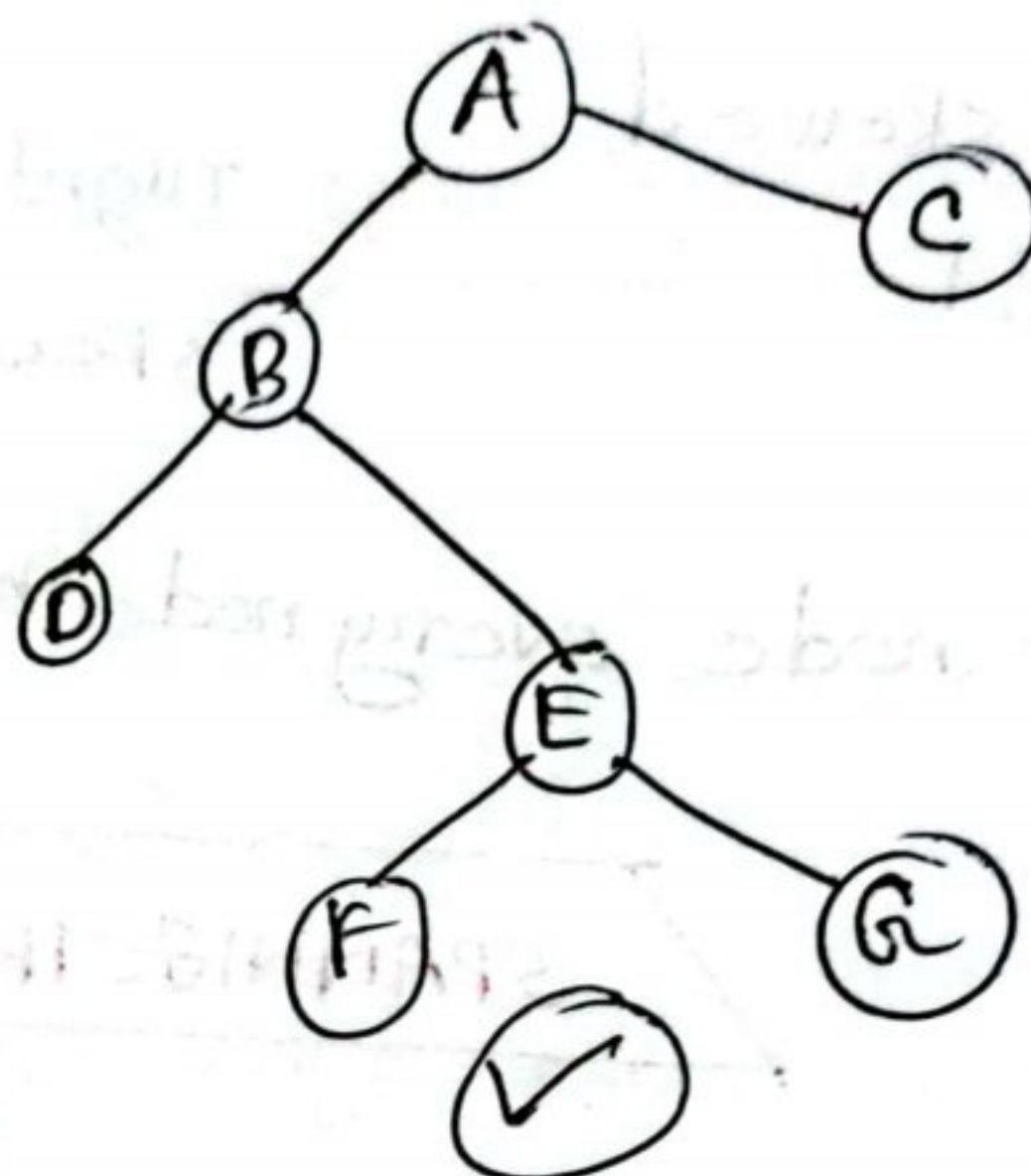
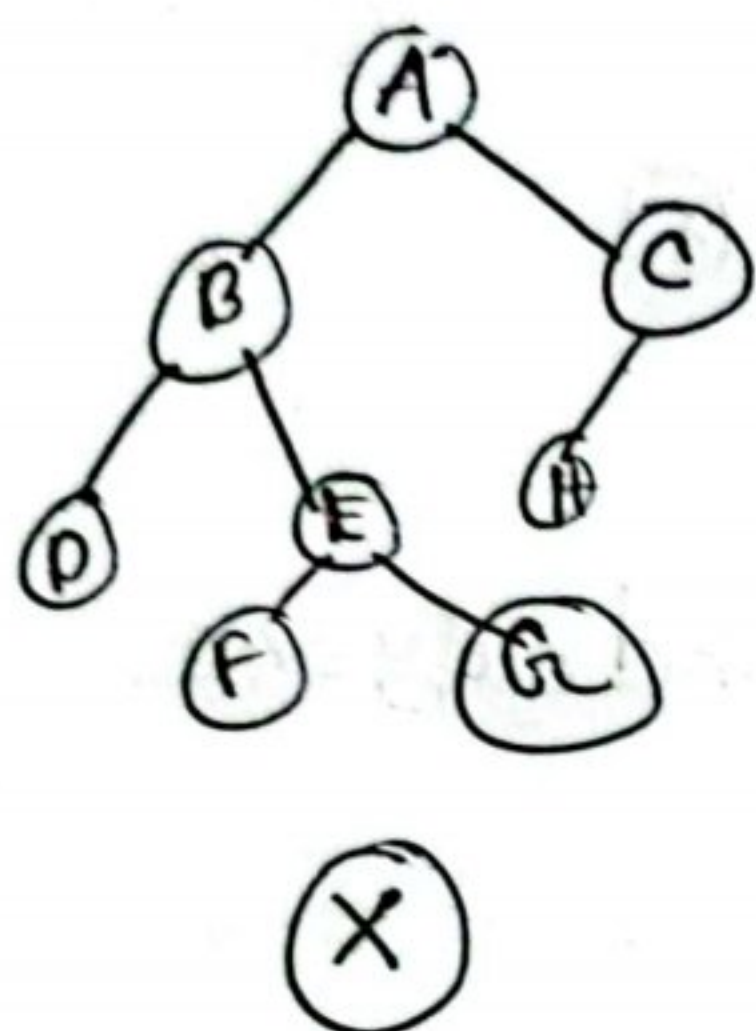
Rooted binary tree

Every node must have two children, and has a root node

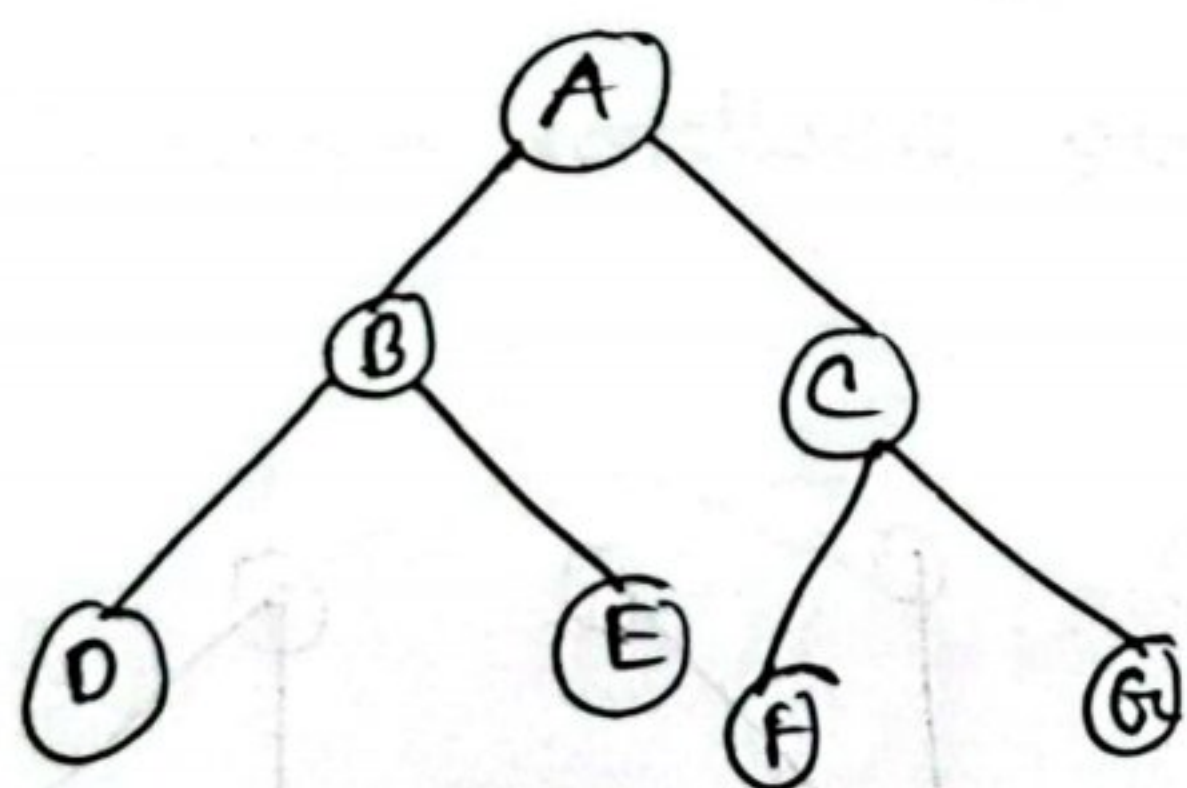


2) full/strictly binary tree

Every node has either 0/2 children

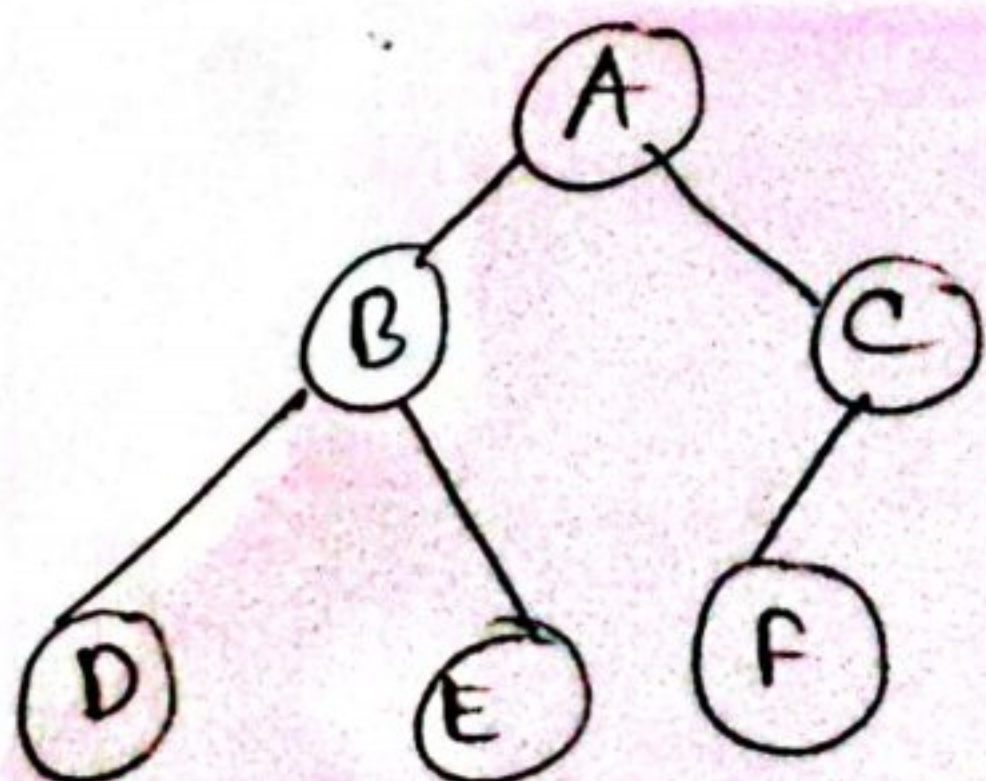


(3) Complete/perfect B.T



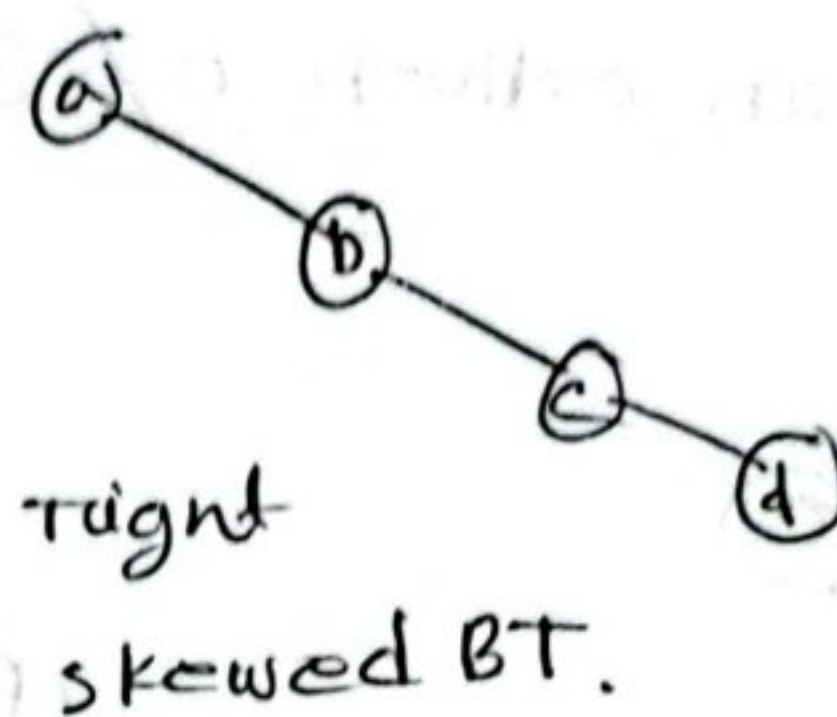
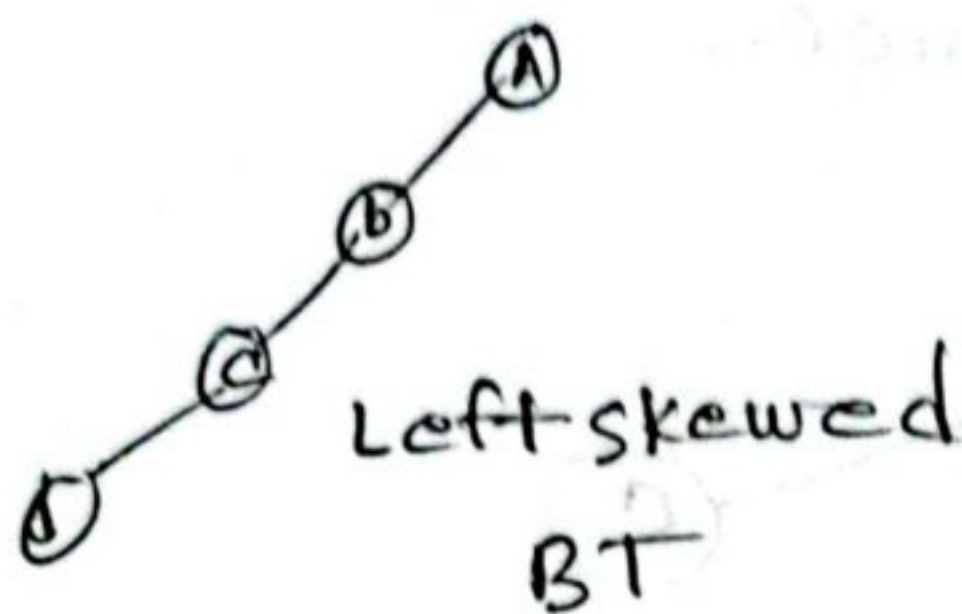
In which every internal node has exactly two children and all the leaf nodes are at same level

(4) Almost complete B.T



All the levels are completely filled except possibly the last level and the last level must be strictly filled from left to right.

Skewed BT

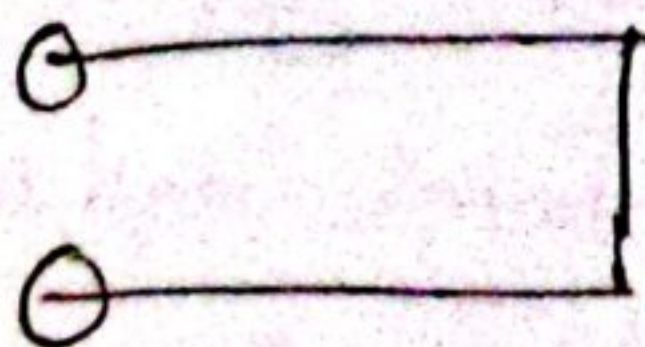
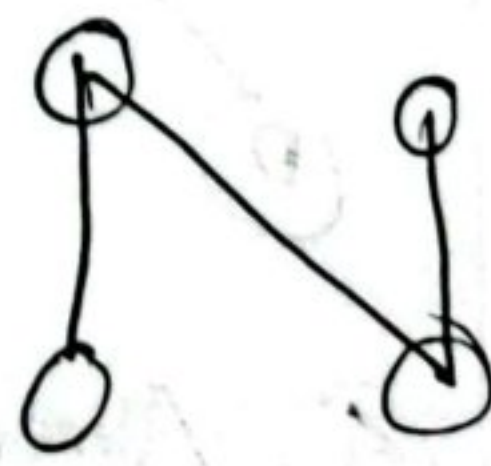
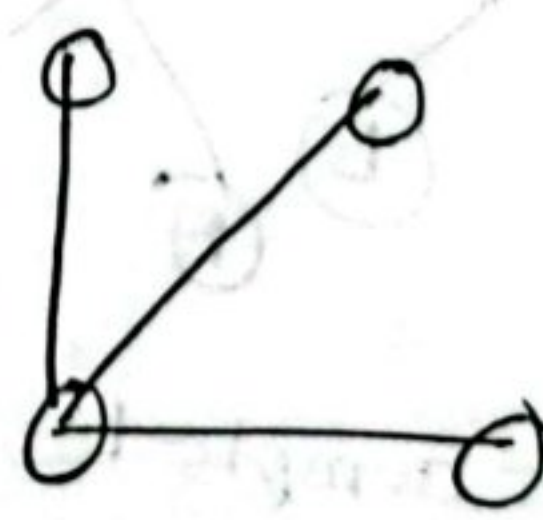
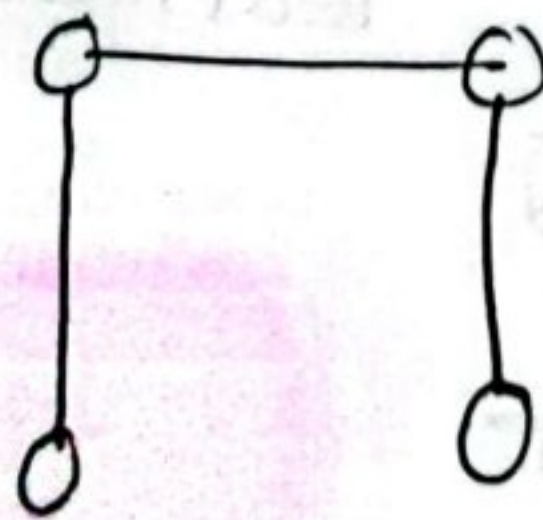
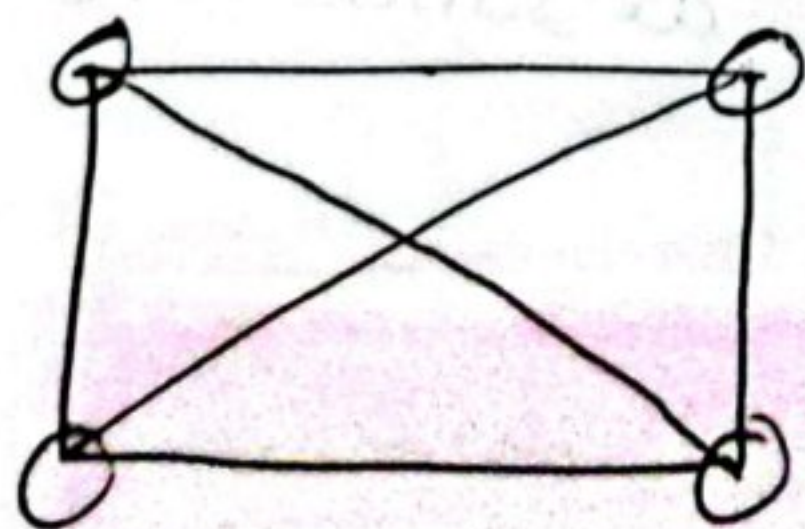


Except one node, every node has one and only one

SPANNING TREE

If G is any connected graph spanning tree in G is a subgroup

T of G , which is a tree



(A - S. tree)

Cayley's Theorem: The complete graph K_n has n^{n-2} different spanning trees.

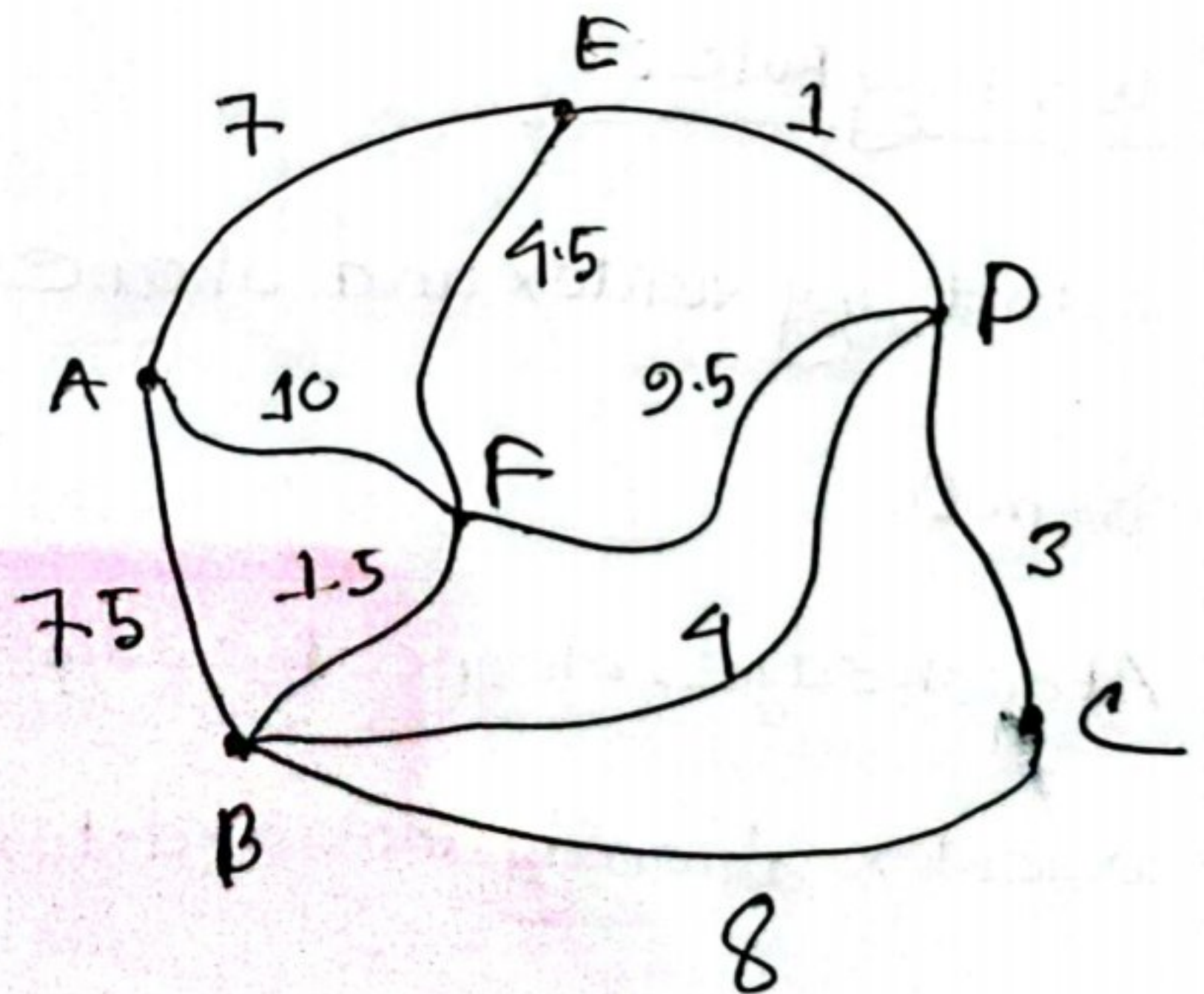
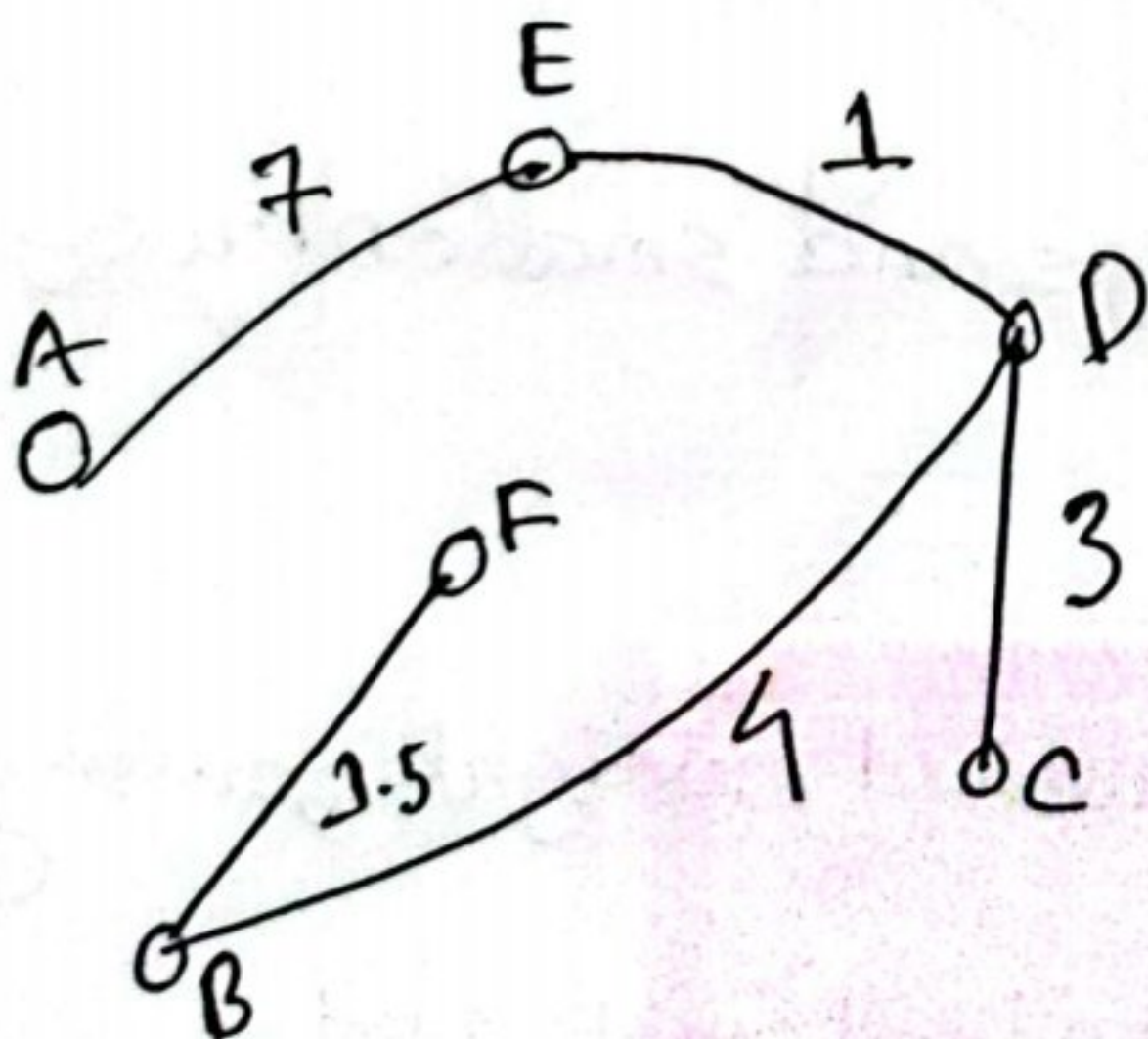
Minimal spanning tree

KRUSKAL'S ALGORITHM

Choose an edge of minimal weight

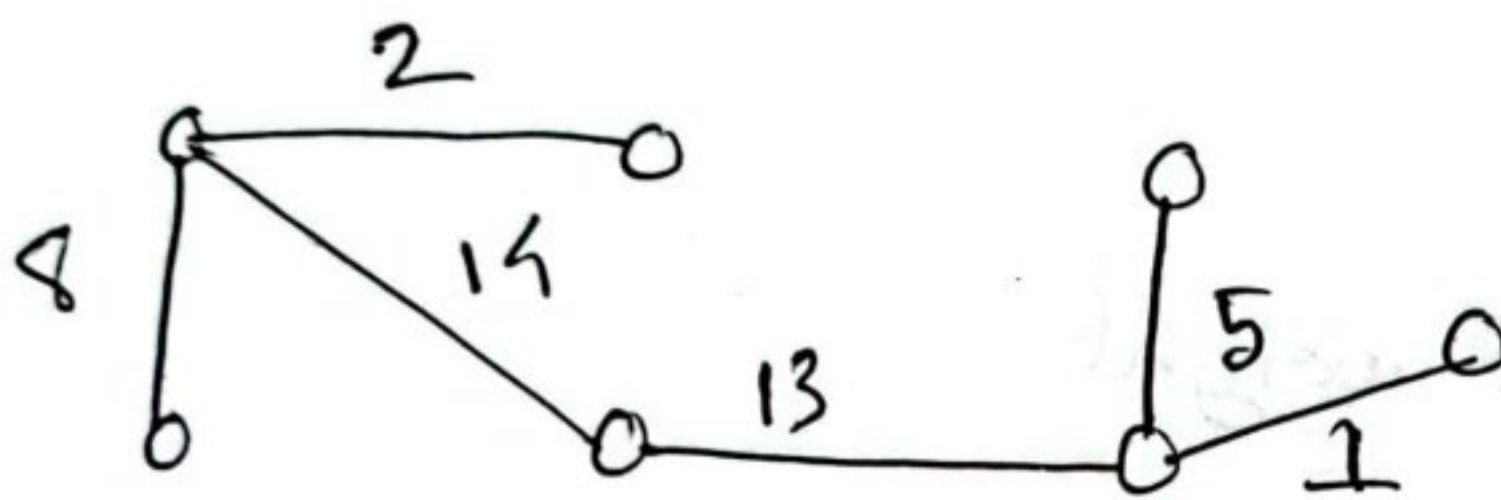
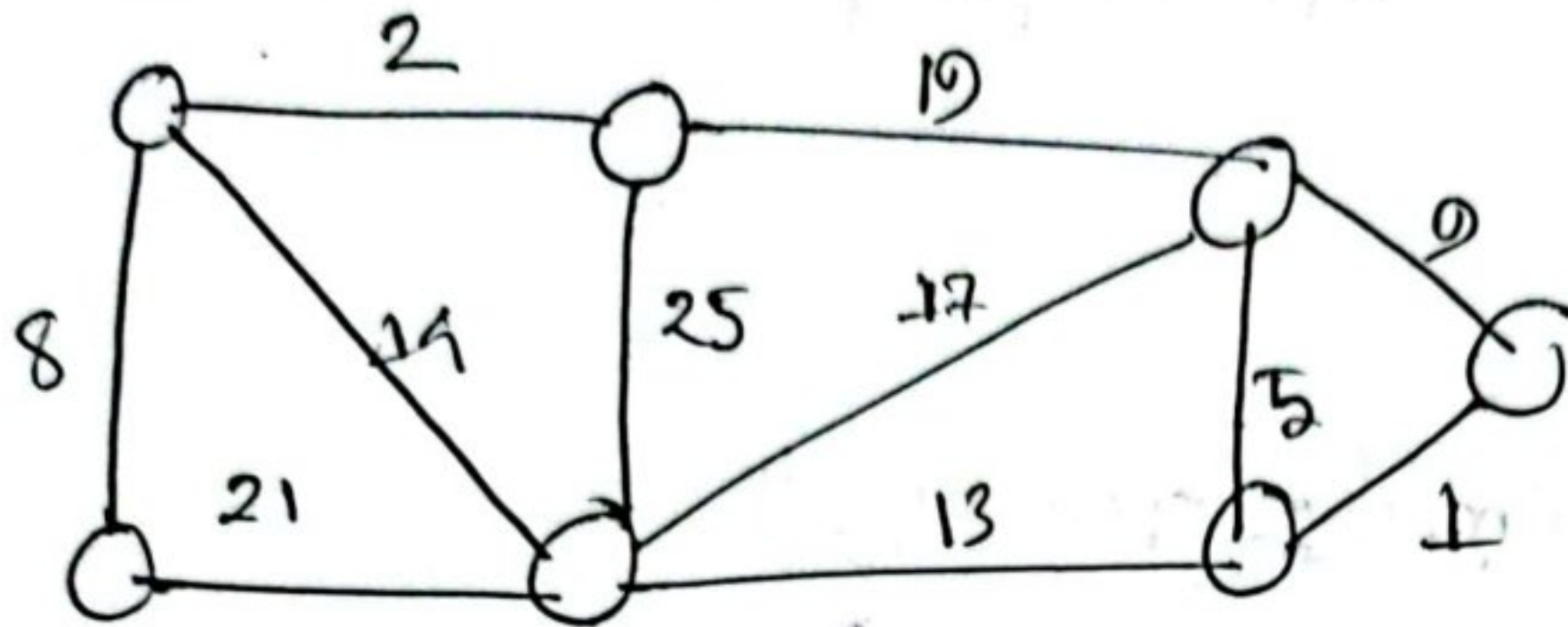
At each step, choose the edge whose inclusion will not create a circuit.

If G has n vertices, stop after $(n-1)$ edges.



Total weight of the tree is $= 1 + 1.5 + 3 + 4 + 7 = 16.5$

Ex-2



weight of the tree = $1+2+5+8+13+14 = 43$

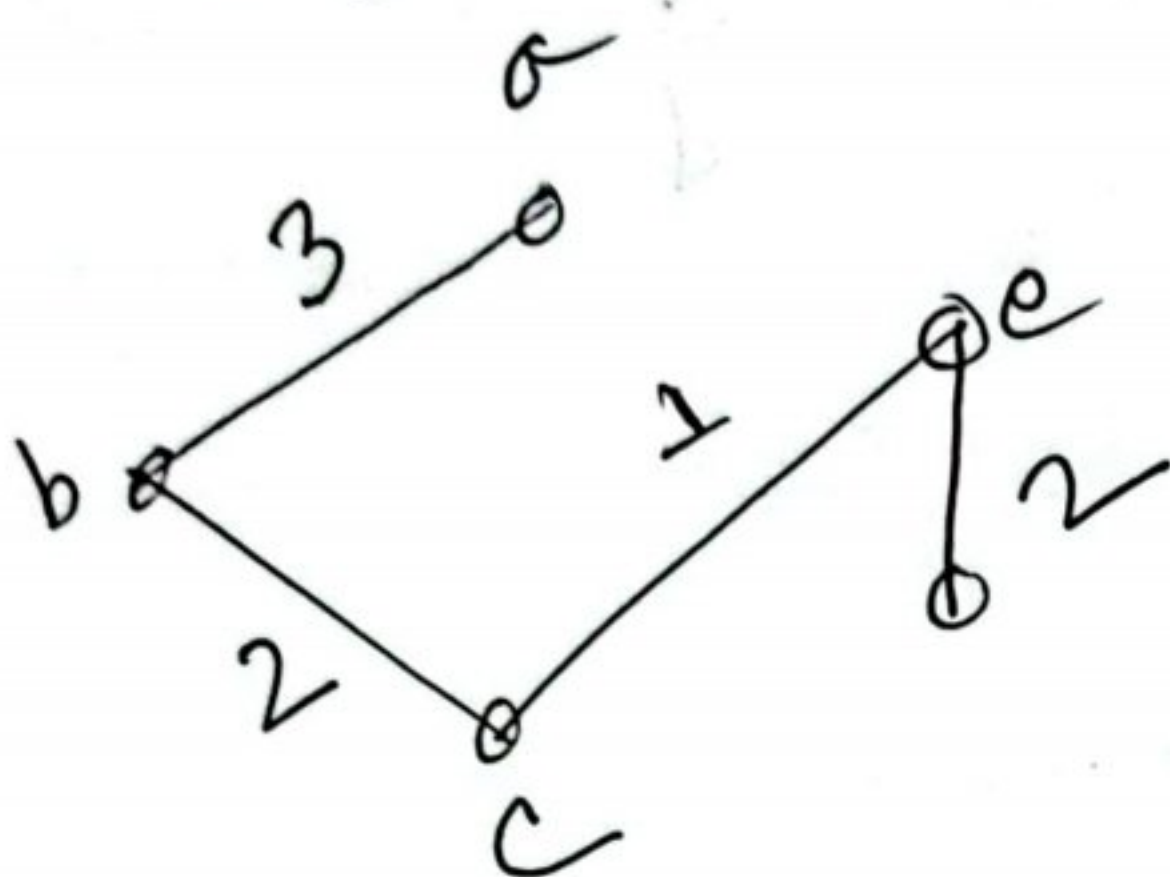
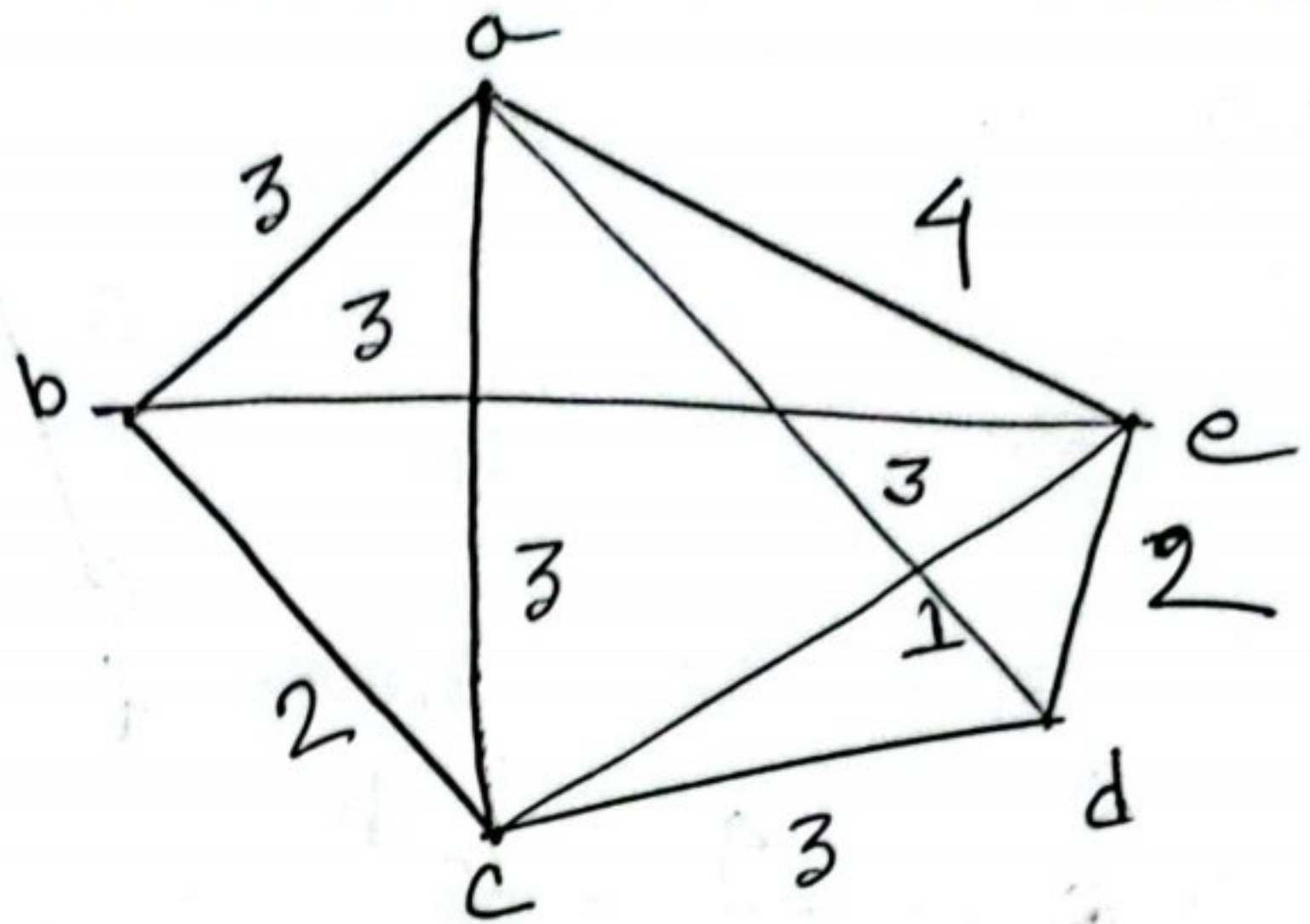
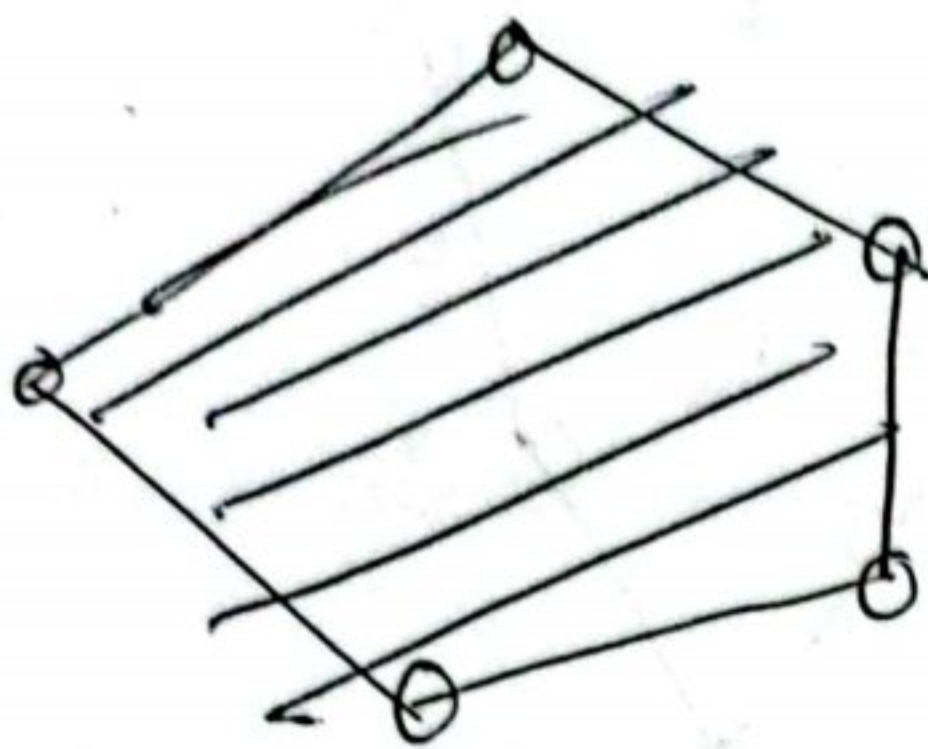
PRIM'S Algorithm

Working Rule:

Select any vertex and choose the edge and smallest weight from it

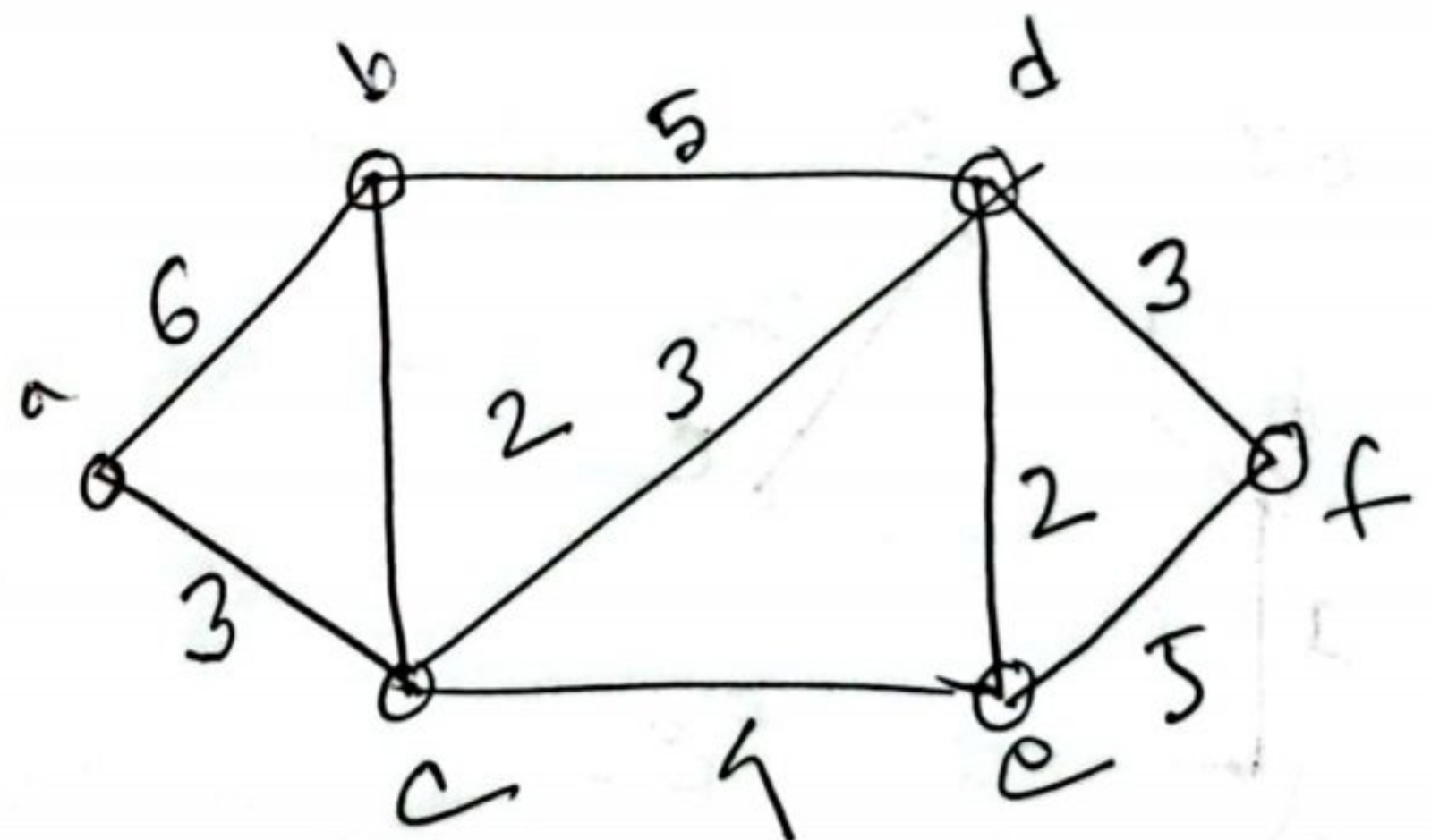
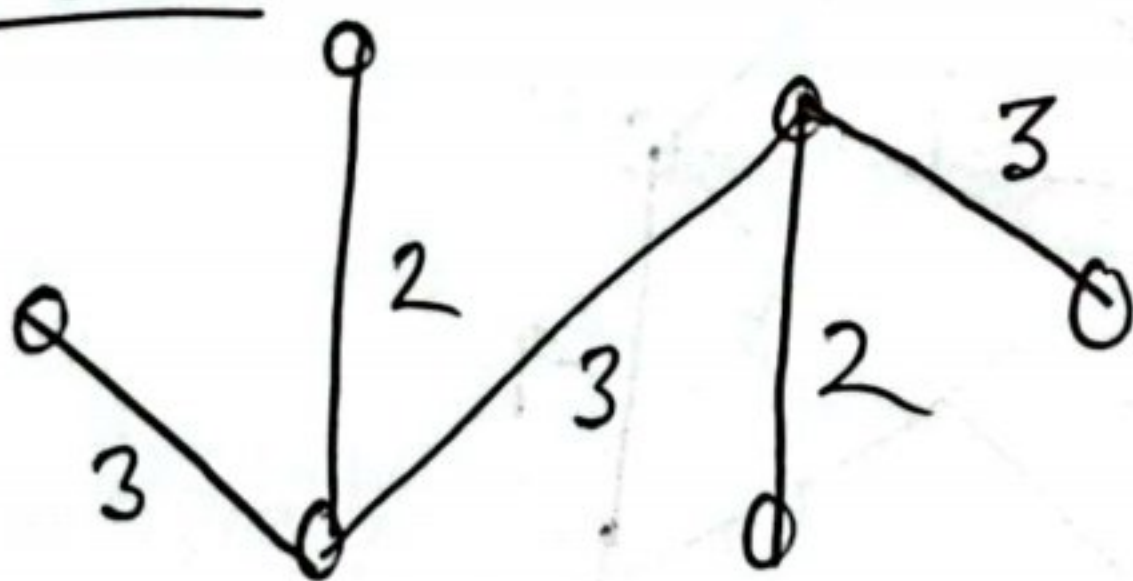
At each edge, choose the edge of smallest weight joining a vertex already included to vertex not yet included

Continue until all vertices are included



$$\text{weight} = 3 + 2 + 1 + 2 = 8$$

Q-1 Kruskal's

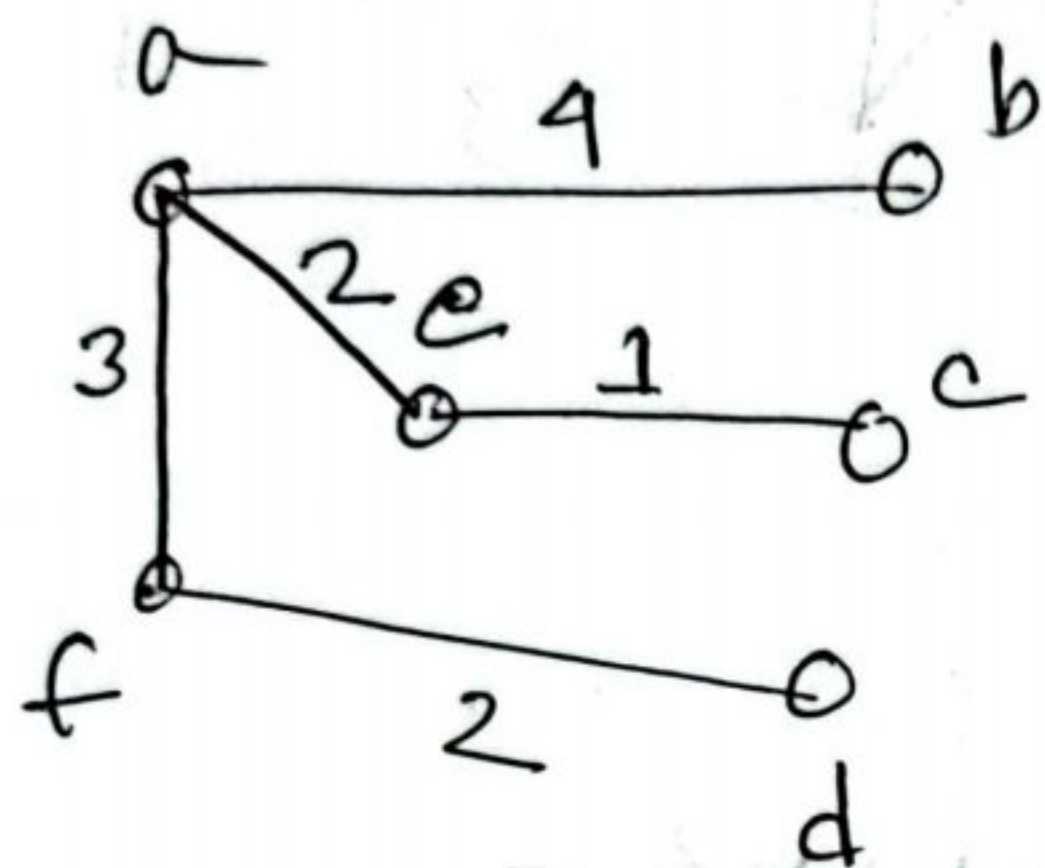


Here 6 vertex and $(6-1)$ edges

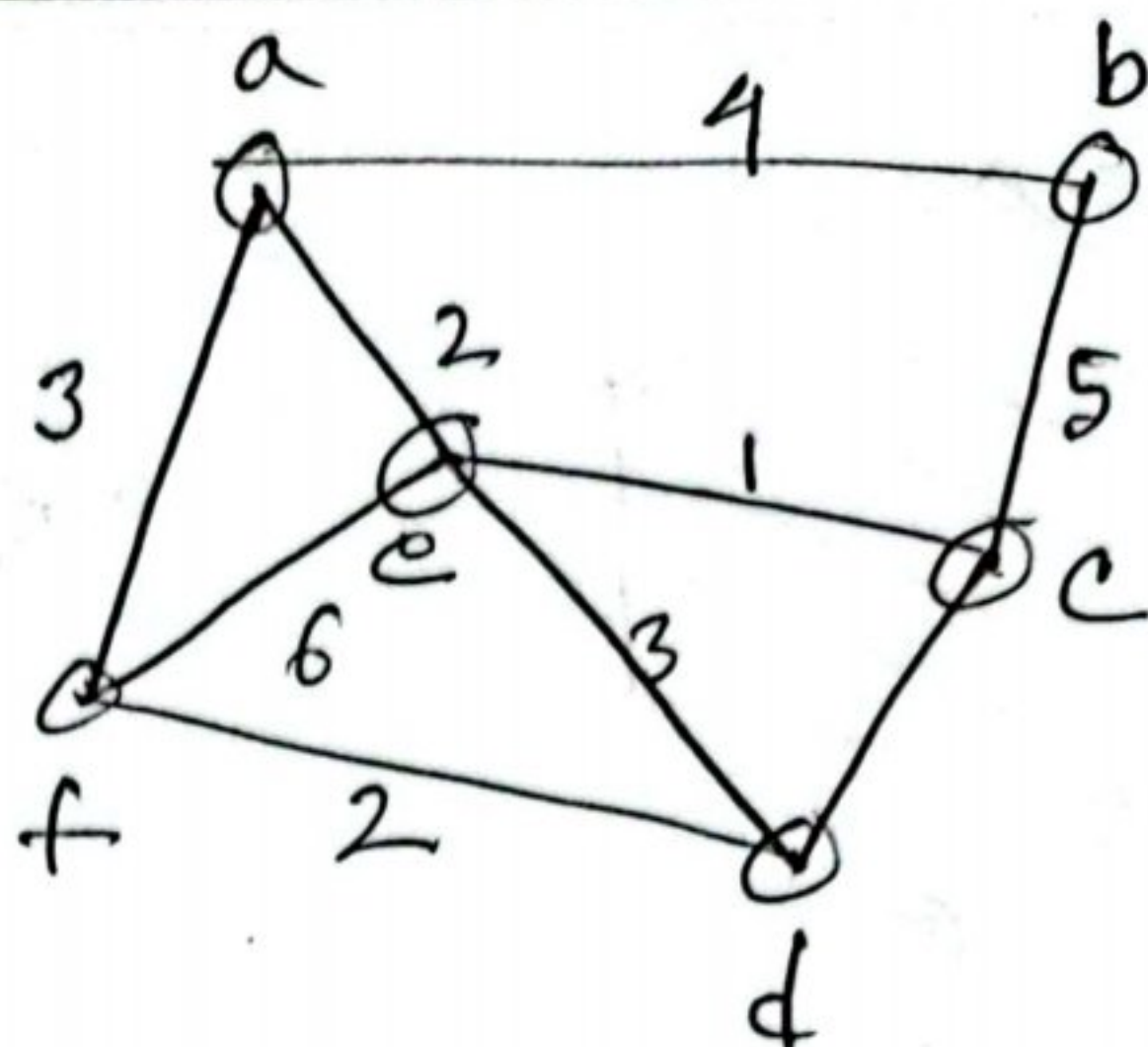
$$\text{Height of the tree} = 3 + 2 + 3 + 2 + 3$$

$$= 13$$

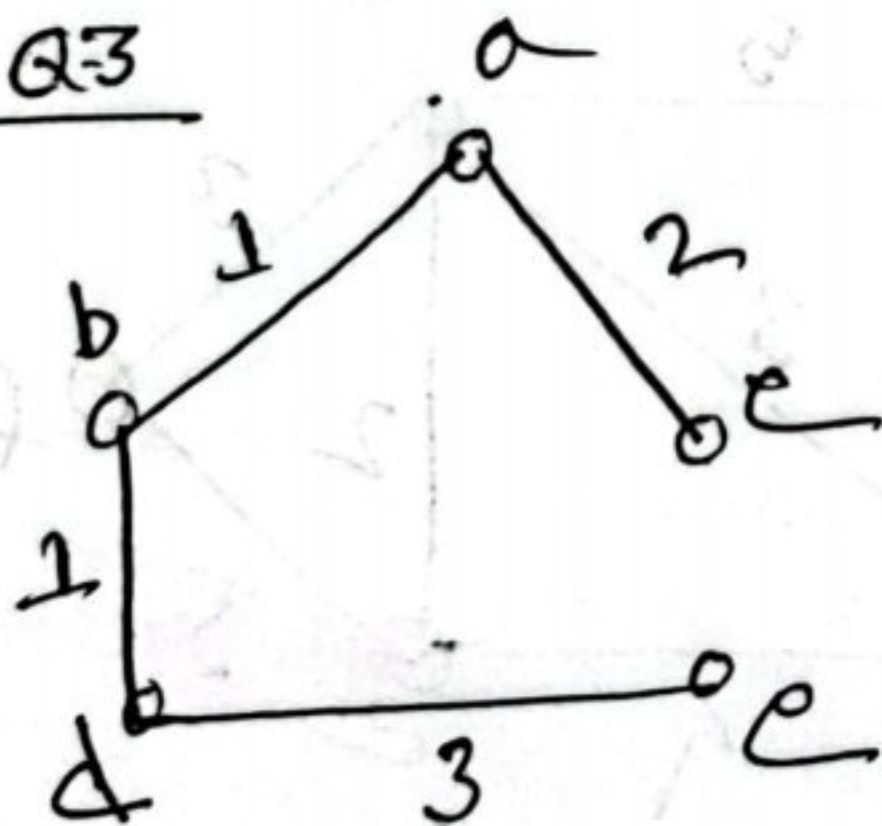
Q-2



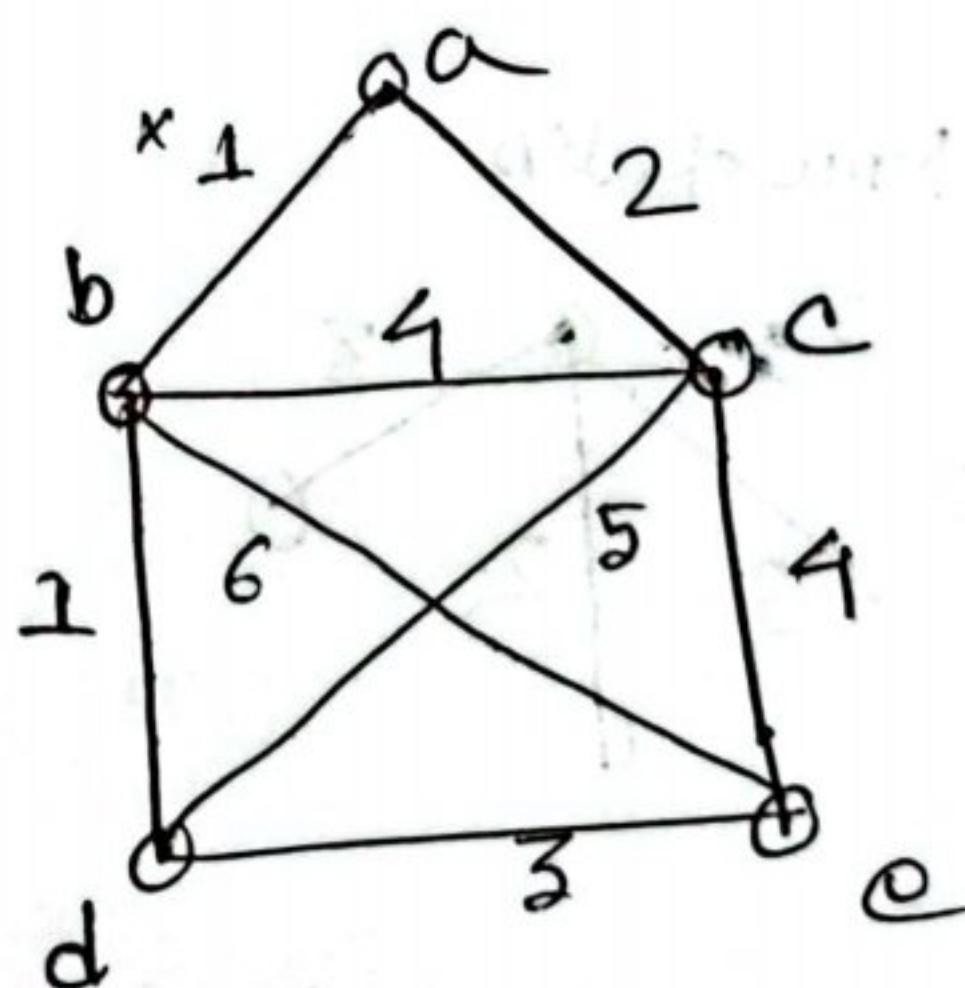
by prim's law. height = $2+3+2+1+4=12$



Q3



height = $1+1+2+3=8$



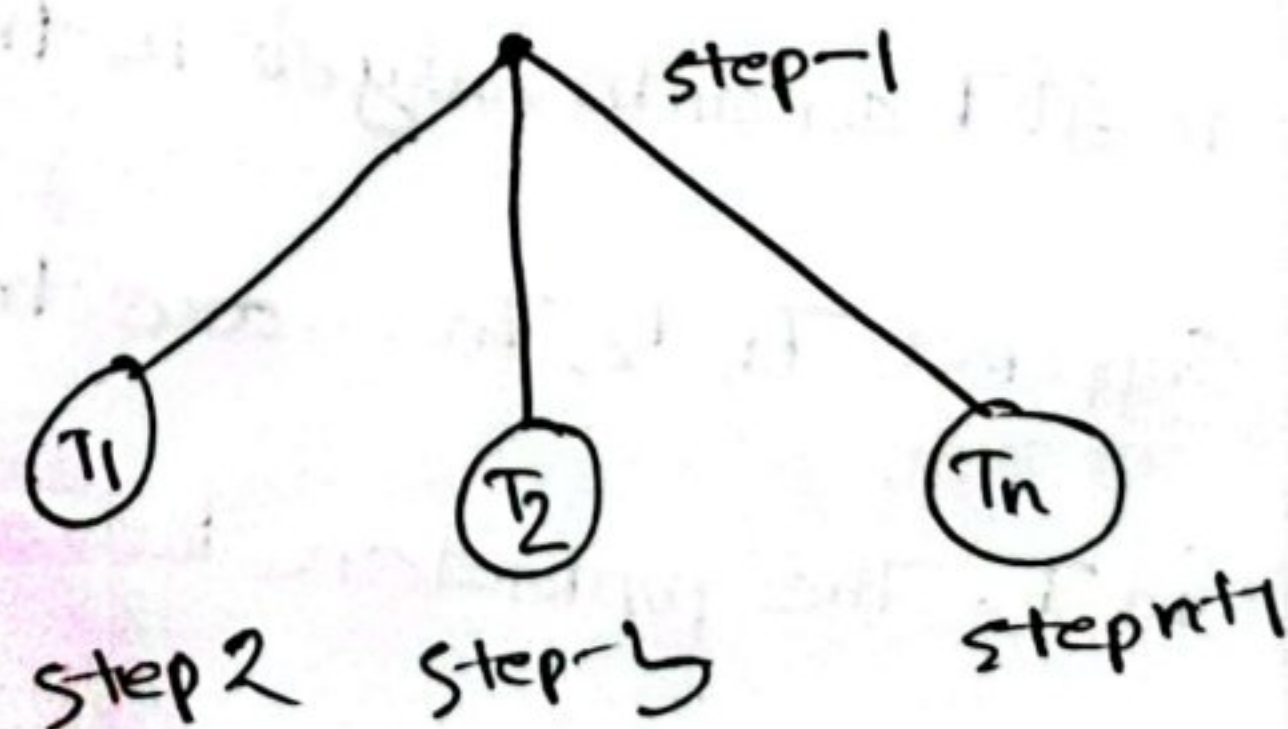
Ans

Tree traversal

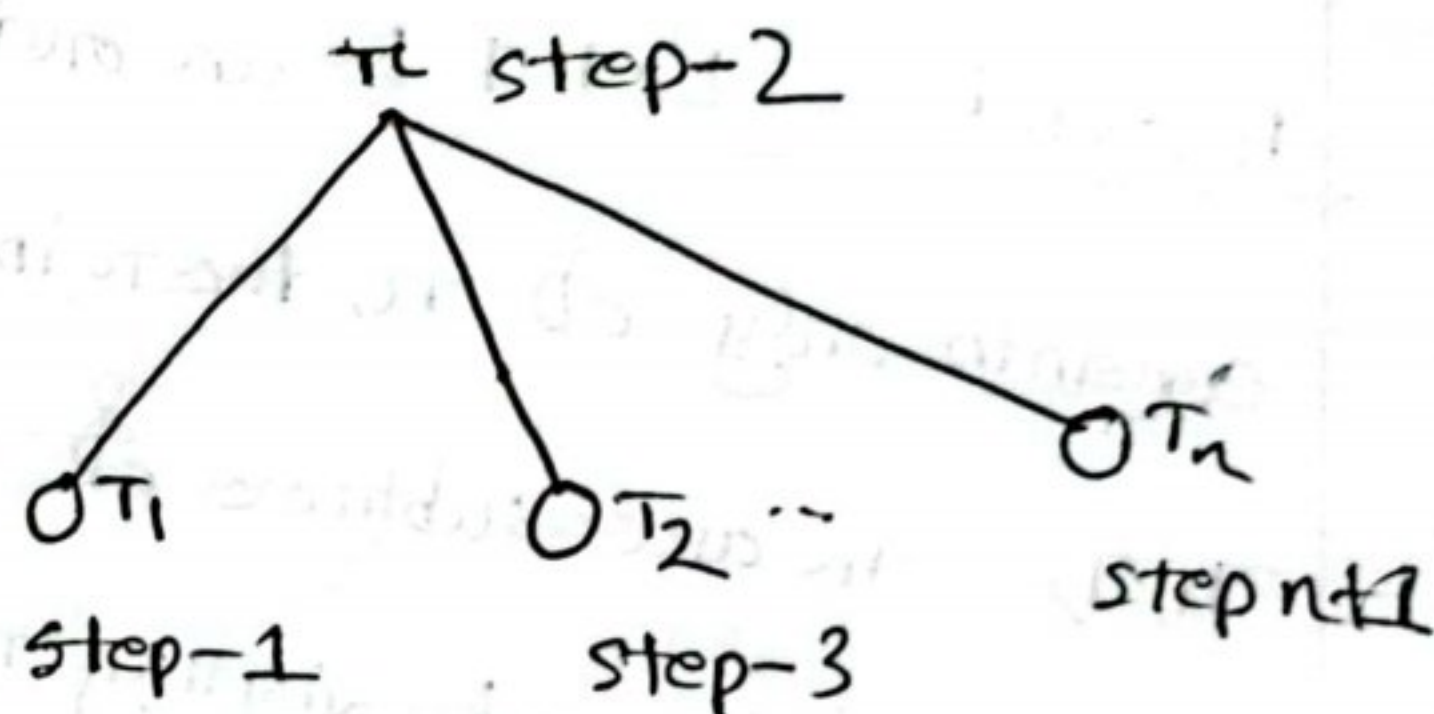
Procedures for systematically visiting every vertex of an ordered tree are called traversals.

The three most used traversals are preorder traversal, inorder traversal and postorder traversal.

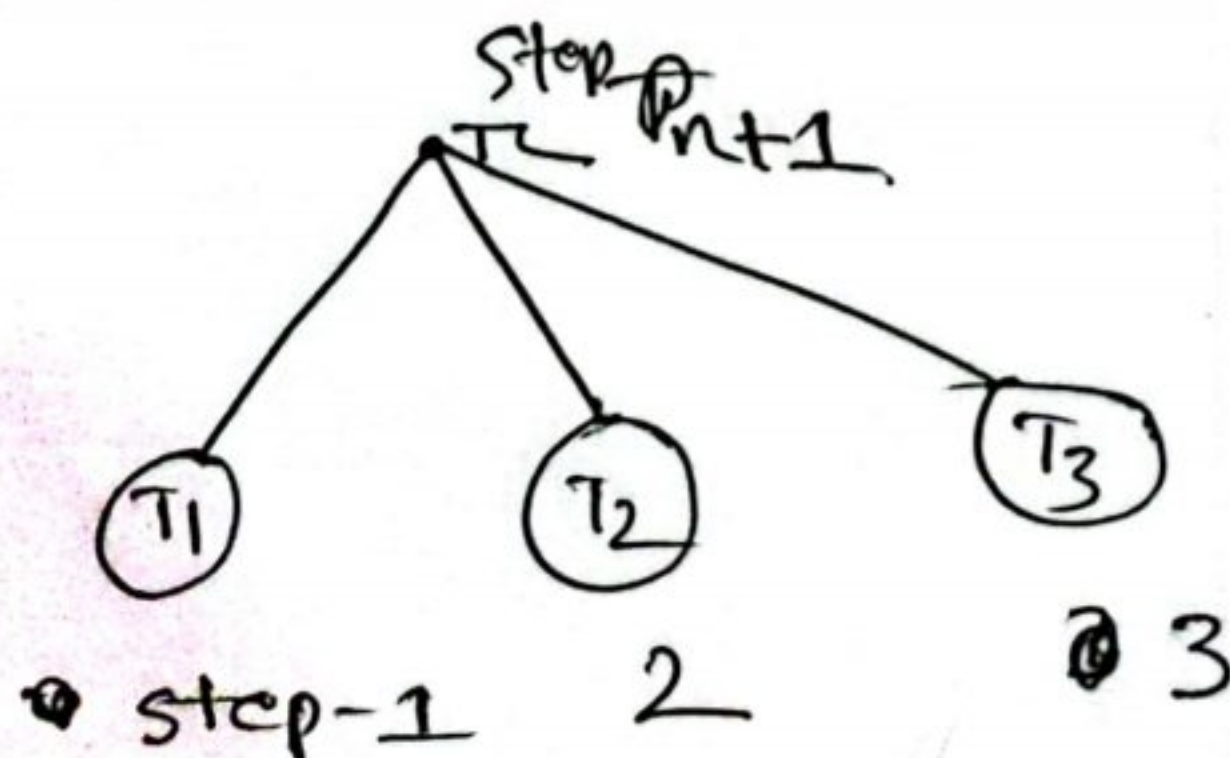
Pre Order Let T be an ordered rooted tree with root r . If T consists only of r , then r is the preorder traversal of T . Suppose $T_1, T_2, \dots, T_n$ are subtrees of r from left to right in T . The preorder traversal begins by visiting r , and continues by traversing T_1 in preorder, then T_2 in preorder, and so on, until T_n is traversed in preorder.

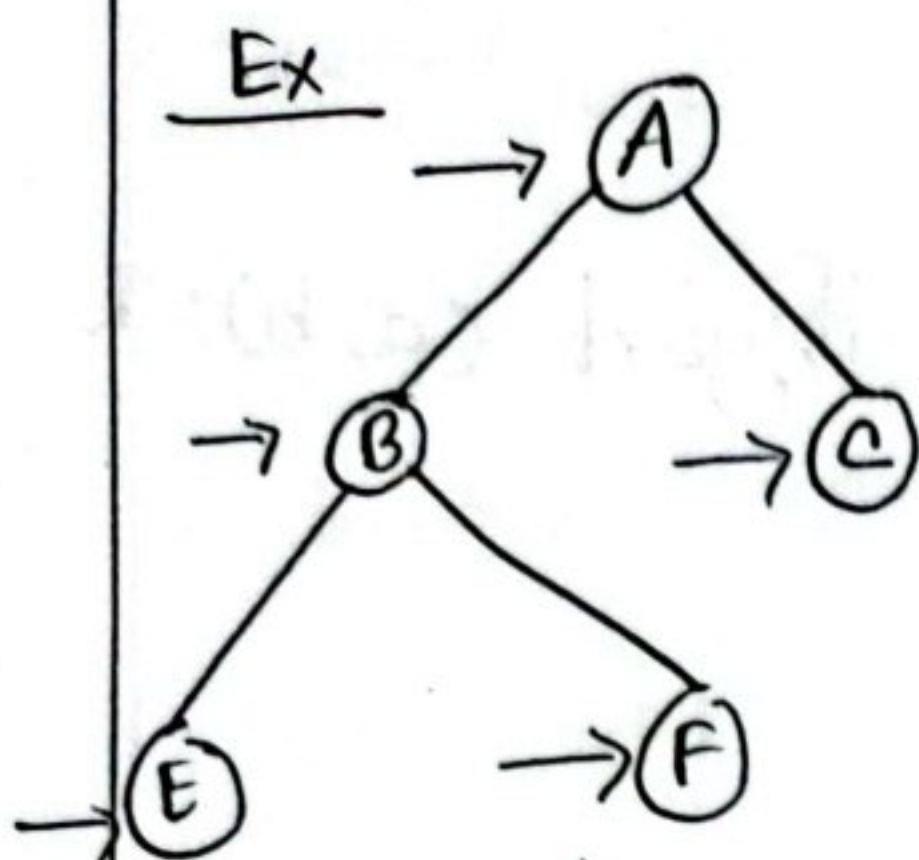


Inorder traversal: Let T be an ordered rooted tree with root π . If T consists only of π , then π is the inorder traversal of T . Suppose, $T_1, T_2, \dots, T_n$ are subtrees of π from left to right in T . The inorder traversal begins by traversing T_1 in order, then visiting π , and continues by traversing T_2 in order, and so on, until T_n is traversed in order.

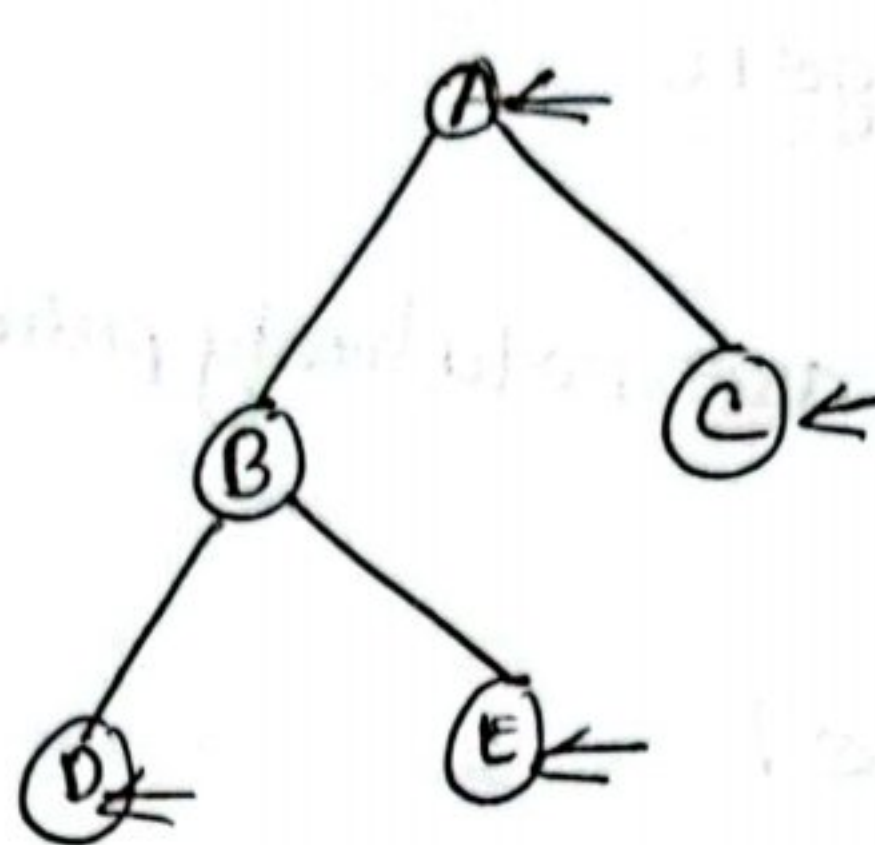


Postorder: Let T be an ordered rooted tree with root π . If T consists only of π , then π is the postorder traversal of T . Suppose, $T_1, T_2, T_n, \dots$ are the subtrees of π from left to right in T . The postorder traversal begins by traversing T_1 , T_2 in postorder, soon, after T_n is traversed in postorder, π is visited.

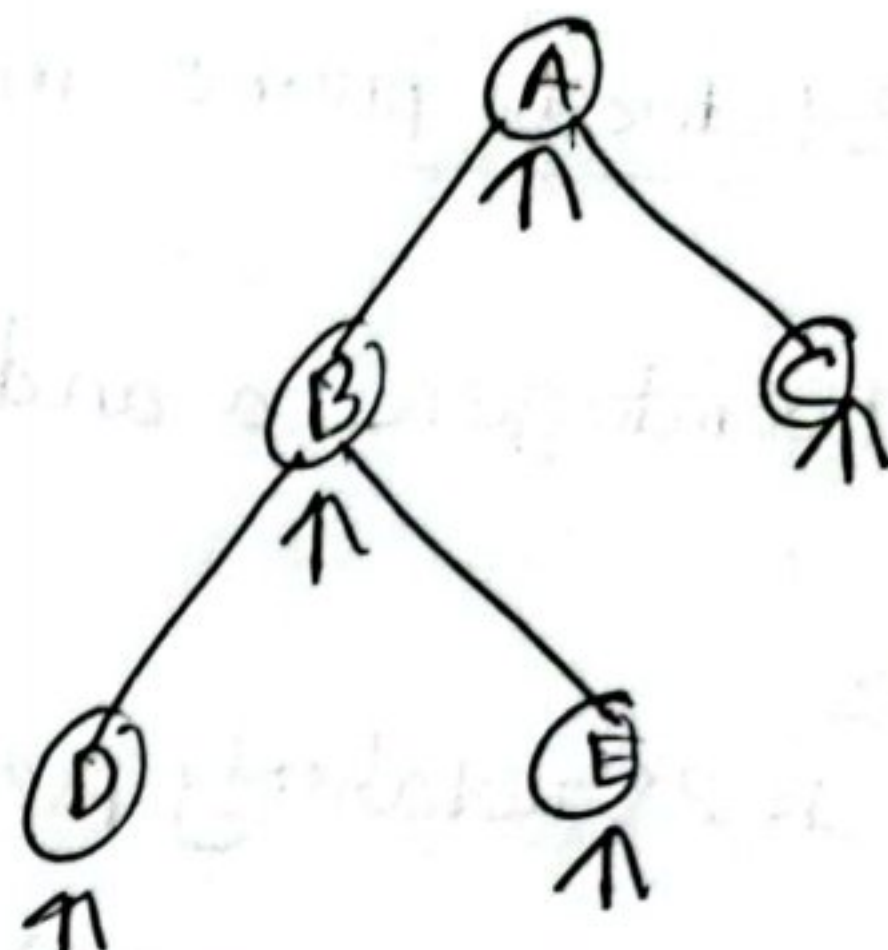




ABDEC
pre



DEDBCA
post



DBEAC
Inorder

Number theory

Integer: Any whole number (+, -, zero) is called an integer. Natural or positive numbers, $N = \{1, 2, 3, \dots\}$

• If $a \leq b$ and $b \leq a$, then $a = b$.

law of trichotomy: for all integers a and b , exactly one of the following holds: $a < b$, $a = b$ or $a > b$.

Relatively prime integers

Two integers a and b are relatively prime if $\gcd(a, b) = 1$.

Ex!

- 15, 28 relatively prime?

Yes, $\gcd(15, 28) = 1$

- 35, 28 relatively prime?

~~No~~ No, $\gcd(35, 28) = 7$.

Least common multiples

The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by a and b .

Examples:

$$\text{lcm}(3, 7) = 21$$

$$\text{lcm}(4, 6) = 12$$

$$\text{lcm}(5, 10) = 10.$$

$$\begin{array}{r} 3 \overline{) 3, 7} \\ 7 \overline{) 3, 7} \\ \hline 1, 1 \end{array}$$

$$3 \times 7 = 21$$

$$\begin{array}{r} 2 \overline{) 4, 6} \\ 2 \overline{) 2, 3} \\ 2 \overline{) 2, 3} \\ \hline 1, 1 \end{array}$$

$$2 \times 2 \times 3 = 12$$

Example

$$a = 60 = 2^2 3^1 5^1$$

$$b = 54 = 2^1 3^3 5^0$$

$$\text{lcm}(a, b) = 2^2 3^3 5^1 = 540$$

Modular Arithmetic

$a \bmod m$ the remainder when a is divided by m

Ex: $9 \bmod 4 = 1$

$$9 \bmod 3 = 0$$

$$9 \bmod 10 = 9$$

$$-13 \bmod 4 = -13$$

Congruences

1 Let a and b be integers and m be a positive number.

■ $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

~~that~~ (2) we say that a is congruent to b modulo m if m divides $a - b$.

(3) $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m .

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Ex: $3 \equiv 3 \pmod{17}$ is True: Any number is congruent to itself ($3-3=0$, divisible by all)

$3 \equiv -3 \pmod{17}$ false. $a-b=6$ isn't divisible by 17

$172 \equiv 177 \pmod{5}$ true: -5 is a multiple of 5

$-13 \equiv 13 \pmod{26}$ is true! 26 divisible by 26.

The Euclidean Algorithm

It finds the greatest common divisor of two integers a and b .

for ex: $(287, 91)$ we divide 287 by 91

$$287 = 91 \cdot 3 + 14$$

we know that for integers a, b, c ,

if $a|b$ and $a|c$, then $a|(a+c)$.

Therefore any divisor of 287, 91 must also be a divisor

$$287 - 91 \cdot 3 = 14$$

Consequently, $\gcd(287, 91) = \gcd(14, 91)$.

In next step we divide 21 by 14.

$$21 = 14 \times 1 + 7,$$

This means $\text{gcd}(14, 21) = \text{gcd}(14, 7)$.

So, we divide 14 by 7 we get,

$$14 = 7 \times 2 + 0$$

We find that $7 \mid 14$, thus $\text{gcd}(14, 7) = 7$

therefore, $\text{gcd} = 7$

$$\begin{array}{r} (21, 14) \\ 21 \\ 14 \end{array}$$

■ In pseudocode, the algorithm can be implemented as follows:

procedure $\text{gcd}(a, b: \text{positive integer})$

$x := a$

$y := b$

while $y \neq 0$

begin

$x := x \bmod y$

$x := y$

$y := x$

end { x in $\text{gcd}(a, b)$ }

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Representation of Integers

Let b a positive integer greater than 1. If n is the integer,

it can be uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0,$$

(k is a non negative integer).

for $b = 10$.

$$\bullet 859 = 8 \cdot 10^2 + 5 \cdot 10^1 + 9 \cdot 10^0$$

for $b = 2$ (binary expression)

$$(10110)_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= (22)_{10}.$$

Example $b = 16$ (Hexadecimal)

$$(3A0F)_{16} = 3 \times 16^3 + 10 \times 16^2 + 0 \times 16^1 + 15 \times 16^0$$

$$= (14863)_{10}.$$

Ex: what is the base 8 expansion of $(12345)_{10}$?

$$12345 = 8 \cdot 1543 + 1$$

$$1543 = 8 \cdot 192 + 7$$

$$192 = 8 \cdot 24 + 0$$

$$24 = 8 \cdot 3 + 0$$

$$3 = 8 \cdot 0 + 3$$

The result is $(12345)_{10} = (30071)_8$.

Addition of integers

Add $a = (1110)_2$ and $b = (1011)_2$

$$a_0 + b_0 = 0 + 1 = 0 \cdot 2 + 1, \text{ so } c_0 = 0 \text{ and } s_0 = 1$$

$$a_1 + b_1 + c_0 = 1 + 1 + 0 = 1 \cdot 2 + 0 \text{ so } c_1 = 1, s_1 = 0$$

$$a_2 + b_2 + c_1 = 1 + 0 + 1 = 1 \cdot 2 + 0 \text{ so } c_2 = 1, s_2 = 0$$

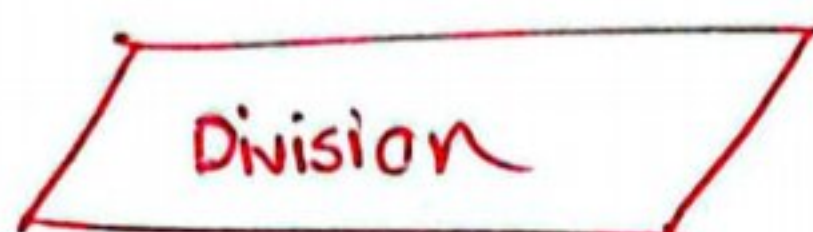
$$a_3 + b_3 + c_2 = 1 + 1 + 1 = 1 \cdot 2 + 1, c_3 = 1, s_3 = 1$$

$$s_4 + s_3 = 1$$

$$s = a + b = (11001)_2$$

$$\begin{array}{r} \# \quad 1 \quad 7583 \\ \quad 4932 \\ \hline 12515 \end{array}$$

$$\begin{array}{r} \boxed{1} \quad (1011)_2 \\ \quad (1010)_2 \\ \hline (10101)_2 \end{array}$$



Let a, b, c be integers such that $a = b \cdot c$

Then b and c are said to divide a , while a is said to be a multiple of b .

" $|$ " denotes divides.

$$b|a \wedge c|a$$

Definition:

If a and b are integers with $a \neq 0$, we say that a divides

b if there is an integer c so that $b = ac$.

When a divides b , we say that a is a factor of b and

that b is a multiple of a .

Then notation $a|b$ means that a divides b .

Ex:

1. $77/7$. False cause small can't be divided by a bigger number

2. $7/77$. True, because $77 = 7 \cdot 11$.

3. $24/242$ True, $\parallel$ $242 = 24 \cdot 10$

4. $0/24 =$ False, only 0 is divisible by 0.

5. $24/0$. True. 0 is divisible by every number

Divisibility theorem

for int a, b and c it is true that,

1. If $a|b$ and $a|c$, then $a|(b+c)$

Ex: $3|6$ and $3|9$ so $3|15$

$$a|b \wedge a|c \rightarrow a|(b+c)$$

$$17|34 \wedge 17|170 \rightarrow 17|(204)$$

2. If $a|b$, then $a|bc$ for all int's etc

$5|10$, so $5|20, 5|30, 5|40$

$$a|b \rightarrow a|bc$$

$$17|34 \rightarrow 17|340$$

3188 $a|b$ and $b|a$, then $a|c$

418 and 8124 so 4124

$$a|b \wedge b|c \rightarrow a|c$$

$$6112 \wedge 121144 \rightarrow 61144$$

Algorithm

Let a be an int and d a positive int. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

In above equation,

$d = \text{divisor}$

$a = \text{dividend}$

$q = \text{quotient}$

$r = \text{remainder}$

Ex 17 by 5

$$17 = 5 \cdot 3 + 2$$

17 dividend

5 divisor

3 quotient

2 remainder

Prime Number

Def: ■ A number $n > 2$ is a prime if it is only divisible by

1 and itself. Example: 2, 3, 5...

■ A number > 2 is called composite number which is not prime.

Ex: 4, 6, 8, 9...

Q-1 If n is a composite number, then n has a prime divisor less than or equal to $\sqrt{n}$. (Spring-18 (1-A))

Proof: If n is a composite number, then it has a factor $1 < a < n$

Hence $n = ab$ where a and b are positive integers > 1 .

We see that, $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$,

since otherwise $ab > \sqrt{n} \cdot \sqrt{n} = n$.

Hence, n has a positive divisor not exceeding $\sqrt{n}$.

This divisor is either prime, or by the fundamental

theorem of arithmetic, has a prime divisor. In

either case, n has a prime divisor

less than or equal to $\sqrt{n}$.

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Mersenne Prime

Def: The prime numbers of the form $2^p - 1$, where p is prime, are called Mersenne prime.

$2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^7 - 1 = 127$ are Mersenne prime

Fundamental theorem of Arithmetic

Theorem: Every positive number greater than 1 can be (int)

written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

• $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$

• $641 = 641$

• $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$

• $1024 = 2 \cdot 2 \cdot \dots = 2^{10}$

Q-2 find the sequence of pseudorandom numbers by the linear congruential method with modulus $m=9$, multiplier $a=7$, increment $c=4$ and seed $x_0=3$. (Answer-17) (almost same)

Soln: Compute the terms of the sequence by successively using the $x_{n+1} = (7x_n + 4) \bmod 9$ with $x_0 = 3$.

$$x_1 = 7x_0 + 4 \bmod 9 = 7 \cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7 \quad (x_1)$$

$$x_2 = 7x_1 + 4 \bmod 9 = 7 \cdot 7 + 4 \bmod 9 = 53 \bmod 9 = 8 \quad (x_2)$$

$$x_3 = 7x_2 + 4 \bmod 9 = 7 \cdot 8 + 4 \bmod 9 = 60 \bmod 9 = 6 \quad (x_3)$$

$$x_4 = 7x_3 + 4 \bmod 9 = 7 \cdot 6 + 4 \bmod 9 = 46 \bmod 9 = 1 \quad (x_4)$$

$$x_5 = 7x_4 + 4 \bmod 9 = 7 \cdot 1 + 4 \bmod 9 = 11 \bmod 9 = 2 \quad (x_5)$$

$$x_6 = 7x_5 + 4 \bmod 9 = 7 \cdot 2 + 4 \bmod 9 = 18 \bmod 9 = 0 \quad (x_6)$$

$$x_7 = 7x_6 + 4 \bmod 9 = 7 \cdot 0 + 4 \bmod 9 = 4 \bmod 9 = 4 \quad (x_7)$$

$$x_8 = 7x_7 + 4 \bmod 9 = 7 \cdot 4 + 4 \bmod 9 = 32 \bmod 9 = 5 \quad (x_8)$$

$$x_9 = 7x_8 + 4 \bmod 9 = 7 \cdot 5 + 4 \bmod 9 = 39 \bmod 9 = 3$$

The sequence generated is,

3, 7, 8, 6, 1, 2, 0, 4, 7, 5, 6, 1, 2, 0, 4, 5, 3,

after generating 9 times

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Ex-1 What is gcd of (48, 72)

common divisor of 48 & 72 = (1, 2, 3, 4, 6, 8, 12 and 24)

$$\text{So gcd} = (48, 72) = 24.$$

Q-2 Finding GCD's as a linear combination: (Autumn 17 1-D)

Express (252, 198) as a linear combination of 252 & 198

Soln: First use the Euclidean algorithm to show gcd

$$(252, 198) = 18.$$

$$\textcircled{i} \quad 252 = 1 \cdot 198 + 54$$

$$\textcircled{ii} \quad 198 = 3 \cdot 54 + 36$$

$$\textcircled{iii} \quad 54 = 1 \cdot 36 + 18$$

$$\textcircled{iv} \quad 36 = 2 \cdot 18$$

• Now, working backwards, from $\textcircled{iii}$ and $\textcircled{iv}$

$$18 = 54 - 1 \cdot 36$$

$$36 = 198 - 3 \cdot 54$$

Substituting the 2nd equation into the 1st yields

$$18 = 54 - 1 \cdot (198 - 3 \cdot 54)$$

$$= 4 \cdot 54 - 1 \cdot 198$$

Substituting ~~5.198 into~~ eqn ① into 1st yield

$$18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198$$

$$= (4 \cdot 252 - 5 \cdot 198)$$

An

Q-3 find an inverse of 3 modulo 7. (spring 17 - 1(0))

Soln: Because, $\gcd(3, 7) = 1$, by Theorem 1, an inverse

of 3 modulo 7 exists

using the euclidian theorem! $7 = 2 \cdot 3 + 1$

from the eqn - we get, $-2 \cdot 3 + 1 \cdot 7 = 1$, and see that

-2 & 1 are the Bézout coefficients of 3 and 7.

hence, -2 is an inverse of 3 modulo 7.

Also every integer congruent to -2 modulo 7 is an inverse of

3 modulo 7, i.e. 5, -9, 12 etc